# Lowered crossover points of Anthropic's subexponential 3-SAT algorithm

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**Abstract**

The October 2026 refutations of the Exponential Time Hypothesis count satisfying assignments in time $2^{o(n)}$, but an asymptotic bound does not say from which input size, the *crossover point*, the algorithms beat the best known ones. We compute the crossover point of the algorithm of [1], and we show why it is so large. That algorithm contracts a complete-graph tensor network on $K$ nodes by signed sums of networks, each contracted in its own order, applied recursively at astronomically large $K$.

We make the #3-SAT recursion of [1] explicit and lower its crossover points, computed from the main term of the proved bound, by proved changes (least admissible group sizes, an explicit constant for its pieces, a contraction-tree base case, an exact count of edge bits, a clustering lemma for the top level). The main theorem of [1] has two parts, a bound in the number $m$ of clauses and a bound in the number $n$ of variables. For the bound in $m$, for #3-SAT at clause density one ($m = n$), the crossover point drops to about $2^{420}$ clauses from $2^{33858}$; for the bound in $n$, for general 3-SAT, our estimate drops to $2^{8.6\cdot 10^9}$ variables from $2^{2.2\cdot 10^{12}}$. We price every lever of the construction, to show where the distance comes from. With integer tables the recursion computes, for every $j$, the number of assignments satisfying exactly $j$ clauses directly, in time $2^{O(m/\sqrt{\log m})}$. We observe that this refutes the Sparse Max-3-SAT Hypothesis of Alman and Vassilevska Williams; the refutation already follows from either refutation of the Exponential Time Hypothesis by the circuit reduction of Impagliazzo, Paturi and Zane, but neither paper states it. At clause density one we lower the crossover point of the spectrum computation against the best known Max-3-SAT algorithms to about $2^{1016}$ clauses from $2^{68364}$.

The distance is structural, not a matter of constants. A polylogarithmic decay of the exponent per edge bit in $K$ forces crossover points exponential in a power of $1/\beta$, where $\beta$ is the exponent per edge bit needed to win, while a polynomial decay would make them logarithmic in $1/\beta$; the constants we improve change neither. We define a rewrite-and-merge class that contains the recursion and all our constructions, and prove that every grid algorithm in it, the recursion as written included, has exponent at least $1/\log_2 K$ per edge bit, so through the reduction of [1] no such algorithm can gain for general 3-SAT below about $2^{6000}$ variables (main-term model). For two balanced homogeneous merges we prove that the exponent is at least $(\log_2 K)^{-3}$, which excludes running times $2^{n^{1-c}}$. The key step uses quantum functionals to show that configurations on at most three cells of two merges cannot trade cost between them. The class leaves one door open, configurations that meet three or more groups of a merge; we argue that a usable crossover point needs it, and that opening it is not enough.

Independently of SAT, we formalise the algorithm of [1] as a model of computation, Signed Network Sums, which contains rank-one methods and contraction trees as special

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cases. On three tables it collapses to rank-one methods, but at $K = 4$ one-level constructions contract the network in $N^{23/6}$ if $\omega = 2$ and in about $N^{3.93}$ on the current (preprint) matrix-multiplication curve, below the bound $N^4$ of every rank-one method and every contraction tree.

***Use of AI.*** *This paper was written with Anthropic's Claude models, Claude Opus 5.5 and Claude Fable 5.1, through Claude Code, in October 2026. Under the author's direction they searched the literature, searched for and developed the proofs, wrote and ran the scripts behind the computed values, drafted and revised the text, and checked it against the cited sources; the research questions, the choice among the directions proposed and the decision on every statement are the author's. The author takes full responsibility for the content.*

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# 1 Introduction

In October 2026 two papers refuted the Exponential Time Hypothesis [1, 2]. Both count the satisfying assignments of a 3-CNF formula in time $2^{o(n)}$, and both say plainly that their algorithms are not meant to be run. The algorithm of [1] reduces counting to a single algebraic problem, which we call the *clique sum*: given $K$ nodes, a string of $\lambda$ bits on every edge of the complete graph $K_K$, and an arbitrary table at every node indexed by the strings on its edges, compute

$$Z = \sum_{\text{all edge strings}} \prod_{a} W_a(\text{strings at } a).$$

Equivalently, $Z$ is the contraction of the complete-graph tensor network with bond dimension $N = 2^\lambda$. The paper contracts it by what we call *Signed Network Sums*: it rewrites the product of the equality tests on a set of edges as an exact signed sum of terms, each of which is cheap to contract in its own order, merges groups of nodes term by term, and recurses on the merged instance. One level squares the number of nodes, and the exponent per edge bit falls by a constant factor per level, which yields a running time $2^{O(n \log\log n/\sqrt{\log n})}$ for 3-SAT.

**Why an explicit threshold.** A bound $2^{o(n)}$ refutes a hypothesis about the asymptotic exponent, but it says nothing about any particular input size: it does not tell whether the algorithm beats the best known algorithm for 3-SAT at a size one could ever write down, or only beyond every such size. The authors of [1] remark that their bound in the number of clauses beats trying all assignments only for $\log m > 8000$; against the best known algorithms, and for general 3-SAT, the distance is much larger. The size from which the proved bound wins, which we call the *threshold* (the crossover point of the bound against the best known algorithm), is the quantity that measures how far the refutation is from a usable algorithm. We compute it from the main term of the bound, which gives a lower bound on it (Definition 3.1). Computing it explicitly does two things that an $O$-bound cannot. It tells which constant of the construction contributes how much of the distance, so that one can see which improvements matter and which do not. And it turns the question whether a cleverer algorithm of the same kind could have a small threshold into a precise question, which we answer for the grid algorithms, a class that contains all known constructions.

**Main results.** First (Section 3), we make the recursion of [1] explicit and lower its thresholds by changes that are each proved. The main theorem of [1] has two parts, a bound $2^{O(km/\sqrt{\log m})}$ in the number $m$ of clauses and a bound $2^{O(k^2 n \log\log n/\sqrt{\log n})}$ in the number $n$ of variables, which covers general formulas through sparsification. For the bound in $m$, for #3-SAT at clause density one ($m = n$), the threshold drops to about $2^{420}$ clauses from $2^{33858}$ (Table 4); for the bound in $n$,

for general 3-SAT, our estimate drops to $2^{8.6\cdot10^9}$ variables from $2^{2.2\cdot10^{12}}$ (Table 5).[1] The algorithm of OpenAI [2] has larger crossover points still: from the constants in its text we estimate about $2^{4.4\cdot10^{10}}$ clauses at clause density one for its Appendix D route (Section 3.9), and its main-text route needs at least $2^{1.09\cdot10^{10}}$ variables, since below that size it is exhaustive search [3].[2] Its crossover point grows only polynomially in the exponent it must save, so it lies outside the class of our lower bound (Section 4), but its constant is $10^2$ to $10^4$ times too large: even with improvements we could justify but did not prove, it stays near $2^{3.7\cdot10^5}$.

Second (Section 3.8), with integer tables the same recursion computes the Max-3-SAT spectrum, the number of assignments satisfying exactly $j$ clauses for every $j$, in time $2^{O(m/\sqrt{\log m})}$. We observe that this refutes the Sparse Max-3-SAT Hypothesis of Alman and Vassilevska Williams [31]. The refutation already follows from either refutation of the Exponential Time Hypothesis by the circuit reduction of [32, Theorem 5], but neither paper states it. At clause density one we lower the threshold of the spectrum computation against the best known Max-3-SAT algorithms to about $2^{1016}$ clauses from $2^{68364}$ (Table 6).

Third (Section 4), a lower bound explains these sizes. We define a class of rewrite-and-merge algorithms that contains the recursion of [1] and all our constructions, and prove that every grid algorithm in it, the recursion as written included, has exponent at least $1/\log_2 K$ per edge bit on clique sums with $K$ nodes, that is, time $2^{(1-o(1))S/\log_2 K}$ with $S$ edge bits (Theorem 4.13, Corollary 4.19). Through the reduction of [1] this means that no grid algorithm gains for general 3-SAT below about $2^{6000}$ variables (Corollary 4.15, in the main-term model of Section 4.1), whatever its pieces and constants. So the distance is not a matter of constants: a small threshold needs an exponent per edge bit that decays polynomially in $K$ (Proposition 4.1), and we argue that this needs pieces whose configurations meet three or more groups of a merge, which the class excludes and no known construction provides (Section 4.8).

Fourth (Section 2), independently of SAT, we study the clique sum itself at a fixed small number of nodes, that is, the contraction of the complete-graph tensor network on $K$ nodes with bond dimension $N$. We formalise the algorithm of [1] as a model of computation, Signed Network Sums (Definition 2.3), which contains rank-one methods and contraction trees as special cases, and ask whether it is stronger than both. On at most three tables it is not. At $K = 4$ it is, if the current matrix-multiplication curve, which rests on two preprints, holds: one-level constructions contract the network in time $N^{23/6}$ if $\omega = 2$ and in about $N^{3.93}$ on that curve, below the bound $N^4$ that holds for every rank-one method and every contraction tree. On the peer-reviewed curve of [9] they beat the best contraction tree computed from the same bounds, but not $N^4$.

[1] Independently, Tulsyan [4] evaluates the same recursion in a note of 29 October 2026, against exhaustive search, whereas we compare with the best known algorithms. For #3-SAT he obtains, for the counting route of the Lean development of [1] as written (with the polynomial degree of its Turing machine set to one), $2^{33858}$ clauses at density one and $2^{552040}$ at $m = 4.26n$, which agrees with column (P) of Table 4 and so checks it independently; with the free parameters of the same lemmas re-optimised, $2^{7163}$ and $2^{245009}$; and with further changes that he labels heuristic, among them a base case by fast matrix multiplication and $x_0 = 10$, $2^{1161}$ and $2^{26621}$. Our thresholds, all proved, are $2^{417}$ and $2^{10766}$, which is also below his heuristic values. Our tree base case (Lemma 3.7) and $x_0 = 7$ (Lemma 3.4) prove two of the changes he uses heuristically; the exact total $U$, Lemma C (Proposition 3.9), the lower bound of Section 4 and the results of Section 2 are outside the scope of his note. For Appendix D of [2] he reads a condition of its text as a hypothesis of the proof, which puts its crossover beyond $2^{2^{85}}$; Section 3.9 explains why our estimate sets that condition aside.

[2] Szeider [3] computes where the two algorithms of [2] stop being exhaustive search: below $2^{1.09\cdot10^{10}}$ variables for the main-text algorithm and below $2^{131071}$ variables and clauses for that of Appendix D. For the algorithm of [1] he evaluates its level recurrence against exhaustive search and finds $2^{275616}$ clauses at $m = 5n$, a value our model of Section 3.3 reproduces exactly (`calc/r123_szeider_check.py`). We compare with the best known algorithms throughout, which gives larger thresholds.

**The measure.** For an algorithm on the clique sum we write its running time as $2^{e \cdot S}$, where $S = \binom{K}{2}\lambda$ is the number of edge bits, and call $e$ the *exponent per edge bit*; $e(K)$ is the best exponent available at $K$ nodes. Every algorithm reads a node table, so $e(K) \geq 2/K$ (the input floor). A method that bounds tensor rank pays the flattening rank of $K_K$ across a balanced cut, about $1/2$ per edge bit, and a contraction tree pays the width of its widest edge, about $4/9$ per edge bit for large $K$. The reduction from 3-SAT turns a bound $e(K) <$ bar into a running time below that of the best known algorithms, where the *bar* bar is fixed by the number of edge bits per variable; for general 3-SAT it is of order $10^{-4}$. The three parts of the paper look at $e(K)$ at small $K$, at the astronomically large $K$ of the recursion, and at the reason why the two ends do not meet.

**The small end (Section 2).** On at most three tables the model of Signed Network Sums collapses to rank-one methods: it gives nothing beyond matrix multiplication there. At four nodes it already goes beyond rank. One level of the construction (a $2 \times 2$ grid, the eight-square identity on the crossing diagonals, row and column merges, rectangular matrix multiplication for the merged tables) contracts $K_4$ in $N^{23/6}$ if $\omega = 2$, and the pieces of Appendix B of [1] on long blocks lower this further; on the current matrix-multiplication curve, which rests on two 2026 preprints, the best value at a fixed block length is about $N^{3.93}$. Every rank-one method and every contraction tree pays at least $N^4$ at $K = 4$, and the best rank bound known for $K_4$ is $N^{4.638}$ [5]. At $K = 6, 9, 16$ the same one-level constructions provably beat every contraction tree at $\omega = 2$; with the pieces of Appendix B of [1], in the limit of long blocks, the margin stays a roughly constant fraction of the tree's cost at $K = 4, 6, 9$ while the gap to the input floor grows.

**The large end (Section 3).** The thresholds stated above come from changes that leave the structure of the recursion of [1] intact: the least admissible group size per level, the constant $c$ of the paper's Theorem B traced to $c = 30$, a contraction-tree base case instead of brute force, the exact number of edge bits of the clause network in place of the paper's bound, and a clustering lemma for the top level, with the paper's bookkeeping constant $C_0$ bounded by 810. Each change is either proved here or proved in [1] and only counted more carefully; the general-3-SAT values are estimates from the paper's closed level formula, and some of them are conditional on a named lemma or construction. The Max-3-SAT spectrum is subexponential in $n$ only at bounded density: the bound in $n$ for general formulas does not carry over, since it rests on a sparsification that keeps only the satisfying assignments, and dense Max-3-SAT is untouched. We price every lever (per-level factor, balancing, slack schedule, sparsification density, the parity-check kill) and show that the factor $\log \log n$ in the general bound is the logarithm of the sparsification density.

**The gap (Section 4).** Why do the gains at four nodes not become a fast recursion? The organising fact is a dichotomy for thresholds: a polylogarithmic decay of $e(K)$ gives thresholds exponential in a power of 1/bar, a polynomial decay would give thresholds logarithmic in 1/bar; the constants that Section 3 improves change neither. We then prove two bounds in the rewrite-and-merge class. Every grid algorithm (levels on $a \times b$ grids with any pieces on the rectangles; all known constructions are of this kind, including the recursion of [1] as written, for which the bound holds up to a negligible factor) has $1/e(K) \leq \log_2 K$, which would be attained by ideal pieces if they exist; this is the bound behind the threshold $2^{6000}$ above. Every algorithm with two balanced homogeneous merges has $1/e(K) \leq (\log_2 K)^3$, so no running time $2^{n^{1-c}}$ arises. The proof has two new ingredients: configurations of edges that lie in at most three of the four cells

cut out by two merges cannot trade cost between the merges (we prove this with the quantum functionals of Christandl, Vrana and Zuiddam; flattening-rank inequalities alone do not suffice, as a triangle shows), and a counting bound for the configurations that can. Along the way we show that sharing computation across terms cannot beat a table-size floor, and argue that table-size arguments at any fixed $K$ reduce to a question about four-leg pieces, which is the open window at $K = 4$. A four-party rank inequality conjectured by Cadney, Huber, Linden and Winter and proved by Song, Chen, Sun and Hu [7, 8] removes the conjecture from our three-cut size floor at $K = 4$ (the floor itself is computed on a grid of parameters). The class leaves one door open: configurations that meet three or more groups of a merge. We say precisely why opening it is necessary, and why it is not sufficient, for a usable threshold.

**What is and is not claimed.** Every statement is marked as a theorem with proof, a computed value (with the script that computes it; numerical searches give lower estimates of a maximum), an argued claim, or a conditional statement with its condition. All thresholds are *main-term* thresholds (Definition 3.1): lower bounds on the size from which a bound wins, with the lower-order terms bounded separately where we can. Values on the current matrix-multiplication curve rest on preprints; we give the peer-reviewed curve of [9] beside them where it matters. No SAT running time in this paper is usable; the point of the explicit constants is to show where the distance comes from.

**Organisation.** Sections 2, 3 and 4 treat the small end, the large end and the gap; each opens with its own roadmap. Section 3 does not depend on Section 2, and Section 4 uses Section 3 only through the bar and the main-term model. Section 5 collects the open problems. Appendices A, B and C hold the longer tables of these three sections. The computed values come from Python scripts in the repository `https://github.com/Majdoddin/anthropic-3sat-crossover`; script names such as `calc/r123_levers.py` are relative to its top folder, and its README lists which script computes which value. Our appendices are not to be confused with those of [1]: "Appendix B pieces" and "App. B" always mean the pieces of Appendix B of [1], which have per-level factor $\rho = 1/\sqrt{2}$ and are used throughout, not our Appendix B.

# 2 The small end: Signed Network Sums at a fixed number of nodes

This section studies the model of the October 2026 refutations of the Exponential Time Hypothesis at a fixed, small number $K$ of nodes. The question is whether Signed Network Sums are a genuinely new model or rank decompositions and contraction trees in disguise: if they gained nothing at small $K$, the recursion of [1] would owe its subexponential bound to the recursion alone. They do gain, from four nodes on, and the values computed here are also the reference points for the size arguments of Section 4. Section 2.1 defines the problem and three models of computation for it, and places them on a ladder of floors. Section 2.2 shows that on at most three tables the model of the refutations gives nothing beyond rank decompositions. Section 2.3 proves that at $K = 4$ it goes below the floor $N^4$ that binds every rank-one method and every contraction tree. Section 2.4 treats $K = 6, 9, 16$, and Section 2.5 collects the values per edge bit in one table.

## 2.1 The model and the ladder

**The clique sum.** Fix $K \geq 3$ and $\lambda \geq 1$ and put $N = 2^\lambda$. Every edge $\{u, v\}$ of the complete graph on $[K]$ carries a string $z_{uv} \in \{0,1\}^\lambda$. Node $v$ holds a table $W_v$, an arbitrary function (with values in $\mathbb{Q}$, say) of the $K-1$ strings on its edges; write $z_v = (z_{uv})_{u \neq v}$. The $(K, \lambda)$-*clique sum* is

$$Z \;=\; \sum_{(z_{uv})} \prod_{v \in [K]} W_v(z_v). \tag{1}$$

Equivalently, every node holds its own copy of each of its edge strings and the sum carries an equality test $[\, z_v(u) = z_u(v) \,]$ for every edge; we use this form when an identity is applied to equality tests. The instance has $S = \binom{K}{2}\lambda$ edge bits and $K$ tables of $N^{K-1}$ entries each. For $K = 3$, $Z = \operatorname{tr}(ABC)$ for three $N \times N$ matrices.

As a $K$-linear form in the tables, $Z = \langle T_K, W_1 \otimes \cdots \otimes W_K \rangle$, where $T_K$ is the *graph tensor* of the complete graph with bond dimension $N$: the tensor product over all edges $\{u, v\}$ of the $N \times N$ identity matrix placed on the legs of $u$ and $v$ [5].

**Cost exponents.** We count arithmetic operations. An algorithm has *exponent $e$ per edge bit* if, for every fixed $K$ and every $\varepsilon > 0$, it computes $Z$ in time $2^{(e+\varepsilon)S}$ for all tables and all large $\lambda$; equivalently in time $N^{E+\varepsilon}$ with $E = e\binom{K}{2}$. We state values both as exponents $E$ of $N$ and as $e$; brute force has $e = 1$. For a class of algorithms, $e(K)$ denotes the infimum of the exponents of its members.

**Three models.** All three evaluate tensor networks; they differ in how many networks and in how they are contracted.

**Definition 2.1** (Contraction trees)**.** A *contraction tree* for the clique sum is a rooted binary tree whose leaves are the $K$ tables. Each inner node $x$ with leaf set $L(x)$ stands for the tensor obtained by summing the product of the tables in $L(x)$ over all strings on edges inside $L(x)$; it is indexed by the strings on the edges between $L(x)$ and $[K] \setminus L(x)$, and it is computed (materialised) from the tensors of the two children, for instance by a rectangular matrix product. The cost of the tree is at least the largest number of entries of a tensor it materialises. This is the model of Austrin, Kaski and Kubjas [6]; the cost of a step is bounded above by fast rectangular matrix multiplication.

The second model is the one in which fast matrix multiplication and the bounds on graph tensors of [5] live: the clique sum is written as a sum of products of linear forms, one in each table.

**Definition 2.2** (Rank-one methods)**.** A *rank-one method* writes $Z = \sum_{r=1}^{R} \prod_v \ell_{r,v}(W_v)$ with linear forms $\ell_{r,v}$, i.e. it uses a rank decomposition of $T_K$ with $R$ terms, and costs at least $R$. Algorithms obtained from bounds on the border rank or the asymptotic rank of $T_K$ (with interpolation or tensor powers) are included: their cost is at least the border rank, respectively the asymptotic rank, of the tensor they decompose.

The third model is the one of the refutation, which mixes the first two: many networks, each contracted by its own tree.

**Definition 2.3** (Signed Network Sums)**.** A *Signed Network Sum* algorithm writes $Z = \sum_q c_q Z_q$ with fixed scalars $c_q$, where each $Z_q$ is the value of a tensor network whose tensors are computed from the input tables (by re-indexing, merging nodes, multiplying by fixed tensors), and each term is contracted along its own contraction tree. Its cost is at least the total number of entries of all tensors it materialises. This is the model of Anthropic's refutation [1]: the equality tests are expanded by an exact identity into a signed sum of products of pieces, and in each term a different set of nodes is merged.

A rank-one term is a network without edges, and a contraction tree is a Signed Network Sum with one term; so both are special cases of Signed Network Sums. Neither is defined as a part of the other. The following proposition gives the lower bounds that each model has to beat; we call them floors, and their order is the ladder of the subsection's title.

**Proposition 2.4** (The ladder)**.** *For every $K \geq 3$:*

*(a) every algorithm that computes $Z$ for all tables (in particular every Signed Network Sum algorithm) has $e \geq 2/K$ (*input floor*);*

*(b) every rank-one method has $e \geq \lfloor K^2/4 \rfloor / \binom{K}{2}$, and this holds for rank, border rank and asymptotic rank alike (*flattening floor*);*

*(c) every contraction tree has $e \geq c(K-c)/\binom{K}{2}$ with $c = \lceil K/3 \rceil$ (*width bound*).*

*Proof.* (a) $Z$ is linear in each table and every entry of every table has a nonzero coefficient (set all other tables to the all-ones table). An arithmetic computation that never reads an input entry computes a value independent of it, so it reads all $KN^{K-1}$ entries and performs at least $KN^{K-1}/2$ operations. Per edge bit this is $(K-1)\lambda/(\binom{K}{2}\lambda) = 2/K$.

(b) For a set $A \subset [K]$ consider the flattening of $T_K$ that groups the legs of the nodes in $A$ against the rest. Every edge inside $A$ or inside its complement contributes a factor of rank one, and every edge across contributes an $N \times N$ identity; since rank is multiplicative under Kronecker products, the flattening has rank $N^{|A|(K-|A|)}$, which is $N^{\lfloor K^2/4 \rfloor}$ for $|A| = \lfloor K/2 \rfloor$. A decomposition with $R$ rank-one terms gives $R \geq N^{\lfloor K^2/4 \rfloor}$. Border rank is bounded below by every flattening rank because matrix rank is lower semicontinuous; asymptotic rank is, because flattening ranks are multiplicative under tensor powers.

(c) Let $x$ be a node of the tree with $|L(x)| \geq K/3$ none of whose children has this property; it exists, since the root has it. If $x$ is a leaf then $K \leq 3$ and $s := |L(x)| = 1$. Otherwise both children have fewer than $K/3$ leaves, so $s < 2K/3$. In both cases $s \geq c$ and $K - s \geq c$. The tensor at $x$ is indexed by the $s(K-s)$ strings between $L(x)$ and its complement, and since the tables are arbitrary it has $N^{s(K-s)}$ entries. As $s \mapsto s(K-s)$ is symmetric about $K/2$ and increasing below it, $s(K-s) \geq c(K-c)$. □

The floors at the values of $K$ used below are in Table 1. Anthropic computes the limit $4/9$ of the width bound [1, Appendix G]; [6] prove the lower bound $\Omega(N^{\mathrm{bw}})$ for trees in terms of branchwidth, of which (c) is the special case we need. The tree and rank-one floors coincide at $K = 4$, where every $2|2$ cut is crossed by four edges: both are $N^4$, $e = 2/3$. No lower bound above the input floor is known for Signed Network Sums at any fixed $K$, and Anthropic's recursion shows that $e(K) \to 0$ for Signed Network Sums as $K \to \infty$ [1]. The ladder therefore reads

$$\tfrac{2}{K} \;\leq\; e_{\mathrm{NS}}(K) \;\leq\; \min\bigl(e_{\mathrm{tree}}(K),\, e_{\mathrm{rank}}(K)\bigr),$$

| $K$ | edge bits per $\lambda$ | input floor $2/K$ | width bound (trees) | flattening floor (rank-one) |
|---|---|---|---|---|
| 3 | 3 | 2/3 | 2/3 | 2/3 |
| 4 | 6 | 1/2 | 2/3 | 2/3 |
| 6 | 15 | 1/3 | $8/15 \approx 0.533$ | 3/5 |
| 9 | 36 | $2/9 \approx 0.222$ | 1/2 | $5/9 \approx 0.556$ |
| 16 | 120 | 1/8 | 1/2 | $8/15 \approx 0.533$ |
| $\infty$ | | 0 | 4/9 | 1/2 |

Table 1: Floors per edge bit (Proposition 2.4). The width bound is $c(K-c)/\binom{K}{2}$ with $c = \lceil K/3 \rceil$: $2, 4, 8, 18, 60$ edges for $K = 3, 4, 6, 9, 16$.

$$e_{\text{tree}}(K) \geq \tfrac{c(K-c)}{\binom{K}{2}} \to \tfrac{4}{9}, \quad e_{\text{rank}}(K) \geq \tfrac{\lfloor K^2/4 \rfloor}{\binom{K}{2}} \to \tfrac{1}{2}.$$

Reaching the tree floor needs good rectangular matrix multiplication at every step of the tree; reaching the flattening floor needs the asymptotic rank of $T_K$ to equal its largest flattening rank, which for $K = 3$ is the statement $\omega = 2$.

### 2.2 Three tables: the model is rank-one

A crossing, the object that Anthropic's identity rewrites, is a pair of equality tests on two disjoint pairs of nodes; it needs four nodes. On three tables there is none, and the per-term choice of contraction order is empty: a network on three nodes is contracted by multiplying two of its tensors and pairing the result with the third. The following theorem makes this precise. It covers every problem that is a trilinear form in three tables, such as the $K = 3$ clique sum, (min,+)-type products after the usual encodings, and pruned matrix products, in which the output is restricted to a given support.

Let $\mathcal{T}(W_1, W_2, W_3)$ be a trilinear form in three tables. A *three-node network term* is given by finite index sets $I, J, L$ and linear maps $\alpha, \beta, \gamma$ that send $W_1, W_2, W_3$ to matrices $A = \alpha(W_1) \in \mathbb{Q}^{I \times J}$, $B = \beta(W_2) \in \mathbb{Q}^{J \times L}$, $C = \gamma(W_3) \in \mathbb{Q}^{L \times I}$; its value is $\operatorname{tr}(ABC)$. Re-indexing, splitting an index into several edges, parallel edges, fixed factors that depend on the strings of one node, and restriction to a support are all linear maps of this kind. An index shared by all three nodes (a hyperedge with values in a set $H$) is also covered: replace $I, J, L$ by $I \times H$, $J \times H$, $L \times H$ and the three tensors by block-diagonal matrices, whose triple product has trace $\sum_h \operatorname{tr}(A_h B_h C_h)$. The tensors of the network are images of the three input tables; no table is factored into several nodes.

**Theorem 2.5** (Three-table collapse)**.** *Let an algorithm compute $\mathcal{T} = \sum_q c_q \operatorname{tr}(A_q B_q C_q)$ as a signed sum of three-node network terms, where in each term the first contraction (the product of two of the three matrices) is computed by a bilinear algorithm with $R_q$ multiplications, and the final pairing with the third matrix is arbitrary. Then*

$$\mathcal{T} = \sum_q \sum_{r=1}^{R_q} c_q \, u_{q,r}(W_1) \, v_{q,r}(W_2) \, w_{q,r}(W_3)$$

*with linear forms $u, v, w$. Hence $\mathcal{T}$ has tensor rank at most $\sum_q R_q$, the algorithm is a rank-one method for $\mathcal{T}$ with at most as many terms as it performs multiplications, and its cost is at least*

*the tensor rank of $\mathcal{T}$. For the $K = 3$ clique sum the cost is at least $R(\langle N, N, N\rangle) \geq N^\omega$, so $e \geq \omega/3$, and a single contraction tree attains $e = \omega/3$.*

*Proof.* Fix a term and suppose, after renaming, that it first computes $AB$. A bilinear algorithm with $R_q$ multiplications for the map $(A, B) \mapsto AB$ is an identity $AB = \sum_{r=1}^{R_q} \phi_r(A)\psi_r(B)M_r$ with linear forms $\phi_r, \psi_r$ and fixed matrices $M_r \in \mathbb{Q}^{I\times L}$. Then $\operatorname{tr}(ABC) = \sum_r \phi_r(A)\psi_r(B)\operatorname{tr}(M_rC)$, and with $u_{q,r} = \phi_r \circ \alpha_q$, $v_{q,r} = \psi_r \circ \beta_q$, $w_{q,r} = \operatorname{tr}(M_r\, \gamma_q(\cdot))$ every term is a sum of $R_q$ products of linear forms, one in each table. The final pairing contributes no further product of inputs: it is bilinear in $(AB, C)$, so the identity holds however the pairing is computed. Summing over $q$ gives the displayed identity, which holds for all tables because the algorithm is exact; so it is a rank decomposition of $\mathcal{T}$ with $\sum_q R_q$ terms. Each multiplication is an operation, so the cost is at least $\sum_q R_q \geq R(\mathcal{T})$.

For the $K = 3$ clique sum, $\mathcal{T}$ is the matrix multiplication tensor $\langle N, N, N\rangle$ (as the trilinear form $\operatorname{tr}(ABC)$), and $R(\langle n, n, n\rangle) \geq n^\omega$ for every $n$: otherwise taking tensor powers of a decomposition would give $\omega < \omega$. One contraction tree, a single matrix product followed by a pairing, costs $N^{\omega+\varepsilon}$ for every $\varepsilon > 0$. □

The theorem has two hypotheses, bilinear first contractions and nodes that are images of the input tables; the following remark says why neither loses much and what the theorem does not say.

**Remark 2.6.** (1) The per-term choice of order is immaterial also in a weaker sense: the three trees of a term compute the three products $AB$, $BC$, $CA$ of the same triple, and the trilinear form $\operatorname{tr}(ABC)$ of $\langle n_1, n_2, n_3\rangle$ is invariant under cyclic shifts and, by transposition ($\operatorname{tr}(ABC) = \operatorname{tr}(C^\top B^\top A^\top)$), under reversal; so the rank, and hence the exponent, of a rectangular product does not depend on which pair is multiplied first. (2) The restriction to bilinear first contractions loses little: every algorithm for matrix multiplication with exponent below $\omega + \varepsilon$ that we use comes from a rank decomposition, and in general the bilinear complexity of a bilinear map is at most twice its multiplicative complexity [17]. (3) The theorem is relative to the rank-one model: it says that on three tables the model of Signed Network Sums cannot beat the best rank decomposition of the target, not what that rank is. (4) It needs the network's nodes to be images of the input tables. A table that could be factored into several nodes would give a network on more nodes, which is a different problem.

At $K = 3$ the ladder collapses: input floor, flattening floor and width bound are all $2/3$, and all three models achieve exactly $\omega/3$.

### 2.3 Four nodes: below the rank floor

Four nodes are the first case in which the identity of [1] applies, and the first in which rank-one methods and contraction trees both pay a floor, $N^4$, above the input floor $N^3$. This subsection shows that one level of the construction of [1] goes below $N^4$: first at $\omega = 2$ with the eight-square identity (Theorem 2.7), then on the known matrix-multiplication curves with three further constructions (Corollary 2.8), and it ends with what is open between $N^3$ and these values.

**Setting.** Name the nodes $a, a'$ (top row) and $b, b'$ (bottom row). The side edges are $aa'$ and $bb'$, the column edges $ab$ and $a'b'$, and the two diagonals $ab'$ and $a'b$. Write $x$ and $v$ for the copies of

the string of $ab'$ held at $a$ and at $b'$, and $y$ and $u$ for the copies of the string of $a'b$ held at $a'$ and at $b$; the other four edges carry one string each. Then

$$Z = \sum W_a(z_{aa'}, z_{ab}, x)\, W_{a'}(z_{aa'}, z_{a'b'}, y)\, W_b(z_{bb'}, z_{ab}, u)\, W_{b'}(z_{bb'}, z_{a'b'}, v)\, [\, x = v\,][\, y = u\,], \quad (2)$$

and the factor $[\, x = v\,][\, y = u\,]$ is the *crossing*. The three 2|2 cuts are rows $\{a, a'\}|\{b, b'\}$, columns $\{a, b\}|\{a', b'\}$ and diagonals; the crossing crosses the first two.

By Proposition 2.4 the input floor is $N^3$, and every rank-one method and every contraction tree costs at least $N^4$. The best contraction tree merges $a$ with $a'$ by summing $z_{aa'}$, which is the product $\langle N^2, N, N^2\rangle$, does the same for $b, b'$, and pairs two tables of $N^4$ entries: cost $N^{2\omega(1,1/2,1)+\varepsilon}$. If $\alpha \geq 1/2$, this is $N^{4+\varepsilon}$. The best rank bound known for $T_4$ is that of Christandl, Vrana and Zuiddam: exponent per edge at most $\log_7(9/2) \approx 0.7729$ for every $K \geq 4$ [5, Cor. 1.2.6], i.e. $N^{4.638}$ at $K = 4$, independent of $\omega$.

**The eight-square identity.** The construction rewrites the crossing as a signed sum of terms, each of which factors cheaply across one of the two cuts that the crossing crosses, so that each term can be contracted by merging rows or by merging columns. Cut $x, y, u, v$ into $t = \lambda/3$ blocks of three bits ($x = (x_1, \ldots, x_t)$ and so on). Anthropic's form of the eight-square identity [1, Lemma in §2.4, eq. (12)] provides a sign function $\sigma : \{0,1\}^3 \times \{0,1\}^3 \to \{\pm 1\}$ (the multiplication table of the octonions in a basis $e_p$ with $e_p e_q = \sigma(p,q) e_{p\oplus q}$) with

$$2\,[\, x_s = v_s\,][\, y_s = u_s\,] = R_s + C_s, \qquad \begin{aligned} R_s &= [\, x_s \oplus y_s \oplus u_s \oplus v_s = 0\,]\, \sigma(x_s, y_s)\, \sigma(v_s, u_s), \\ C_s &= [\, x_s \oplus y_s \oplus u_s \oplus v_s = 0\,]\, \sigma(x_s, u_s)\, \sigma(v_s, y_s). \end{aligned} \quad (3)$$

Multiplying over the blocks, $2^t [\, x = v\,][\, y = u\,] = \sum_{Q \subseteq [t]} \prod_{s \notin Q} R_s \prod_{s \in Q} C_s$. The *row choice* $R_s$ factors across the row cut through three bits: $R_s = \sum_{k \in \{0,1\}^3} \big([\, x_s \oplus y_s = k\,]\sigma(x_s, y_s)\big)\big([\, u_s \oplus v_s = k\,]\sigma(v_s, u_s)\big)$, the first factor a function of the top row's strings and the second of the bottom row's. Symmetrically the *column choice* $C_s$ factors across the column cut through three bits. Across its own wrong cut each choice keeps six bits on each side.

**Theorem 2.7** ($K = 4$ at $\omega = 2$)**.** *If $\omega = 2$, the $(4, \lambda)$-clique sum can be computed in $N^{23/6+\varepsilon}$ operations for every $\varepsilon > 0$. Thus $e(4) \leq 23/36 \approx 0.639$ for Signed Network Sums, against the floor $2/3$ of every rank-one method and every contraction tree.*

*Proof.* Write $Z = 2^{-t} \sum_Q Z_Q$, where $Z_Q$ is (2) with the crossing replaced by $\prod_{s \notin Q} R_s \prod_{s \in Q} C_s$. Fix $Q$ with $|Q| = at$ and $a \leq 1/2$; for $a > 1/2$ exchange the roles of rows and columns, which is the mirror image (in the column arrangement the column choice is the cheap one).

*Merge the rows.* Using the factorisation of $R_s$, $Z_Q = \sum T_Q \cdot B'_Q$ where the top table

$$T_Q(z_{ab}, z_{a'b'}, (k_s)_{s \notin Q}, (x_s, y_s)_{s \in Q}) = \sum_{z_{aa'}} \sum_{\substack{x_s, y_s\ (s \notin Q) \\ x_s \oplus y_s = k_s}} W_a W_{a'} \prod_{s \notin Q} \sigma(x_s, y_s)$$

has $N \cdot N \cdot 8^{t-at} \cdot 64^{at} = N^{3+a}$ entries, the bottom table $B_Q$ is defined in the same way from $W_b, W_{b'}$ with $\sigma(v_s, u_s)$ and indices $(u_s, v_s)_{s \in Q}$, and $B'_Q$ is $B_Q$ with the $64 \times 64$ kernel $C_s$ applied on each block $s \in Q$. Applying the $at$ kernels one block at a time costs $64\, t\, N^{3+a}$, and the final pairing costs $N^{3+a}$.

*Build $T_Q$ by one rectangular product.* Split the row-choice blocks $[t] \setminus Q$ into halves $P_1, P_2$. For $s \in P_1$ substitute $x_s = k_s \oplus y_s$, for $s \in P_2$ substitute $y_s = k_s \oplus x_s$. Then $T_Q = \sum F \cdot G$, summed over the *inner* index $(z_{aa'}, (y_s)_{s\in P_1}, (x_s)_{s\in P_2})$, where

$$F = W_a \prod_{s\in P_1} \sigma(k_s \oplus y_s, y_s) \quad \text{indexed by } (z_{ab}, k_{P_1}, x_Q;\ \text{inner}),$$
$$G = W_{a'} \prod_{s\in P_2} \sigma(x_s, k_s \oplus x_s) \quad \text{indexed by } (\text{inner};\ z_{a'b'}, k_{P_2}, y_Q).$$

Every argument of $W_a$ and $W_{a'}$ is determined by these indices, so $F$ and $G$ are signed re-indexings of the input tables, built in time equal to their size. In exponents of $N$, the outer dimensions are $p = 1 + \frac{1-a}{2} + a = \frac{3}{2} + \frac{a}{2}$ on each side and the inner dimension is $q = 1 + (1-a) = 2 - a$; $F$ and $G$ have $N^{p+q} = N^{7/2-a/2}$ entries. If $\omega = 2$, the product $\langle N^p, N^q, N^p\rangle$ costs $N^{\max(2p,p+q)+\varepsilon}$ (cut the larger dimensions into square blocks). The bottom table is built in the same way.

*Count.* The cost of term $Q$ is $N^{c(a)+\varepsilon}\mathrm{poly}(\lambda)$ with $c(a) = \max(3 + a, \frac{7}{2} - \frac{a}{2})$, and there are $\binom{t}{at} \le N^{\mathrm{H}(a)/3}$ terms of type $a$, with H the binary entropy. On $[1/3, 1/2]$, $c(a) = 3 + a$ and $\mathrm{H}(a)/3 + 3 + a$ is increasing, with value $1/3 + 7/2 = 23/6$ at $a = 1/2$. On $[0, 1/3]$, $\mathrm{H}(a)/3 + \frac{7}{2} - \frac{a}{2} \le \frac{1}{3} + \frac{7}{2} = 23/6$. Summing over the $t + 1$ values of $at$ gives $N^{23/6+\varepsilon}\mathrm{poly}(\lambda)$. □

The octonion multiplication tensor, through which the row-choice blocks of $T_Q$ factor, plays no role here: the substitution $x_s = k_s \oplus y_s$ moves its three bits into the inner dimension of a matrix product. At exponents $\omega > 2$ the same construction gains only if the asymptotic rank of the octonion tensor is small (below about 10.03 on the VXXZ curve and 12.34 on the current curve), and the best bound we know on that rank, Alman and Li's, is far above (about 21.94 at $\omega = 9/4$) [15].

**Three further constructions.** All three start from the same row/column structure; they are stated with their status.

*Grouping counted by ranks.* Anthropic groups the terms with $|Q| = i$ into one piece $A_i$ named by extra bits [1, eq. (20), Lemma 2]. Counted by bits this changes nothing at $K = 4$: the merged table of $A_i$ is the disjoint union of the tables $T_Q$, $|Q| = i$ (proved). Counted by ranks it helps. As $64 \times 64$ matrices (rows $(x, y)$, columns $(v, u)$) one has $R_s^2 = 8R_s$ and $C_s = 2I - R_s$, so $P = R_s/8$ is a projector of rank 8, $P^\perp = I - P$ has rank 56, and $A_i = \sum_{|Z|\le i} c_{|Z|,i}\, P_{[t]\setminus Z} \otimes P^\perp_Z$ with $c_{z,i} = 2^z\binom{t-z}{i-z}8^{t-i}(-6)^{i-z}$. The row rank of $A_i$ is $\sum_{z\le i}\binom{t}{z}56^z 8^{t-z}$, below the bit count. Building one table per $Z$ and contracting gives $N^{3.8179}$ at $\omega = 2$, with table-size floor $N^{3.8012}$ (computed; the identity is checked exactly at $\lambda = 3, 6, 9, 12$).

*The covering hybrid.* A term $Q$ (rows merged) can be served by the table $T_D$ of any $D \supseteq Q$, in which the blocks of $D$ keep $(x_s, y_s)$; $T_Q$ follows from $T_D$ by applying $(x_s, y_s) \mapsto x_s \oplus y_s$ with sign $\sigma(x_s, y_s)$ on the blocks of $D \setminus Q$. Take a family of $dt$-sets covering all $at$-sets (the greedy covering has at most $\binom{t}{at}\binom{dt}{at}^{-1}(1 + \ln\binom{dt}{at})$ members [18]), compute each $T_D$ by the substitution of Theorem 2.7, derive the $T_Q$ blockwise, and contract each $T_Q$ as before. The exponent is

$$\max_a \min_{d\ge a} \max\Big[\tfrac{\mathrm{H}(a) - d\,\mathrm{H}(a/d)}{3} + \max\big(C_w(d),\, 3 + d + g\big),\ \tfrac{\mathrm{H}(a)}{3} + 3 + a\Big],$$

where $C_w(d)$ is the cost of the rectangular product of Theorem 2.7 with $a$ replaced by $d$ under the matrix-multiplication curve $w$, and $g = 0.0321$ bounds the intermediate arrays of the derivation. This uses neither the octonion rank nor long blocks. At $\omega = 2$ it gives $23/6$; its gain is at

| curve | status | best tree | covering hybrid | App. B ($m \to \infty$) | below $N^4$? |
|---|---|---|---|---|---|
| $\omega = 2$ | hypothetical | 4 | 23/6 | 3.547 (3.631, $m = 256$) | yes |
| current | two preprints | 4 | 3.9305 | 3.9142 | yes |
| OpenAI | claimed, Lean-claimed | 4.0264 | 3.9963 | 3.9720 | yes |
| VXXZ | peer-reviewed | 4.0860 | 4.0790 | 4.0802 (4.140, $m = 256$) | no |

Table 2: Exponents of $N$ at $K = 4$ (upper bounds; input floor 3, flattening floor and width bound 4, CVZ19's rank bound 4.638). At $\omega = 2$ also: Theorem 2.7 gives $23/6 \approx 3.833$ for the plain construction and grouping counted by ranks gives 3.8179. "Best tree" is the tree $N^{2\omega(1,1/2,1)}$ evaluated on the same curve, an upper bound; the lower bound for trees is $N^4$ on every curve. The App. B column is a limit $m \to \infty$ (argued) except where $m$ is given (computed).

exponents $\omega > 2$ (Table 2). The maximum over $a$ is at the grid point $a = 1/2$ exactly (the objective is non-decreasing in $a$), and every grid point in $d$ and in the bypass fraction is a valid algorithm, so the printed values are upper bounds given the curve (computed; derivation and bypass checked exactly at $\lambda = 3, 6, 9, 12$).

*Appendix B pieces on long blocks.* Anthropic's Appendix B gives, for each block of $m$ bits, an identity $\sum_{s=-2h}^{2h} A_s = h!\,[x = v][y = u]$ with $h = h_m$ the least $h$ with $p(h) \geq 2^m$ ($p$ the partition function), in which piece $A_s$ factors through $r_s = \log_2 \sum_{i=0}^{2h} p(s+i)$ bits across the rows and $c_s = \log_2 \sum_i p(2h - s - i)$ bits across the columns [1, App. B, (54)–(55)]. Used as one piece over all $\lambda$ bits it does not fit $K = 4$: we know no algorithm for the resulting bilinear map below $N^4$ (argued). Used per block, with $\lambda/m$ blocks, the product expands into $(4h+1)^{\lambda/m}$ terms, a factor $N^{\log_2(4h+1)/m}$, and each block is fused into the rectangular product or bypassed as in Theorem 2.7, now at a re-indexing cost $2^{O(m)}$ per block. As $m \to \infty$, $r_s/m \to \sqrt{2+u}$ and $c_s/m \to \sqrt{2-u}$ with $u = s/h$; the worst type is $u = 0$ and costs $2 + 4/(4-\sqrt{2}) \approx 3.547$ at $\omega = 2$ (closed form for pure types; mixtures of types computed, never worse), against the table-size floor $2 + \sqrt{2} \approx 3.414$. The limit $m \to \infty$ is argued from Anthropic's (46), not proved. At fixed $m$, with exact partition numbers, the bound (maximum cost over pure piece types plus the entropy term; mixtures checked only in the limit) is 3.796 for $m = 64$ and 3.631 for $m = 256$ at $\omega = 2$ (computed).

**Values per matrix-multiplication curve.** The constructions use rectangular matrix multiplication, so their exponents depend on the bound used for $\omega(1, k, 1)$. Each cost is a non-decreasing function of the curve: a better proved curve can only lower it. We give the values for four curves (Table 2): $\omega = 2$ (hypothetical); the *current curve* $\omega(1, k, 1) = 2$ for $k \leq 1/2$ and $\leq 1 + k + 1/(4k)$ for $k \geq 1/2$ (so $\alpha \geq 1/2$ and $\omega \leq 9/4$), proved in two independent preprints by Amanov, van den Berg and Zuiddam [10] and by Uznański [11], not peer-reviewed; OpenAI's bounds $\omega \leq 9/4$, $\alpha > 0.465$, $\omega(1, 0.709, 1) < 2.092$ [16], also not peer-reviewed; and the peer-reviewed curve of Vassilevska Williams, Xu, Xu and Zhou [9], interpolated linearly between the entries of its table (valid by convexity).

The corollary collects the values of Table 2 that are below $N^4$ and says on which curve and with which status each holds.

**Corollary 2.8** (Beyond rank at four nodes)**.** *On the current curve (resting on the preprints [10, 11]) the $(4, \lambda)$-clique sum can be computed in $N^{3.9305}$ operations by the covering hybrid, with three-bit blocks and no assumption on the octonion tensor (value computed); if $\omega = 2$, in $N^{23/6+\varepsilon}$ (Theorem 2.7) and in $N^{3.631}$ with Appendix B pieces on 256-bit blocks (value computed*

*over pure piece types). In both cases this is below the flattening floor $N^4$, which binds every rank-one method, whether it uses rank, border rank or asymptotic rank, and below the width bound $N^4$, which binds every contraction tree. On the VXXZ curve the covering hybrid beats the tree $N^{2\omega(1,1/2,1)} = N^{4.0860}$ computed from the same bounds, but not $N^4$.*

With Appendix B pieces in the limit $m \to \infty$ the current-curve value is 3.914, the best we have, but it rests in addition on the argued convergence in $m$, and the margin over the tree needs $m$ in the tens of thousands; at a fixed block size the covering hybrid is the best.

**Remark 2.9** (What is open at $K = 4$)**.** Between the input floor $N^3$ and $N^{3.547}$ (at $\omega = 2$) nothing is known in either direction. Every lower-bound tool we know stops at $N^3$ for Signed Network Sums: quantum functionals of $T_4$ are at most $N^3$, and the flattening ranks $N^4$ bound only pieces that are cheap across one fixed cut (argued). A bound above $N^3$ would be a superlinear lower bound for an explicit degree-4 polynomial, which is out of reach. On the upper side, a laser-type analysis of tensor powers gives nothing beyond the constructions above: inside the rank model it is bounded by the flattening floor, and inside the model of Signed Network Sums its type-and-entropy half is what the constructions already do (argued). The gap is decided by the per-crossing pieces: a family whose pieces cost $(2, 0)$ and $(0, 2)$ per bit on the row and column cuts, the least the pigeonhole principle allows, would reach the input floor $N^3$ by table sizes alone; the known pieces cost $\sqrt{2}$ per bit on both cuts in the worst type (Appendix B), or $(1, 2)$ and $(2, 1)$ (eight-square), and building their tables costs more than their size.

### 2.4 Six, nine and sixteen nodes

At $K = 4$ the construction beats every contraction tree. This subsection asks whether such a margin survives at larger $K$, where the recursion of [1] would use such levels: it sets up one-level grid layouts, explains the two kinds of numbers that occur (certified and search values), and gives the results at $K = 9$, 16 and, computed exactly over all profiles, $K = 6$.

**One-level grid layouts.** For $K = ab$ place the nodes on an $a \times b$ grid. Every pair of rows and pair of columns spans a rectangle whose two diagonals form a crossing; expand each crossing per block (eight-square, grouped by ranks, or Appendix B pieces), and in each term merge either all rows or all columns. The side edges inside a merged row are summed while the row's table is built, by rectangular products with the substitution of Theorem 2.7. The merged instance is a clique sum on the $a$ merged rows (or $b$ merged columns) with unequal edge weights, which is contracted by its best contraction tree, or at $K = 16$ alternatively by the $K = 4$ construction with Appendix B pieces. This is Anthropic's grid level [1, §2.5] with the recursion replaced by trees at the top. The adversary chooses the profile of the term (the piece type on every rectangle), the algorithm the merge direction, and the exponent is

$$\Phi(w) = \max_{\text{profiles}} \min_{\text{rows, columns}} \max\bigl(\text{top contraction},\ \text{row tables}\bigr)$$

under a matrix-multiplication curve $w$. Every term in $\Phi$ is non-decreasing in $w$, so $\Phi$ is non-decreasing in $w$ (proved).

Two kinds of numbers occur. A *certified* value is an upper bound on $\Phi$ over all profiles, from closed-form bounds (for instance at $K = 9$, $\Phi_{\text{top}} \le (5 + 2\bar{\rho})w(1)$, with $\bar{\rho}$ the largest average cost per rectangle and bit, $\bar{\rho} = \sqrt{2}$ for Appendix B pieces) and a certified grid for the row tables. A *search* value is the largest $\Phi$ found by numerical search over profiles; it is a lower estimate of the

construction's exponent, so a margin below the tree computed from it is an upper bound on the true margin.

**Results.** At $K = 9$ ($3 \times 3$) and $K = 16$ ($4 \times 4$) the width bound of Proposition 2.4(c) is 18 and 60 ($e = 1/2$), and it binds every tree on every curve. The best trees found under the curve are 18/20.25/21.344 at $K = 9$ and 60/64/65.376 at $K = 16$ for $\omega = 2$ / current curve / VXXZ; these are upper bounds from the same curve.

- $\omega = 2$, $K = 9$: Appendix B pieces give $\Phi \le 15.725$ ($e \le 0.437$) as $m \to \infty$, and $\le 16.456$ at $m = 256$; the eight-square constructions give at most 17.721 (certified). All are below the width bound 18.
- $\omega = 2$, $K = 16$: grouping counted by ranks gives exactly 59.229, block-free, below 60 by 0.77; Appendix B pieces give $\le 57.95$ as $m \to \infty$ and $\le 59.718$ at $m = 512$ (certified).
- Current curve: at $K = 9$ Appendix B pieces give $\le 17.614 < 18$ in the limit $m \to \infty$, but 18.019 at $m = 512$, so below the width bound only in the limit (argued convergence); every construction beats the best tree 20.25 at fixed block size. At $K = 16$ Appendix B pieces give $\le 59.95 < 60$ in the limit (margin 0.05), 61.72 at $m = 512$; all constructions beat the best tree 64.
- VXXZ curve: every construction beats the best tree at both $K$; none is certified below the width bound.

At $K = 16$ the certified Appendix B range with a tree on top is $[49.94, 57.95]$; a search from structured profiles finds only the uniform profile (49.94, $e = 0.416$), so the upper end may be loose. With the $K = 4$ construction on top instead of a tree, the search gives $\ge 46.862$ ($e \ge 0.390$) at $\omega = 2$ and $\ge 48.883$ ($e \ge 0.407$) on the current curve (lower estimates).

At $K = 6$ ($2 \times 3$; with rows merged the top is a dot product, with columns merged a triangle) the maximum over profiles can be computed rather than searched. A profile is the vector $u \in [-2, 2]^3$ of piece types on the three rectangles, and a rectangle of type $u$ costs $\rho = \sqrt{2 + u}$ per bit with rows merged and $\kappa = \sqrt{2 - u}$ with columns merged. The cost of each direction is non-decreasing in its three per-rectangle costs, because every step of the row tables and of the top is a maximum of sizes and of terms $\Omega_w$ that grow with them, and $\Omega_w$ is non-decreasing in each argument when $w$ is non-decreasing with $w(k) - kw'(k) \ge 1$, which holds for the three curves. So on a box of profiles the cost is at most the minimum of the rows cost at the box's largest $\rho$ and the columns cost at its largest $\kappa$, and branch and bound over $[-2, 2]^3$ gives

$$\Phi \le 7.360 \quad (e \le 0.4907) \text{ at } \omega = 2, \qquad \Phi \le 8.017 \quad (e \le 0.5345) \text{ on the current curve,}$$

within 0.002 of the largest values found at single profiles (7.357 and 8.014), against the trees 8 and 9. A rectangle whose blocks carry a mixture of piece types costs the mean of their $(\rho, \kappa)$, a point under the quarter circle $\rho^2 + \kappa^2 = 4$ and hence below a pure type in both coordinates, so mixtures are covered. At $\omega = 2$, where every product with sorted dimensions $M \ge m \ge s$ costs $M + m$ (each $\omega(a, b, c)$ taken equal to its trivial lower bound $\max(a + b, b + c, a + c)$), the construction therefore beats the width bound 8 and with it every contraction tree. On the current curve it beats the best tree 9 found under the same curve, but not the width bound: at the worst profile the cost 8.014 is above 8. On the VXXZ curve the bound is 8.442 against the tree 9.397. These are limits $m \to \infty$, but they hold at a fixed block length as well: with the exact partition

| $K$ | input $2/K$ | ideal pieces | App. B, $\omega = 2$ | tree, $\omega = 2$ | App. B, current | tree, current |
|---|---|---|---|---|---|---|
| 4 | 0.5 | 0.5 (exact) | 0.591 (lim) | 0.667 | 0.652 (lim) | 0.667 |
| 6 | 0.333 | 0.444–0.449 (c) | 0.4904–0.4907 (c) | 0.533 | 0.5342–0.5345 (c) | 0.6 |
| 9 | 0.222 | $\geq 0.386$ (s) | 0.434–0.437 (c, lim) | 0.5 | $\leq 0.490$ (c, lim) | 0.5625 |
| 16 | 0.125 | not computed | $[0.416, 0.483]$ (c, lim); $\geq 0.390$ (s, $K_4$ top) | 0.5 | $\leq 0.4996$ (c, lim); $\geq 0.407$ (s, $K_4$ top) | 0.533 |

Table 3: Exponent per edge bit $e = E/\binom{K}{2}$ of one-level layouts ($2 \times 2$, $2 \times 3$, $3 \times 3$, $4 \times 4$ grids). (c) certified upper bound or certified interval over all term profiles; (s) numerical search, a lower estimate of the construction's exponent; (lim) in the limit of long blocks $m \to \infty$, convergence argued. "Tree" is the best contraction tree found under the curve; at $\omega = 2$ it equals the width bound. The current curve rests on two preprints. At $K = 4$ the App. B entries are 3.547/6 and 3.914/6 from the accounting of Section 2.3 (worst type closed form, mixtures computed).

numbers of Appendix B at $m = 128$, and the count of terms paid separately, the bounds are 7.818 at $\omega = 2$ and 8.492 on the current curve, still below 8 and 9. On the current curve the best tree at $K = 6$ costs 9 although its largest table has $N^8$ entries: its final product is the binding step. Grouping counted by ranks gives 8.116 at $\omega = 2$ at a single profile, already worse than the tree.

### 2.5 The exponent per edge bit as a function of $K$

Table 3 puts the values side by side. Besides the input floor and the constructions it has a column for *ideal pieces*: the value of the same layout, counted by table sizes alone (no entropy, no cost of building tables, each term contracted by its best tree), if every crossing had a two-piece family with costs exactly $(2, 0)$ and $(0, 2)$ per bit on the row and column cuts. Such pieces are allowed by every rank constraint we know, including the four-party rank inequality of Song, Chen, Sun and Hu [8]; whether they exist is the open question of Remark 2.9. The column shows what one-level layouts could give at best with pair pieces.

**The picture.** Two trends are visible, both measured on one level of grid layouts. First, the constructions stay a roughly constant fraction below the tree: with Appendix B pieces at $\omega = 2$ the ratio to the tree is about 0.89 at $K = 4$, 0.92 at $K = 6$ and 0.87 at $K = 9$, and with ideal pieces it is 0.75, 0.83–0.84 and at least 0.77. Second, they move away from the input floor: the ideal-piece value divided by $2/K$ is 1.00, 1.33 and at least 1.74 at $K = 4, 6, 9$. So a single level with pair pieces does not approach the input floor as $K$ grows; even with ideal pieces it is held near the tree's order of magnitude by the cost of the merged top instance, which is a generic clique sum solved by a tree. Section 4 makes the corresponding statements for the rewrite-and-merge class precise. At fixed $K$, by contrast, no size argument separates the model from the input floor: we argue in Section 4.6 that the per-block piece question at any $K$ reduces to the four-node question of Remark 2.9.

# 3 The large end: the explicit cost of the subexponential 3-SAT recursion

The algorithm of [1] counts the satisfying assignments of a $k$-CNF with $m$ clauses in time $2^{O(km/\sqrt{\log m})}$, and of a $k$-CNF on $n$ variables in time $2^{O(k^2 n \log\log n/\sqrt{\log n})}$. Both bounds come from one recursion on clique sums whose number of nodes squares from level to level. The paper states that its bound in $m$ beats trying all assignments only for $\log m > 8000$ at clause density one ($m = n$) ([1, end of App. A]). In this section we make the recursion explicit: we fix every constant of the main term, we say which constants remain in the lower-order terms, and we compute from which input size on the proved bound is below the best known algorithm for the same problem. We then lower these sizes by a sequence of changes, each of which is either proved here, or proved in [1] and only counted more carefully, or marked as conditional with the condition named. The changes do not alter the scale: the smallest main-term threshold we can prove is about $2^{420}$ clauses at clause density one, and for general 3-SAT the thresholds are of the form $2^{10^9}$ to $2^{10^{10}}$ variables. No running time in this section is usable; the point of the explicit constants is to show where the distance comes from and which named obstacle each further improvement must remove.

Section 3.1 recalls the recursion. Section 3.2 says what an explicit threshold is. Section 3.3 gives the threshold tables. Sections 3.4–3.7 prove the changes behind them: the least admissible group size, the constant $c$ of the pieces, the base case by a contraction tree and a clustering lemma for the top level; Section 3.8 applies the recursion to Max-3-SAT. Section 3.9 prices every lever, including the ones we cannot pull, and Section 3.10 treats general 3-SAT, sparsification as a parameter and the factor $\log\log n$.

## 3.1 The recursion of [1]

This subsection recalls only what the thresholds depend on: the networks and pieces of [1], the recursion (5) with its three per-level costs, and the two top levels, for bounded density and for general formulas, which decide how large the instance must be for a given number of nodes.

We use the notions of [1, §3]. A *network* on a node set $J$ has, for each node $a$, a set of settings $S_a$ (bit strings of length at most $P$) and a table $W_a\colon S_a \to \mathbb{Z}$, and for each pair of nodes $a \neq b$ a factor $\Psi_{a,b}$ that factors through $c_{a,b}$ bits, i.e. is a sum over $\theta \in \{0,1\}^{c_{a,b}}$ of a product of a function of $z_a$ and a function of $z_b$. Its value is the sum over all settings of the product of all tables and factors, its *total* is $U = \sum_{a\neq b} c_{a,b}$ (every shared bit counted from both ends), and $\sum_b c_{a,b} \le P$ for every $a$ ([1, Definition 2]). The $(K,\lambda)$-*clique* is the network on $[K]$ in which every pair of nodes shares a string of $\lambda$ bits and the tables are arbitrary functions of the $K-1$ strings at a node ([1, Definition 3]); trying all strings evaluates it with $2^{\lambda K(K-1)/2}$ terms. We measure the cost of an algorithm for $(K,\lambda)$-cliques by the exponent $e$ in $2^{eK^2\lambda}$, so brute force has $e = 1/2$; in Section 2 the same cost is measured per edge bit, which is $2e$.

**Pieces and one level.** A *family of pieces with parameters* $(\rho, c)$ [1, Definition 7] is, for every $\lambda \ge 1$, a decomposition $\sum_{i\in I} A_i = d\,[x = v][y = u]$ of the crossing of two equalities on $\lambda$-bit strings into integer functions $A_i$, each of which factors through $c_i^{\mathrm{row}}$ bits along rows and $c_i^{\mathrm{col}}$ bits along columns with $c_i^{\mathrm{row}} + c_i^{\mathrm{col}} \le 4\rho\lambda + c\log_2(2\lambda)$, with $|I| \le (2\lambda)^c$, all numbers below $2^{(2\lambda)^c}$, and one algorithm computing every value within $2^{c\lambda}$ operations. [1, Theorem B] gives pieces with $\rho = 1/\sqrt{2}$ and "an absolute constant $c$" (we make $c$ explicit in Section 3.5).

One level of the recursion takes a $(g^2, \lambda)$-clique, arranges its nodes in a $g \times g$ grid, writes every rectangle's crossing as a sum of pieces, and for every choice of pieces evaluates the cheaper of the row network (rows merged) and the column network (columns merged); each is a network on $g$ nodes. Lemma 8 of [1] then groups the $g$ nodes into $K$ groups of $g/K$ nodes each and solves the resulting $(K, L)$-clique by the level below. In its case (ii), a uniformly random balanced grouping and Hoeffding's inequality for sampling without replacement show that a good grouping exists provided

$$K \mid g, \qquad c_{a,b} \le P/g \;\; (a \neq b), \qquad 2\zeta^2 g \;\ge\; K \ln(2gK), \tag{4}$$

where $\zeta$ is the slack: the clique below has $K^2 L \le U + 2\zeta g P + K^2$ bits. With $g = sK$ and $K' = g^2$ the paper's recursion [1, (40)] reads

$$K_{\ell+1} = g_\ell^2 = (s_\ell K_\ell)^2, \qquad e_{\ell+1} = (\rho + 3\zeta_\ell)\, e_\ell + 3/K_\ell, \tag{5}$$

with the paper's choices $K_0 = 2^{x_0}$, $x_0 \ge 18$, $\zeta_\ell = 2^{-\ell-5}$, $s_\ell = 2^{3\ell+x_0}$ and the base case $e_0 = 1/2$ (brute force). The $\rho$ is the per-level factor, $3\zeta_\ell$ the price of balancing, and $3/K_\ell$ the cost of the merged groups' own bits ([1, §2.6]). Since $\log K$ roughly doubles per level while $e$ falls by $\rho$, $e_\ell = \Theta((\log K_\ell)^{-\log_2(1/\rho)})$, which is $(\log K)^{-1/2}$ for $\rho = 1/\sqrt{2}$.

**The top level, bounded density.** For a $k$-CNF with $m$ clauses, the network of [1, §6] has as nodes $g$ clusters of clauses; a setting of a cluster is a proposed value at each occurrence of a variable in it, and two clusters share one bit for each *equality test*, a pair of consecutive occurrences of one variable in a fixed order of its occurrences. So $U < 2km$, and [1, Theorem A(c)] evaluates it by one call of Lemma 8 in case (i) (Chebyshev balancing, $g = 3K^8$), followed by the clique algorithm at level $\ell$ with $K_\ell \le m^{1/9}$. The logarithm of the number of operations is [1, (45)]

$$2e_\ell km + O\Big(km/K + K^8 \log K + K^2 \log(km)\Big), \tag{6}$$

where the $O$ hides the constants $C_0$ (of Theorem A(b)) and $C_1$ (of Lemma 8), which depend on $c$. We write $q$ for the exponent with $m \ge K^q$ that the top level needs: $q = 9$ in [1].

**The top level, general formulas.** For formulas of unbounded density [1, Theorem D] first sparsifies into at most $2^{\varepsilon n}$ formulas in which every variable occurs in at most $\Delta(\varepsilon)$ clauses, and then evaluates for each a network whose nodes hold variables. The condition that no clause is violated is expanded by inclusion–exclusion over sets of marked clauses, cut off at $\bar{n} \approx n/\log\Delta$ marks; assignments that violate more clauses are killed by $O(n)$ sparse parity checks, each found in an explicit family in which every large set of clauses has odd intersection with at least a quarter of the members. The total is $U = O(kn \log\Delta)$, the top level uses case (i) of Lemma 8 with $K \le n^{1/17}$ ($q = 17$), and the time is $2^{O(ekn\log\Delta)}$ per sparsified formula ([1, (68)–(70)]).

### 3.2 What an explicit threshold is

The bounds of [1] are of the form $2^{O(km/\sqrt{\log m})}$, and the question from which size on such a bound beats a known algorithm has an answer only once three things are fixed: the known algorithm we compare with, the part of the bound whose constants we make numeric, and the size measure. The following definition fixes them.

**Definition 3.1** (bar, main term, threshold)**.** A *bar* for a class of formulas is the exponent of the best known algorithm for it, written per unit of the size measure. For bounded density we take the class of 3-CNFs with $m = \Delta n$ clauses and the bar $\min\{0.5283\,m,\ 0.711\,n\}$, the better of the clause-parameterised bound $O^*(1.4423^m)$ of Peng, Sheng and Xiao [19] and the variable-parameterised bound $O^*(1.637^n)$ for #3-SAT [20]; since the main term of (6) is $2ekm = 6em$, the bar on the clique exponent is

$$\bar{e}(\Delta) = \min\{0.5283,\ 0.711/\Delta\}/6, \tag{7}$$

which is 0.0881, 0.0592, 0.0395, 0.0278 for $\Delta = 1, 2, 3, 4.26$. The factor $6 = 2k$ comes from the paper's bound $U < 2km$. The network's total is exact: a variable with $d$ occurrences has $d - 1$ tests and each test puts one bit at each of its two ends, so $U = 2(km - n)$ when every variable occurs (a variable that does not occur contributes a factor 2 and can be removed), and by the definition of $L$ in [1, Lemma 8] the main term is $e\,U$ up to the lower-order terms already in (6). With this count the bar is

$$\bar{e}_U(\Delta) = \min\{0.5283,\ 0.711/\Delta\}/(6 - 2/\Delta), \tag{8}$$

which is 0.1321, 0.0711, 0.0444, 0.0302 for $\Delta = 1, 2, 3, 4.26$: a factor 3/2 at density one, falling to 1.08 at density 4.26. For general 3-SAT we take the deterministic decision bound $2^{0.4092n}$ of Liu [22]. The *main term* of a bound is its leading term ($2e_\ell km$ in (6), $\varepsilon n + e_\ell U$ for Theorem D). The *main-term threshold* $m^*$ (or $n^*$) is the least size from which the main term, with the top level chosen as the construction prescribes, is below the bar; we report $\log_2 m^*$. It is the crossover point of the main term against the best known algorithm, and a lower bound on the crossover point of the proved bound itself.

The bar for bounded density rests on two published results: the clause bound of [19] and Kutzkov's deterministic $O^*(1.637^n)$ for #3-SAT [20, Theorem 3], which improves his earlier $O(1.6423^n)$ [21], i.e. $0.7157\,n$. Our algorithm counts, so comparing it with a decision algorithm in the general case is the stronger comparison; against the counting bar 0.7157 or the randomized decision bar 0.3865 [23] the general thresholds move by one or two levels, i.e. by a factor of at most about four.

**Why main-term thresholds, and what they omit.** A main-term threshold is a lower bound on the size from which the proved bound wins, since the omitted terms are positive. It is the right quantity for two reasons. First, at the sizes in question the omitted terms are smaller than the main term by a factor $m^{-\Omega(1)}$: in (6) with $K \le m^{1/9}$ every term after $2ekm$ is $O(km^{1-1/90})$ [1, after (45)], and with $\log_2 m$ in the thousands such terms are negligible as soon as their constants are numbers. Second, the constants of the omitted terms are not numbers in [1]: $C_0$, $C_1$, $C_4$, $C_5$ are "constants that depend only on $c$", and $c$ itself is "large enough". So a fully explicit theorem needs (a) every main-term constant numeric, and (b) every lower-order constant bounded by a number. This section does (a) completely. For (b) it makes $c$ numeric (Section 3.5), which also removes the only place where $c$ touches the main term, and bounds $C_0$ by 810 (Proposition 3.10), far below the $2^{51}$ that the thresholds need (Proposition 3.9).

**Status labels.** In the tables below, *proved* means that the main term rests only on theorems proved here or in [1]; *computed* values come from a numerical optimisation over admissible schedules and are exact for the model (each reported schedule can be checked level by level, Appendix B.1); *estimate* marks values obtained from a cruder level formula; *conditional* names the missing lemma or construction.

### 3.3 The threshold tables

Table 4 is the headline of this section. Each column adds one change to the previous one; the changes are proved in the sections cited in the column heads.

| density $\Delta$ | (P) paper as written | (G) group size | (X) $x_0 = 7$ ($c = 30$) | (T) tree base case | (U) exact total $U$ | (C) Lemma C top, brute base | (U+C) all |
|---|---|---|---|---|---|---|---|
| 1 | 33858 | 28761 | 22264 | 5396 | 1538 | 1685 | **417** |
| 2 | 68364 | 59309 | 46597 | 11684 | 6610 | 6059 | **1551** |
| 3 | 275616 | 122506 | 98247 | 25161 | 23117 | 19775 | **5227** |
| 4.26 | 552042 | 246356 | 197966 | 51204 | 47986 | 41546 | **10766** |
| status | proved | proved | proved | proved | proved | proved | proved |

Table 4: Main-term thresholds $\log_2 m^*$ for counting satisfying assignments of 3-CNFs with $m = \Delta n$ clauses, against the bar (7); $\rho = 1/\sqrt{2}$ (pieces of [1, App. B]) throughout. (P): recursion (5) with the paper's schedule, $x_0 = 18$, top level $q = 9$. (G): the same with the least group size $s_\ell$ admitted by (4) and the slack schedule optimised (Proposition 3.2). (X): also $x_0$ optimised, which gives 7; admissible because $c = 30$ (Lemma 3.4). (T): also the base case by a contraction tree, $e_0 = \omega(1, \frac{1}{2}, 1)/8 = 0.2554$, which makes $x_0 = 8$ optimal (Lemma 3.7); top level still $q = 9$. (U): also the exact total $U = 2(km - n)$, i.e. the bar (8). (C): columns (X) and (U) with the top level by Lemma C (Proposition 3.9; $q = 2.10$ at $\Delta = 1$), brute-force base case; it needs $C_0 \le 2^{61}$ and $C_0 \le 810$ by Proposition 3.10. (U+C): all changes, tree base case, exact total and Lemma C ($q = 2.41$ at $\Delta = 1$, two levels).

The first column reproduces the paper's own remark: with brute force as the competitor at $\Delta = 1$ (bar 1/6) the same model gives $\log_2 m^* \ge 8046$ at level 4, against the paper's "$\log m > 8000$". Column (U+C) is the best value; the factor from (P) to (U+C) is 81 at density 1 and 51 at density 4.26. Without Lemma C, column (U) is the best value; Lemma C divides it by 3.7 at density one and by 4.5 at density 4.26. The exact count matters most at density one, where the larger bar saves two levels (5396 → 1538), and little at density 4.26. The order of the columns follows what each change needs: (G), (X) and (U) only re-read the paper's proof, with the bound on $C_0$ of Proposition 3.10 a count of its bookkeeping, (T) adds a contraction tree with fast rectangular matrix multiplication, and (C) adds a new lemma.

For general 3-SAT the same changes act on a much smaller target exponent, because the parity-check apparatus of Theorem D costs about 2500 bits per variable where the clause network at density one costs about 4 (Section 3.10). Table 5 lists the thresholds.

Three remarks fix the status of Table 5. (i) All rows are main-term estimates from the closed level formula of [1, Theorem A(a)], not from an admissible schedule. At $x_0 = 18$ that formula is the paper's schedule. At $x_0 = 7$ it is not admissible (the paper's schedule needs $x_0 \ge 18$ for (4)), and it underestimates $\log K$: at density one and six levels it gives $27 \cdot 2^6 = 1728$ where the admissible optimum is 2474. So the rows G3, G4 and G7 are optimistic, by a factor that we have not computed. (ii) Row G4 needs a clustering lemma for the $n$-network of Theorem D. Lemma 3.8 is proved for the clause network of [1, §6]; the $n$-network has variables as nodes and clause and parity-check marks as shared bits, and whether its marks are shared pairwise along chains, as the occurrences of §6 are, has not been checked. After sparsification its test multigraph has degree $O(k\Delta)$ instead of $2k$, which would cost only $O(\log \Delta)$ additional bits in $\log_2 n^*$ if the lemma holds. (iii) Rows G5–G7 need a parity-check family with kill rate near 1/2. Such a family

| row | accounting and levers | $\log_2 n^*$ | status |
|---|---|---|---|
| G0 | paper's inequalities, $x_0 = 18$, $q = 17$ | $2.2 \cdot 10^{12}$ | proved (main term) |
| G1 | same construction counted sharply, Calabro's $\Delta(\varepsilon)$, $x_0 = 18$, $q = 17$ | $3.4 \cdot 10^{10}$ | estimate |
| G2 | G1 with the tree base case | $8.6 \cdot 10^{9}$ | estimate |
| G3 | G2 with $x_0 = 7$ | $3.9 \cdot 10^{9}$ | estimate, see text |
| G4 | G1 with $x_0 = 7$, top level $q = 2.2$ | $2.0 \cdot 10^{9}$ | cond.: $n$-network clustering |
| G5 | G1 + optimised $\varepsilon$ + kill rate 1/2, $x_0 = 18$, $q = 17$ | $8.6 \cdot 10^{9}$ | cond.: check family |
| G6 | G5 with the tree base case | $2.1 \cdot 10^{9}$ | cond.: check family |
| G7 | G5 with $x_0 = 7$, $q = 2.2$ | $5.0 \cdot 10^{8}$ | cond.: both |

Table 5: Main-term thresholds $\log_2 n^*$ for general 3-SAT (counting algorithm, deterministic decision bar 0.4092), $\rho = 1/\sqrt{2}$. "Counted sharply" replaces the inequalities [1, (59), (61), (68)] by the actual numbers of marks and checks of the same construction. "Kill rate 1/2" is a parity-check family that gives odd intersection with probability near 1/2 instead of 1/4; see the text for its status. All rows use the level formula of [1, Theorem A(a)], $\log K_\ell = (3x_0 + 6)2^\ell - 6\ell - 2x_0 - 6$ and $e_\ell \le 0.85\rho^\ell$; since the latter is an upper bound, a level count may be one level too large.

can be built by pairwise-independent subsampling at rate $\mu/v$, which gives odd intersection with probability at least $(1 - 1/\mu)/2$ by Chebyshev's inequality, at the price of $\log_2 \mu$ additional bits per mark; the rows have not been recomputed with these bits, so they are lower estimates for that construction.

### 3.4 The least admissible group size and a free slack schedule

The paper's schedule $s_\ell = 2^{3\ell+x_0}$, $\zeta_\ell = 2^{-\ell-5}$, $x_0 \ge 18$ is one convenient way to satisfy the hypotheses of Lemma 8 at every level. The proof of Theorem A(b) uses it only through four inequalities, so any schedule that satisfies them gives the same theorem.

**Proposition 3.2** (Theorem A(b) for an admissible schedule)**.** *Suppose that there are pieces with parameters $(\rho, c)$. Let $x_0 \ge 1$, $K_0 = 2^{x_0}$, $e_0 = 1/2$, and for $\ell \ge 0$ let $s_\ell \ge 1$ be integers and $\zeta_\ell \in (0, 1]$ reals; define $g_\ell = s_\ell K_\ell$ and $K_{\ell+1}$, $e_{\ell+1}$ by* (5)*. Call the schedule* admissible *if for every $\ell$*

*(S1)* $2\zeta_\ell^2 g_\ell \ge K_\ell \ln(2 g_\ell K_\ell)$*;*

*(S2)* $1/g_\ell \le \zeta_\ell$*;*

*(S3)* $(c + 5)K_\ell^2 \le g_\ell^4/2$*;*

*(S4)* $e_\ell \le 1$*.*

*Then for an admissible schedule the conclusion of [1, Theorem A(b)] holds: one deterministic algorithm, given $\ell$, $\lambda, B \ge 1$ and query access to tables with $|W_w| < 2^B$, outputs the value of the $(K_\ell, \lambda)$-clique within $2^{e_\ell K_\ell^2 \lambda} (K_\ell(\lambda + B))^{C_0 K_\ell^2}$ operations, with $C_0 = 2(c + C_4 + 1) + C_5$ as in [1].*

*Proof.* We follow the proof of [1, Theorem A(b)] (l. 1617–1671) and list the places where the schedule (40) enters. The base case is brute force and does not involve the schedule. In the

inductive step: case (ii) of Lemma 8 needs $K \mid g$, which holds as $g = sK$, the bound $c_{a,a'} \le P/g$, which is shown there for every $g$, and $2\zeta^2 g \ge K \ln(2gK)$, which is (S1) (the paper verifies it from $x_0 \ge 18$); the step from $U + 2\zeta N$ to $(\rho + 3\zeta)N$ in (42) uses $1/g \le \zeta$, which is (S2); the bound (43) on $T$ uses $e \le 1$, which is (S4); the inequality $c\lambda \le N/K$ used in (44) follows from (S3), since $N/K = g^4\lambda/K \ge 2(c+5)K\lambda$; and the final absorption of the exponent of $\Xi$ into $C_0 g^4$ uses $(c+5)K^2 \le g^4/2$, which is (S3) (the paper derives it from $2^{x_0} \ge c+5$). Nothing else in the inductive step depends on $s_\ell$, $\zeta_\ell$ or $x_0$. □

Condition (S3) holds for all $\ell$ as soon as $K_0^2 \ge 2(c+5)$, because $g_\ell \ge K_\ell \ge K_0$; with $c = 30$ (Lemma 3.4) this is $x_0 \ge 4$, and at $x_0 = 7$ it allows every $c \le 8187$. So once $c$ is numeric, $c$ no longer enters the main term at all.

**The computed schedules.** For a given bar, $\rho$ and $x_0$ we choose at every level the least integer $s_\ell$ with (S1) and (S2) for a slack $\zeta_\ell$ from the grid $2^{-j/2}$, $2 \le j \le 17$, and optimise over the grid by dynamic programming on the Pareto front of $(\log K_\ell, e_\ell)$; the reported value is $\log_2 m^* = q \log_2 K_L$ for the first level $L$ with $e_L$ below the bar. Every reported schedule is admissible: (S1), (S2) hold by the choice of $s_\ell$, (S3) by $x_0 \ge 4$, and (S4) because the exponents decrease from level to level along every reported schedule (Appendix B.1). Two observations explain where the gain of column (G) comes from.

**Remark 3.3** (the paper's slack schedule is nearly optimal)**.** Minimising $\log K_L$ over slack schedules with the group size fixed by (S1) is, to first order, the allocation of a budget $\sum_\ell \zeta_\ell$ against costs $\sum_\ell 2^{L-\ell} \cdot 2\log(1/\zeta_\ell)$, whose Lagrange solution is $\zeta_\ell = \gamma 2^{-\ell}$. The closed form is within 2% of the optimum in $\log_2 K$ (density one: 2529 against 2474; density two: 5258 against 5177; density 4.26: 22343 against 21996), with best constants $\gamma = 0.031$, 0.026, 0.019. At density one $\gamma = 1/32$, which is the paper's $2^{-\ell-5}$. So the gain from column (P) to (G) does not come from the slack schedule but from the group size: the paper takes $s_\ell = 2^{3\ell+x_0}$, while (S1) needs only $s \ge \ln(2sK^2)/(2\zeta^2)$, about $2^{2\ell+10}\log K$.

### 3.5 The constant $c$ of the pieces

The constant $c$ of [1, Theorem B] enters the main term only through the condition $2^{x_0} \ge c+5$ of (40), i.e. through (S3), and the lower-order terms through $B_1 = gB + g^2(2\lambda)^c$, $\tau = 2^{c\lambda}\Xi^{C_2}$ and $C_0$. The paper gives no value; its proof of Theorem B ends "Definition 7 holds with $\rho = 1/\sqrt{2}$ if $c$ is large enough" (l. 2018), and it counts the operations of its algorithm only as $\Delta^{O(1)}$. Definition 7(c) is a property of an algorithm, so a value of $c$ is a statement about a particular algorithm and a particular way of counting. We state it for the algorithm described in the proof of Theorem B, with the operation count below.

**The objects.** For $\lambda \ge 1$ let $h \ge 2$ be least with $p(h) \ge 2^\lambda$ ($p$ the partition function), $I = \{-2h, \dots, 2h\}$, $d = h!$; the piece $A_s$ factors along rows through $\lceil \log_2 R_s^{\mathrm{row}} \rceil$ bits and along columns through $\lceil \log_2 R_s^{\mathrm{col}} \rceil$ bits, with $R_s^{\mathrm{row}} = \sum_{i=0}^{2h} p(s+i)$ and $R_s^{\mathrm{col}} = \sum_{i=0}^{2h} p(2h-s-i)$ [1, (54), (55)]. The algorithm lists the partitions of $0, 1, \dots, 4h$, at most $\Delta \le (4h+1)p(4h)$ of them, computes the tables of the operators of [1, (O2)] on these partitions (entries at most $2^{12h^2}$ in absolute value), and evaluates each $A_s$ as a sum of at most $\Delta$ products of two factors.

**Operation count.** We count one arithmetic operation on the integers that occur (at most $24h^2 + O(h \log h)$ bits) as one operation. Building one table of at most $\Delta^2$ entries takes at most $h$ steps of (O2) with at most $8h$ terms per entry, there are at most $\Delta$ tables, and the evaluation uses at most $(2h+1)\Delta$ pairs with $\Delta$ look-ups each; we bound the total generously by $64h^3\Delta^3$. This is our reconstruction of the paper's description; a different but reasonable count moves the result by a few units.

**Lemma 3.4** ($c = 30$)**.** *For the algorithm of the proof of [1, Theorem B], with operations counted as above, the pieces satisfy Definition 7 of [1] with $\rho = 1/\sqrt{2}$ and $c = 30$.*

*Proof.* For each $\lambda$ we compute the least $c$ that meets each condition of Definition 7: (b) $\max_s(\lceil \log_2 R_s^{\mathrm{row}} \rceil + \lceil \log_2 R_s^{\mathrm{col}} \rceil) \le 2\sqrt{2}\lambda + c \log_2(2\lambda)$; (c1) $4h + 1 \le (2\lambda)^c$; (c2) every value below $2^{(2\lambda)^c}$, the largest being the bound $(2h+1)\Delta^3\, 2^{3h}\, h!\, 2^{24h^2}$ on $|A_s|$; (c3) $64h^3\Delta^3 \le 2^{c\lambda}$. For $1 \le \lambda \le 271$ (where $4h \le 24000$) all partition numbers are computed exactly; the largest requirement is 27.198, from (c3) at $\lambda = 1$ ($h = 2$, $\Delta = 67$), and it decreases to 6.39 at $\lambda = 271$ (Table 11 in the appendix). For $\lambda \ge 271$ we use the Hardy–Ramanujan bounds

$$c_{\mathrm{HR}}\sqrt{E} - \log_2 E - C_{\mathrm{HR}} \;\le\; \log_2 p(E) \;\le\; c_{\mathrm{HR}}\sqrt{E}, \qquad c_{\mathrm{HR}} = \pi\sqrt{2/3}/\ln 2,\ C_{\mathrm{HR}} = 3.84, \quad (9)$$

the upper one for $E \ge 0$ [24, Thm. 14.5], the lower one for $E \ge 1$ by Lemma 3.5. With them the requirements are at most 9.588 on the sampled range $271 \le \lambda \le 10^{14}$; beyond it we argue as follows (not computed): at the defining point $c_{\mathrm{HR}}\sqrt{h-1} = \lambda + \log_2 h + C_{\mathrm{HR}}$, the excess $c^{\mathrm{row}} + c^{\mathrm{col}} - 2\sqrt{2}\lambda$ is $(2\sqrt{2}+2)\log_2 h + O(1)$ with $\log_2 h = 2\log_2\lambda + O(1)$, so (b) tends to $4 + 4\sqrt{2} = 9.657$, while (c1), (c2), (c3) tend to 2, 4, 6. Hence $c = \lceil 27.198 \rceil = 28$ meets (b)–(c3) for every $\lambda$, and the replacement of longer factorizations by ones through $2\lambda$ bits ([1, l. 1527–1530]) adds 2: $c = 30$. □

The proof used the lower Hardy–Ramanujan bound with an explicit constant $C_{\mathrm{HR}}$; the asymptotic formula for $p(E)$ does not give one, so we derive it from Lehmer's remainder bound.

**Lemma 3.5** (explicit $C_{\mathrm{HR}}$)**.** *For every integer $E \ge 1$, $\log_2 p(E) > c_{\mathrm{HR}}\sqrt{E} - \log_2 E - 3.84$.*

*Proof.* Let $\mu = \frac{\pi}{6}\sqrt{24n-1}$. Lehmer's remainder bound for Rademacher's series, with one term ($N = 1$), as quoted by DeSalvo and Pak [13, (4), (5)], gives, after dropping the positive term $e^{-\mu}$ of the main part, bounding $\sinh\mu < e^{\mu}/2$ and dropping $-1/\mu^2$ in the remainder,

$$p(n) > \frac{\pi^2}{6\sqrt{3}}\,\frac{e^{\mu}}{\mu^2}\Big[1 - \frac{4}{\mu} - \mu^2 e^{-\mu}\Big].$$

The bracket increases for $\mu \ge 2$ and equals 0.5456 at $\mu = 9$; for $n \ge 13$, $\mu \ge \frac{\pi}{6}\sqrt{311} = 9.234$, so the bracket is at least $1/2$. With $\mu^2 \le \frac{2\pi^2}{3}n$ this gives $p(n) > e^{\mu}/(8\sqrt{3}\,n)$. Since $\sqrt{24n-1} \ge \sqrt{24n} - 1/\sqrt{24n}$, we have $\mu \ge \pi\sqrt{2n/3} - 0.02965$ for $n \ge 13$, and taking $\log_2$, $\log_2 p(n) > c_{\mathrm{HR}}\sqrt{n} - \log_2 n - (3.79248 + 0.04277) = c_{\mathrm{HR}}\sqrt{n} - \log_2 n - 3.8353$. For $1 \le n \le 12$ exact values give the inequality with any constant above 3.7007 (the value $n = 1$ forces the constant to exceed $c_{\mathrm{HR}} = 3.7007$). □

The value $c = 30$ depends on how operations are counted. The following remark says how it moves under the paper's own cost model and under built-in tables, and why the main term does not care.

**Remark 3.6** (other counts)**.** Lemma 3.4 counts arithmetic operations on integers of up to $24h^2 + O(h \log h)$ bits as single operations. The paper's own model counts operations on words that hold an address (l. 756), in which an operation on such integers costs a number of word operations polynomial in their length; at $\lambda = 1$, where (c3) binds, this adds a few bits per $\lambda$ to the requirement, and a hand estimate (not checked) puts $c$ in the high thirties in that model. Conversely, Definition 7(c) allows built-in tables, and with the pieces for $\lambda \le 12$ stored in the algorithm the binding requirements become (c3) at $\lambda = 13$ (about 10) and the limit 9.66 of (b), so $c \approx 13$. None of this matters for the main term: every $c \le 8187$ is admissible at $x_0 = 7$.

### 3.6 The base case by a contraction tree

Theorem A(b) of [1] starts its induction with brute force at $K_0$ nodes, $e_0 = 1/2$. A contraction tree with one fast matrix product per merge is cheaper at every $K_0$.

**Lemma 3.7** (tree base case)**.** *Let $K_0$ be divisible by 4 and let $\omega_{1/2} = \omega(1, \frac{1}{2}, 1)$ be the exponent of multiplying an $N \times N^{1/2}$ by an $N^{1/2} \times N$ matrix. For every $\delta > 0$ the $(K_0, \lambda)$-clique can be evaluated with $O_\delta(2^{(\omega_{1/2}+\delta)K_0^2\lambda/8})$ arithmetic operations on integers of $O(K_0^2(\lambda + B))$ bits. With $\omega_{1/2} \le 2.042994$ [9] this is the base case $e_0 = 0.2554$ in place of $1/2$.*

*Proof.* Split $[K_0]$ into quarters $Q_1, \dots, Q_4$ of $K_0/4$ nodes and halves $H_1 = Q_1 \cup Q_2$, $H_2 = Q_3 \cup Q_4$. For a quarter $Q$, its table $T_Q$ is a function of the strings on the edges leaving $Q$, obtained by summing the product of the $W_w$, $w \in Q$, over the strings on the edges inside $Q$; it has $2^{\lambda(K_0/4)(3K_0/4)}$ entries and costs $2^{\lambda(K_0^2/32+3K_0^2/16)} = 2^{7K_0^2\lambda/32}$ terms. The table $T_{H_1}$, a function of the strings on the $K_0^2/4$ edges between $H_1$ and $H_2$, is $T_{H_1}(y_1, y_2) = \sum_z T_{Q_1}(y_1, z) T_{Q_2}(y_2, z)$, where $z$ ranges over the strings on the $K_0^2/16$ edges between $Q_1$ and $Q_2$ and $y_i$ over those on the $K_0^2/8$ edges from $Q_i$ to $H_2$ (other arguments of $T_{Q_i}$ are fixed and range over $y_i$ too). This is one product of an $N \times N^{1/2}$ and an $N^{1/2} \times N$ matrix with $N = 2^{\lambda K_0^2/8}$, at cost $O_\delta(N^{\omega_{1/2}+\delta})$; likewise $T_{H_2}$. Finally the value is $\sum_y T_{H_1}(y) T_{H_2}(y)$ over $N^2$ strings $y$, and $2 \le \omega_{1/2}$. All other costs are smaller. So the exponent per $K_0^2\lambda$ is $(\omega_{1/2} + \delta)/8$, and $2.042994/8 = 0.25537$. □

In the induction of Proposition 3.2 the base case enters only through its exponent and its polynomial factor, so the lemma gives the recursion with $e_0 = 0.2554$. Since $e_\ell$ falls by about $\rho$ per level, halving $e_0$ saves two levels of $\rho = 1/\sqrt{2}$, and as $\log K$ roughly doubles per level the thresholds fall by a factor of about four (column (T) of Table 4: 3.9 to 4.1). If $\alpha \ge 1/2$ for the dual exponent of matrix multiplication, as claimed in two recent preprints [10, 11], then $\omega_{1/2} = 2$, $e_0 = 1/4$ and the thresholds change by 1–2% (density one: 5323 against 5396). Two caveats. The constant in $O_\delta$ of the rectangular matrix multiplication algorithm is not given in [9]; it enters the lower-order factor, alongside $C_0$, and has to be bounded for a fully explicit theorem. And the base case is the slot where a fixed-$K$ algorithm such as those of Section 2 could enter; but the recursion needs $K_0 \ge 2^4$ by (S3) and in practice $K_0 = 2^7$ or $2^8$ by (S1) at level 0 (with $x_0 = 2$, (S1) forces $\zeta_0$ near 0.8, so that $\rho + 3\zeta_0 > 1$), so the gain at the base case comes from the tree and not from constructions at $K = 4$, 9 or 16.

### 3.7 A clustering lemma for the top level

Theorem A(c) of [1] uses Lemma 8 in case (i) at the top, with $g = 3K^8$ clusters, hence $m \ge K^9$. Case (ii) would need only $g$ a polylogarithmic factor above $K$, but it requires $c_{a,b} \le P/g$ for every pair of clusters, and with an arbitrary split of the clauses all tests of a cluster may go to

one other cluster. The following lemma supplies a split with this property at a constant-factor cost in $gP$, and the point of its use is that $gP$ enters Lemma 8 only multiplied by $\zeta$ or by $1/K$.

Let $\Phi$ be a $k$-CNF with $m$ clauses, with the equality tests of [1, §6] fixed. Let $\Gamma$ be the multigraph on the clauses with one edge for each test whose two occurrences lie in different clauses. For $h\colon \Phi \to [g]$ let $n_a = |h^{-1}(a)|$ and, for $a \neq b$, let $c_{a,b}$ be the number of edges of $\Gamma$ between $h^{-1}(a)$ and $h^{-1}(b)$.

**Lemma 3.8** (Lemma C, clustering)**.** *Let $k \geq 1$, $g \geq 2$, $r = 4k-1$, $\kappa = 1/\ln 2$, and*

$$x_1 = \Big\lceil \kappa \, \frac{2}{g^2} \Big\lceil \frac{km}{r} \Big\rceil + \log_2(g^2 r) \Big\rceil, \qquad y = \Big\lceil \kappa \, \frac{m}{g} + \log_2(2g) \Big\rceil, \qquad P = g\, r\, x_1.$$

*There is a map $h$ from the clauses to $[g]$, computed deterministically in time polynomial in $km$, such that in the network of [1, §6] on the clusters $h^{-1}(a)$ every cluster has at most $ky \leq P$ occurrences, $\sum_{b \neq a} c_{a,b} \leq P$, and $c_{a,b} \leq P/g$ for all $a \neq b$. Moreover $gP \leq 2\kappa(km + r) + rg^2(\log_2(g^2 r) + 1)$, and if $m \geq (4 + 2\ln 2 \cdot g^2(\log_2(4kg^2) + 1))/\varepsilon$ for some $\varepsilon > 0$, then $gP \leq 2\kappa(1+\varepsilon)km$.*

*Proof. Matchings of equal size.* A clause has at most $k$ occurrences and each lies in at most two tests, so $\Gamma$ has maximum degree $2k$ and an edge meets at most $4k-2$ others; the greedy edge colouring uses at most $r$ colours, with classes $M_1, \ldots, M_r$ that are matchings. While $|M_i| \geq |M_j| + 2$ for some $i, j$, the components of $M_i \cup M_j$ are paths and even cycles alternating between the classes, so some path has one more edge of $M_i$; swapping the classes on it keeps both matchings and moves one edge from $M_i$ to $M_j$. The sum of $|M_t|^2$ decreases, so this stops with $|M_t| \leq s_1 := \lceil km/r \rceil$ for all $t$ (McDiarmid's equitable edge colouring).

*Tail per matching.* Let $h$ be uniform. For $a \neq b$ and $t$, let $X_t^{ab}$ be the number of edges of $M_t$ between clusters $a$ and $b$. The edges of a matching are vertex-disjoint, so their indicators are independent with probability $2/g^2$ each, and $\mathbb{E}2^{X_t^{ab}} \leq \exp(2s_1/g^2)$. Likewise $\mathbb{E}2^{n_a} = (1+1/g)^m \leq e^{m/g}$.

*Pessimistic estimator.* Let $\Psi(h) = \sum_{a<b} \sum_{t=1}^{r} 2^{X_t^{ab}(h) - x_1} + \sum_a 2^{n_a(h) - y}$. By the choice of $x_1, y$, $\mathbb{E}\Psi \leq \binom{g}{2} r/(g^2 r) + g/(2g) < 1$. If $\Psi(h) < 1$, then $X_t^{ab} \leq x_1 - 1$ and $n_a \leq y - 1$ for all $a, b, t$.

*Conditional expectations.* Assign $h(E_1), h(E_2), \ldots$ one at a time, each time minimising the conditional expectation of $\Psi$, which is at most the average over the $g$ choices; so $\Psi(h) < 1$ at the end. Each conditional expectation is a sum of $\binom{g}{2} r + g$ products over the edges of a matching (resp. the clauses) of factors $2^{I_e}$ (both ends assigned), $1 + 1/g$ or $1$ (one end assigned, inside or outside $\{a, b\}$), $1 + 2/g^2$ (none), and of $2^{\#\text{assigned to } a}(1 + 1/g)^{\#\text{unassigned}}$: rationals with polynomially many bits. So the procedure takes time polynomial in $km$ (we may assume $g \leq m$).

*The network conditions.* $c_{a,b} = \sum_t X_t^{ab} \leq r(x_1 - 1) < P/g$; hence $\sum_{b \neq a} c_{a,b} < P$; cluster $a$ has at most $k(y-1)$ occurrences, and $ky \leq P$ because $r \geq k$, $g \geq 2$ and $P \geq 2\kappa km/g + 2k\log_2(2g) \geq ky$. The remaining parts of Definition 2 are as in [1, §6].

*Size of $gP$.* $gP = g^2 r x_1 \leq 2\kappa r s_1 + rg^2(\log_2(g^2 r) + 1) \leq 2\kappa(km + r) + rg^2(\log_2(g^2 r) + 1)$, and with $r < 4k$ the last claim follows by rearranging. □

With the clustering in hand, the top level can use case (ii) of Lemma 8 of [1] with $g$ only a polylogarithmic factor above $K$, instead of $g = 3K^8$. The next proposition is the resulting bound; its main term is the $2ekm$ of (6) times $1 + 2\kappa\zeta(1+\varepsilon)$, plus a term of order $km/K$.

**Proposition 3.9** (top level with Lemma C)**.** *Let $(K, e) = (K_\ell, e_\ell)$ be a level of an admissible schedule (Proposition 3.2), with $K \geq 16$. Let $0 < \zeta \leq 1$, $0 < \varepsilon \leq 1$, $s \geq 1$ an integer with $2\zeta^2 s \geq \ln(2sK^2)$, $g = sK$, and suppose*

$$m \geq (4 + 2\ln 2 \cdot g^2(\log_2(4kg^2) + 1))/\varepsilon. \tag{T1}$$

*Then the number of satisfying assignments of a $k$-CNF with $m$ clauses is computed with a number of operations whose logarithm is at most*

$$2km\Big[e(1+2\kappa\zeta(1+\varepsilon))+\frac{2\kappa(1+\varepsilon)}{K}\Big]+eK^2+C_0K^2\log_2(2Kkm)+g\log_2 K+O(\log(km)).$$

*Proof.* Cluster by Lemma 3.8 and apply Lemma 8 of [1] in case (ii) with $B=1$, which is allowed since $K \mid g$, $c_{a,b}\le P/g$ and $2\zeta^2 g \ge K\ln(2gK)$ (as $g\ge K$, $\ln(2gK)\le\ln(2sK^2)$). In Lemma 8, $P$ enters only through $L=\lceil(U+2\zeta gP)/K^2\rceil$, $B'=\lceil 2gP/K\rceil+g^2B$ and the factor $2^{2gP/K}$, and the Hoeffding ranges in its proof are $P$ and $P/g$, which are exactly what the lemma's hypotheses supply; $U<2km$ does not involve $P$. With $gP\le G:=2\kappa(1+\varepsilon)km$ and $T$ from Proposition 3.2, the logarithm of the count is at most $e(U+2\zeta G+K^2)+C_0K^2\log_2(K(L+B'))+2G/K+g\log_2 K+O(\log(km))$, and $L+B'\le 2km$ under (T1) for $K\ge 16$, $\varepsilon\le 1$. □

Under (T1), $\log_2 m\ge 2\log_2 K+2\log_2 s+O(\log\log K)$ with $\log_2 s=\log_2\log_2 K+2\log_2(1/\zeta)+O(1)$, so $q=\log_2 m/\log_2 K=2+O\big((\log\log K+\log(1/\zeta)+\log(1/\varepsilon))/\log K\big)$ tends to 2 (we argue this elementary estimate rather than prove it with constants; the computed values below are exact for the model). For the thresholds, the recursion is run to a target slightly below the bar, and at the top the largest $\zeta$ is taken for which $e(1+2\kappa\zeta(1+\varepsilon))+2\kappa(1+\varepsilon)/K+\nu e$ is below the bar, with $\nu=10^{-3}$ reserved for the lower-order terms; $m^*$ is then the least $m$ with (T1). At density one this gives $x_0=7$, six levels, $\log_2 K=2478$, $\zeta=2^{-11.3}$, $\log_2 s=33.3$, $\log_2 m^*=5037$ and $q=2.033$; at density 4.26, nine levels, $\log_2 m^*=44154$, $q=2.006$. The honest top level costs 85 to 122 bits over the crude $2\log_2 K$. The lower-order budget $\nu\cdot 2ekm$ covers the term $C_0K^2\log_2(2Kkm)$ at $m^*$ provided $C_0\le 2^{57.6}$ at density one ($2^{59.3}$, $2^{60.9}$, $2^{76.2}$, $2^{93.0}$ at densities 1.35, 2, 3, 4.26). With the exact total $U=2(km-n)$ the margins are larger still: $C_0\le 2^{51.1}$ suffices in every row of Table 4 (the tightest case is column (U+C) at density one). These budgets are only useful if $C_0$ is a number, and [1] gives none. The next proposition bounds it by counting the steps that the paper calls polynomial.

**Proposition 3.10** (explicit $C_0$)**.** *In the word RAM model of [1], with schoolbook integer arithmetic, Theorem A(b) of [1] holds with $C_0\le 810$.*

*Proof.* The paper sets $C_0=2(c+C_4+1)+C_5$, with $c=30$ by Lemma 3.4. The bracket of [1, (44)] is a sum of four terms; it is at most four times the largest, and each is at most $T2^{2N/K}2^{c\lambda}\Xi^{C_2+(c+4)C_1+C_3+c+1}$, so $C_4\le C_2+(c+4)C_1+C_3+c+3$. Constant factors are absorbed by raising an exponent by one, since $\Xi=g^2(\lambda+B)$ and $K_0(\lambda+B)$ exceed every constant used ($g\ge 2^{18}$). We count the steps that the paper calls polynomial. (i) Base case: one term of the brute-force sum queries $K_0$ tables, multiplies $K_0$ integers below $2^B$, adds the product to the running sum and advances the enumeration, each in at most $(K_0(\lambda+B))^3$ operations; so $C_5\le 4$. (ii) Lemma 8, per term of an entry: testing a grouping, locating the blocks of an argument of fewer than $KL\le g(gP+1)$ bits, multiplying at most $g^2$ values below $2^B$, and adding a term of at most $B'$ bits each take at most $(g(P+B))^4$ operations; so $C_1\le 5$, and we use $C_1\le 6$. (iii) The parameter $\tau$ of the row and column networks: a merged table value is a product over at most $g^4$ incidences of a table value and of piece-factor values below $2^{(2\lambda)^c}$, each computed in $2^{c\lambda}$ operations [1, Def. 7(c)]; the product has at most $\Xi^{c+2}$ bits, so $\tau\le 2^{c\lambda}\Xi^{2c+7}$ and $C_2\le 2c+7$. (iv) Per $\iota$: $d^{|R|}$ has at most $g^4(2\lambda)^c\le\Xi^{c+2}$ bits and $Z_\iota$ at most $\Xi^3$; computing $d^{|R|}$ by $|R|\le g^4$ multiplications and the final division cost at most $\Xi^{2c+8}$, and the other steps less; so $C_3\le 2c+8$. Hence $C_4\le 67+34\cdot 6+68+33=372$ and $C_0\le 2(30+372+1)+4=810$. □

The counts are crude, and they rest on our reading of the constructions of [1, Lemma 1, Proposition 3], not on a line-by-line re-proof; but against the $2^{51}$ that is needed only the polynomiality of each step matters.

### 3.8 Max-3-SAT through integer tables

The network of [1, §6] checks every clause inside the table of the cluster that contains it, and Definitions 2–4 of [1] allow arbitrary integer tables, with the bit length $B$ of the values as a parameter of every bound. So the same algorithm can weight each assignment instead of testing it. For a CNF $\Phi$ with nonnegative integer clause weights $w_E$ and an assignment $\xi$, let $w(\xi)$ be the total weight of the clauses that $\xi$ satisfies, and let $N_j(\Phi)$ be the number of assignments $\xi$ with $w(\xi) = j$; with all weights 1, $N_j$ counts the assignments that satisfy exactly $j$ clauses, and the largest $j$ with $N_j \neq 0$ is the Max-SAT value.

**Proposition 3.11** (Max-$k$-SAT spectrum)**.** *Let $c_w \geq 0$. One deterministic algorithm computes, for a $k$-CNF $\Phi$ with $m \geq 2$ clauses in which $n$ variables occur and with integer clause weights $0 \leq w_E \leq m^{c_w}$, all the numbers $N_j(\Phi)$, $0 \leq j \leq \sum_E w_E$, as numbers of assignments of these $n$ variables, in time $2^{O(km/\sqrt{\log m})}$, where the $O$ depends on $c_w$ but not on $k$. In particular weighted Max-$k$-SAT with polynomially bounded weights is solved in this time, which is $2^{O(n/\sqrt{\log n})}$ when $k$ and $m/n$ are bounded.*

*Proof.* Let $W = \sum_E w_E$ and $y = 2^{n+1}$. Build the network of [1, §6] (with clusters as in the proof of [1, Theorem A(c)], or as in Lemma 3.8) with the same settings, the same numbers $c_{a,b}$ and the same edge functions $\Psi_{a,b}$, and change only the node tables: $W_a(z_a)$ is $y^j$, where $j$ is the total weight of the clauses of cluster $a$ that have a true occurrence under $z_a$, if $z_a[o] = z_a[o']$ for every test $\{o, o'\}$ inside cluster $a$, and 0 otherwise. As in [1, §6], every test is checked exactly once, by a $W_a$ or by a $\Psi_{a,b}$, so the terms of the value $Z$ that are not zero are those in which all occurrences of each variable carry the same bit; they correspond one to one to the assignments $\xi$ of the $n$ variables, and the term of $\xi$ is $\prod_a y^{w_a(\xi)} = y^{w(\xi)}$, where $w_a(\xi)$ is the weight of the clauses of cluster $a$ that $\xi$ satisfies. Hence

$$Z = \sum_{j=0}^{W} N_j(\Phi)\, y^j.$$

Since $0 \leq N_j \leq 2^n < y$, the $N_j$ are the digits of $Z$ in base $y = 2^{n+1}$, that is, consecutive blocks of $n+1$ bits of $Z$, and they are read off in time linear in the length $(n+1)(W+1)$ of $Z$.

It remains to check that the evaluation of [1, Theorem A(c)] still takes time $2^{O(km/\sqrt{\log m})}$. The settings, the numbers $c_{a,b}$, $P$ and $U$ are those of the 0/1 network, so the main term $2ekm$ and the terms $2gP/K$ and $g \log K$ of [1, (45)] are unchanged. Only $B$ changes: a cluster holds at most $P$ clauses, so every value of $W_a$ is below $2^B$ with $B = (n+1)\, P\, m^{c_w} + 1$, and computing an entry, which is writing one bit at position $(n+1)j$, takes time polynomial in $B$; so $\tau$ is polynomial in $km$, since $n \leq km$. In Lemma 8 of [1], $B$ enters through $B' = \lceil 2gP/K \rceil + g^2 B$ and the factor $(g(P+B))^{C_1}$, and the clique algorithm of [1, Theorem A(b)], which holds for all integer tables with $|W_w| < 2^B$ and every $B \geq 1$, contributes the factor $(K(L+B'))^{C_0 K^2}$. With $g \leq m$, $P \leq 2km$ and $n \leq km$, we have $\log(K(L+B')) = O(\log(km))$ with a constant that depends only on $c_w$, so the term $C_0 K^2 \log(K(L+B'))$ of [1, (45)] is again $O(K^2 \log(km))$ and all terms after $2ekm$ remain $O(km^{1-1/90})$. The arithmetic on integers of poly$(km)$ bits in Lemma 8 and in Theorem A(b) is already counted there, since both are stated for arbitrary $B$. Finally, a variable that does not occur multiplies every $N_j$ by two. □

The proposition does not use the inclusion–exclusion network of [1, Theorem D], and its bound in $n$ does not follow for unbounded density. That bound needs two steps that are specific to satisfiability. First, the sparsification lemma [1, Lemma 4] replaces $\varphi$ by formulas whose satisfying assignments partition those of $\varphi$; it says nothing about assignments that violate clauses, so the number of clauses an assignment satisfies is not preserved. Second, Theorem D writes $[V(\xi) = \emptyset]$ as an alternating sum over sets of marked violated clauses and cuts this sum off at about $n/\log\Delta$ marks, which changes nothing for the assignments with few violated clauses and is corrected by parity checks for the others; the weight $y^{|V(\xi)|} = \sum_{\Pi\subseteq V(\xi)}(y-1)^{|\Pi|}$ has no such cut-off, since all its terms are positive. So the bound $2^{O(m/\sqrt{\log m})}$ is below $2^n$ only for $m = o(n\sqrt{\log n})$. The Max-3-SAT hypothesis, which asks for no $(2-\varepsilon)^n$ algorithm at any number of clauses and whose refutation would follow from a refutation of the 3-uniform hyperclique hypothesis [29], concerns dense formulas, where the proposition gives nothing; at bounded density a time $(2-\varepsilon(\Delta))^n$ was known before [30]. What the proposition adds is a subexponential time at bounded density, together with the whole spectrum $(N_j)_j$ rather than only its largest index, computed by the recursion itself and with an explicit crossover point (Table 6). The subexponential time alone does not need the recursion, and the following consequence belongs to the two refutations: either of them, by the route of [32, Theorem 5] recalled below, refutes the Sparse Max-3-SAT Hypothesis of Alman and Vassilevska Williams [31, Hypothesis 3], that for every $\varepsilon > 0$ some density $c$ makes Max-3-SAT on $cn$ clauses require time $2^{(1-\varepsilon)n}$, and with the same argument for $k = 2$ their sparse Max-2-SAT hypothesis [31, Hypothesis 4]; they introduced the sparse form because no sparsification lemma for Max-$k$-SAT is known. Neither refutation states this consequence, so we record it. The route of [32, Theorem 5] is: a linear-size circuit that tests whether exactly $j$ clauses are satisfied, turned into a 3-CNF with $O(m)$ variables and clauses in which every assignment extends to exactly one satisfying assignment, followed by any algorithm for #3-SAT in time $2^{o(n)}$, that of [1] or of [2]. The algorithm of [2, Corollary 1.2] approximates Max-$k$-SAT to within $\varepsilon m$ clauses in time subexponential in $n$ for every density; it does not give the exact value.

**Thresholds.** The best known exact algorithm for Max-SAT in terms of the number of clauses runs in time $O^*(1.2886^m)$ [27], that is $0.3658\,m$; the best bound in the length, $O^*(1.0911^L)$ [28], gives $0.3773\,m$ for 3-CNFs with $L = 3m$ and is weaker, and we found no better bound for Max-3-SAT in particular. In terms of $n$ we take trying all assignments, $2^n$, which also gives the whole spectrum: the algorithms for sparse Max-SAT beat $2^n$ by a factor $2^{-\mu(\Delta)n}$ with $\mu(\Delta)$ known only up to a constant factor [30], so they give no numeric bar. With the exact total $U = 2(km - n)$ as in (8), the bar for Max-3-SAT at density $\Delta$ is

$$\bar{e}_{\max}(\Delta) = \min\{\log_2 1.2886,\ 1/\Delta\}/(6 - 2/\Delta), \tag{10}$$

which is 0.0915, 0.0732, 0.0625, 0.0424 for $\Delta = 1, 2, 3, 4.26$; against (8) it is lower at density one and higher from density two on, where #3-SAT has the faster competitors. Table 6 gives the main-term thresholds for this bar, in the model of Table 4.

The larger tables change only lower-order terms. In Proposition 3.9 the bound $L + B' \le 2km$ becomes $L + B' \le 4\kappa\, g\, m\,(n+2)$, since a cluster of Lemma 3.8 holds at most $y - 1$ clauses and $B \le (n+1)(y-1)+1$; so $\log_2(K(L+B'))$ is about twice $\log_2(2Kkm)$ at $m^*$. The budget $\nu \cdot 2ekm$ for the term $C_0K^2\log_2(K(L+B'))$ then needs $C_0 \le 2^{47.8}$ in the tightest row (column (U+C) at density three) instead of about $2^{48.8}$, far above the $C_0 \le 810$ of Proposition 3.10.

| density $\Delta$ | bar (10) | (P) as written | (U) tree, exact $U$ | (C) Lemma C, brute base | (U+C) all |
|---|---|---|---|---|---|
| 1 | 0.0915 | 68364 | 4107 | 3671 | **1016** |
| 2 | 0.0732 | 68364 | 6344 | 5845 | **1489** |
| 3 | 0.0625 | 137430 | 11308 | 10113 | **2594** |
| 4.26 | 0.0424 | 275616 | 23770 | 20805 | **5371** |
| status | | proved | proved | proved | proved |

Table 6: Main-term thresholds $\log_2 m^*$ for the Max-3-SAT spectrum of 3-CNFs with $m = \Delta n$ clauses, against the bar (10); columns as in Table 4, column (P) with the paper's bound $U < 2km$, i.e. the bar $\min\{\log_2 1.2886, 1/\Delta\}/6$. At density one the threshold is larger than for #3-SAT (1016 against 417, one level more), because the Max-SAT competitor is faster there; from density two on it is smaller, because the only competitor in $n$ with a numeric constant is $2^n$.

### 3.9 The levers, priced

Table 7 lists every lever of the recursion with its effect on the threshold. We give the factor by which it divides $\log_2 m^*$; values are main-term values of the model, and the rows marked hypothetical assume a constant or a construction that is not available. The bounded-density rows of the second block were computed at the bar 0.0867 (a competitor $2^{0.52n}$ that we used before the clause bound of [19] was identified); the factors, not the absolute values, are the content of those rows.

Three things stand out. First, no lever that is proved or plausible changes the scale: they move $\log_2 m^*$ between about 34000 and 5000 at density one. The shape of the recursion explains this: $\log_2 m^* \approx q(3x_0 + 6)\,(c'/\bar{e})^{1/\alpha'}$ with $\alpha' = \log_2(1/\rho)$, because $\log K$ doubles per level while $e$ falls by $\rho$; $q$, $x_0$, the base case and the overheads act on the constant factors, $\rho$ on the exponent $1/\alpha'$, and only a growth of $K$ by a constant factor per level would turn the double exponential into a single one. Second, the levers interact sub-multiplicatively when they act through the number of levels: a better sparsifier raises the target exponent and saves five levels at $\rho = 1/\sqrt{2}$, but only three at $\rho = 1/2$, so $\rho \to 1/2$ is worth $2000\times$ with Calabro's sparsifier and $500\times$ with $\Delta = 2$; the levers $x_0$ and $q$ multiply exactly. Third, the rows that would change the scale are exactly those that Section 4 shows to be closed inside the class of rewrite-and-merge schemes, or that need an object nobody has constructed.

**Proposition 3.12** (structural floor)**.** *Let $\rho = 1/\sqrt{2}$ and let the base case have $e_0 \le 1/2$. In every admissible schedule (Proposition 3.2) whose first level gains, i.e. $e_1 < e_0$, we have $K_1 > 2^{28}$. Consequently, with any top level that needs $m \ge K^2$, the recursion beyond its base case acts only on inputs with more than $2^{56}$ clauses, for every problem and every competitor.*

*Proof.* $e_1 = (\rho + 3\zeta_0)e_0 + 3/K_0 < e_0$ needs $\rho + 3\zeta_0 < 1$, so $\zeta_0 < (1-\rho)/3 < 0.0977$ and $1/(2\zeta_0^2) > 52.3$; and it needs $3/K_0 < (1-\rho-3\zeta_0)e_0$, i.e. $K_0 > 3/((1-\rho-3\zeta_0)e_0) \ge 6/(1-\rho) > 20$, so $K_0 \ge 32$ as $K_0$ is a power of two. By (S1), $s_0 \ge \ln(2s_0K_0^2)/(2\zeta_0^2) > 52.3\,(\ln 2048 + \ln s_0)$, which forces $s_0 > 740$, i.e. $\log_2 s_0 > 9.5$. Hence $\log_2 K_1 = 2(x_0 + \log_2 s_0) > 2(5 + 9.5) = 29$. □

The floor does not depend on 3-SAT or its bar: it says that no problem, however weak its best known algorithm, is helped by the recursion below $2^{56}$ bits of input. In the computed optimal schedules the first level has $\log_2 K_1 = 38.5$ (brute-force base, $x_0 = 7$) and 40.7 (tree base, $x_0 = 8$). A formula encoding with $L$ bits per variable needs $e < 1/(2L)$, i.e. about $\log(2L)/\log(1/\rho)$ levels with $\log_2 K$ growing like $27 \cdot 2^\ell$, which for 3-SAT ($L = 6\Delta$) is five to seven levels.

**The polynomial-method algorithm.** OpenAI's algorithm [2] reaches the same bound $2^{O(n/\sqrt{\log n})}$ at fixed density by a different method, with no tower of clique sizes. Its main-text route (rational interpolation) has a per-variable exponent free of $n$, but by the tests of equation (2.19) of [2] it is exhaustive search below $2^{1.09\cdot 10^{10}}$ variables, at every density, as Szeider [3] computed; so $\log_2 n^* \geq 1.09 \cdot 10^{10}$ for it. Its Appendix D (Boolean tensor contraction) proves both bounds of its main theorem with a main term $\log_2 T \approx (s/\sqrt{\lambda})(C_w M_0/s + 1)$, $\lambda = \lceil \log_2 s \rceil$, $s = n + m$, so $\log_2 s^* = ((C_w M_0/s + 1)/\bar{e})^2$; reading the constants off the text gives $C_w = 18535$ bits per root wire, dominated by $1/\sqrt{\varepsilon} = 362$, and thresholds $\log_2 s^* \approx 4.4 \cdot 10^{10}$ (density one) and $1.4 \cdot 10^{12}$ (density 4.26). Read literally, its parameter bounds also ask for $u \geq \lambda^{4/5}$ on every active call, which holds only for $\lambda \geq 2^{85}$, and Tulsyan [4] takes this as a hypothesis of the proof. The condition serves only to let $2^u$ absorb fixed powers of $\lambda$, for which $u$ of a few hundred suffices, and at $\lambda = 4.4 \cdot 10^{10}$ the smallest active $u$ is about $5 \cdot 10^3$; so we set it aside, and our values for Appendix D are estimates in this sense. With every improvement that we could justify, but did not prove, $C_w$ falls to 106–134 and the thresholds to $\log_2 s^* \approx 3.7 \cdot 10^5$–$5.9 \cdot 10^5$ and $3.7 \cdot 10^6$–$5.8 \cdot 10^6$, still 17 to 26 times above column (X) of Table 4; our thresholds at the two densities, column (U+C) of that table, are 417 and 10766. So the polynomial method has the better shape, polynomial in $1/\bar{e}$, but a constant $10^2$ to $10^4$ times too large, and in the general case it uses the same parity tests.

### 3.10 General 3-SAT, sparsification, and the factor $\log\log n$

The thresholds for general 3-SAT in Table 5 are larger than those for bounded density by many orders of magnitude in $\log_2 n^*$. This subsection says where that factor comes from (the parity-check apparatus of Theorem D), shows that no function of the number of violated clauses can replace the checks (Remark 3.13) and that rank arguments cannot show the checks to be necessary (Remark 3.14), compares the cut-off with an exact test from parameterized complexity, and identifies the factor $\log\log n$ of the general bound with the density of the sparsifier (Proposition 3.15).

**Where the bits go.** For a 3-CNF on $n$ variables, the main term of Theorem D is $\varepsilon n + eU$ per sparsified formula, and $U$ has two parts. The marks of the inclusion–exclusion, cut off at $\bar{n} \approx n/\log(2\Delta)$ violated clauses, cost about $2(k-1)n = 4n$ bits, independent of $\Delta$; this is the same as the clause network at density one. The rest is the parity-check apparatus: $M$ marked incidences, each shared along a path of $k$ tests and each naming one of about $\log_2 \Delta + \log_2\log_2(2\Delta) + O(1)$ bits' worth of incidences. So $U/n \approx 2k(M/n)(\log_2 \Delta + c') + 4$ with $c' \approx 8$–$14$. With the paper's inequalities ($M \leq 14n$, crude $\Delta$) this is about 16300 bits per variable; counted sharply ($M \approx 7.2n$, Calabro's $\Delta$, $\log_2 \Delta \approx 48$) about 2500. The target exponent $e^*$ is then $2.5 \cdot 10^{-5}$ resp. $1.6 \cdot 10^{-4}$, 31 resp. 25 levels, which is the gap between Tables 4 and 5. Inside the paradigm (truncated inclusion–exclusion, parity checks, 2-adic cut-off) these factors have floors: at least about $2.2n$ checks, $\log\Delta + \log\log\Delta$ bits per mark, path length $k - 1$, which together give $U/n \approx 700$–$900$.

**Remark 3.13** (no count-based kill)**.** The checks are needed because no function of the number of violated clauses alone can do their job. Theorem D needs a factor $G$ with $G = 1$ on satisfying assignments and $G = 0$ when more than $\bar{n}$ of the $m$ clauses are violated. Suppose $G = \sum_{i \leq d} c_i \binom{v}{i}$ with integer $c_i$, $v = |V(\xi)|$, and $G(v) \equiv 0 \pmod{2^N}$ for $v = \bar{n} + 1, \ldots, m$. Re-expanding around $\bar{n} + 1$, $G(\bar{n} + 1 + j) = \sum_{i \leq d} c'_i \binom{j}{i}$ with integer $c'_i$ (Vandermonde's identity), and the values at $j = 0, \ldots, m - \bar{n} - 1$ determine $c'_0, \ldots, c'_{m-\bar{n}-1}$ by a unitriangular integer system; so if $d < m - \bar{n}$,

all $c'_i \equiv 0 \pmod{2^N}$ and $G \equiv 0$ everywhere, including $G(0)$. Hence $d \geq m - \bar{n}$: there is no symmetric truncated kill, even modulo $2^N$, and a kill must use which clauses are violated, as the parity checks do.

Whether the assignments that violate more than $\bar{n}$ clauses can be killed with $O(n)$ shared bits is open. The following remark shows that rank arguments cannot decide it in the negative.

**Remark 3.14** (rank bounds cannot force the factor $\log \Delta$)**.** Consider a network on the nodes $[g]$ in which node $a$ holds a block $X_a$ of at most $\lceil n/g \rceil$ variables: a setting $z_a$ consists of the values $\xi_a$ of these variables and of auxiliary bits (marks), the tables $W_a$ and the functions $f_{a,b}$ of the factorizations of $\Psi_{a,b}$ through $c_{a,b}$ bits are integer valued, as in [1, Definition 2]. For $\xi \in \{0,1\}^n$ let $H(\xi)$ be the sum of the terms of the network's value in which the variables are set to $\xi$, and let a *representation* be a sum of $L$ such networks on the same blocks. The networks of Step 3 of Theorem D for $F(\Lambda)$, summed over the $L = 2^{O(g^2 \log n)}$ admissible profiles, are such a representation with

$$H(\xi) \equiv [\xi \text{ satisfies } \Phi] \pmod{2^{n+1}} \qquad \text{for every } \xi$$

once the checks are certified ($X(\Lambda) = 0$): the proof of [1, Lemma 12] works term by term in $\xi$, and [1, (63)] holds term by term because the terms of $X(\Lambda)$ are nonnegative. Its total is $U = O(kn \log \Delta) + 2g^2$ [1, (68)], with $\log \Delta = O(k \log \log n)$ after sparsification.

*(a) Upper bounds on ranks (proved).* For a partition $\mathcal{P} = \{P_1, \ldots, P_t\}$ of $[g]$ into $t \geq 2$ groups let $c(\mathcal{P})$ be the sum of $c_{a,b}$ over the pairs $\{a, b\}$ in different groups (the largest such sum among the $L$ networks), and view a function of $\xi$ as a $t$-party tensor whose $i$-th index is the restriction of $\xi$ to $X_{P_i} = \bigcup_{a \in P_i} X_a$. Then the tensor rank of $H$ over $\mathbb{Q}$ is at most $L\, 2^{c(\mathcal{P})}$, and so is the tensor rank of $[\xi \text{ satisfies } \Phi]$ over $\mathbb{F}_2$. Indeed, write each $\Psi_{a,b}$ as the sum over $\theta_{a,b} \in \{0,1\}^{c_{a,b}}$ of $f_{a,b}(z_a, \theta_{a,b}) f_{b,a}(z_b, \theta_{a,b})$; for fixed $\theta_{a,b}$ on the pairs between groups, the sum over the auxiliary bits and over the $\theta_{a,b}$ inside the groups is a product of $t$ integer functions, the $i$-th depending only on $\xi$ restricted to $X_{P_i}$; reducing modulo 2 gives the statement over $\mathbb{F}_2$. For $t = 2$ this is the flattening (cut) bound, for $t = g$ (singletons) it reads rank $\leq L\, 2^{U/2}$, since $U$ counts every shared bit from both ends.

*(b) The caps (proved, elementary).* Every $t$-party tensor whose $i$-th index takes $2^{x_i}$ values has tensor rank at most $2^{x_1 + \cdots + x_t - \max_i x_i}$ over every field (fix all indices but the largest one). So the logarithm of every quantity that is bounded by the tensor rank of the grouping $\mathcal{P}$ (a flattening rank, border rank, slice rank, and over $\mathbb{C}$ the quantum functionals of [5]) is at most $\mathrm{cap}(\mathcal{P}) = n - \max_i |X_{P_i}|$, whichever integer function $H$ is used; for $t = g$ the cap is at most $n - n/g$. Even a tight tensor-rank lower bound therefore gives no more than $U \geq 2(n - n/g - \log_2 L)$.

*(c) All caps at once (proved).* Let $c_{a,b} = w = \lceil 2\lceil n/g \rceil / g \rceil$ for all pairs. For a partition with largest group of $s$ nodes, at least $g(g-s)/2$ pairs lie in different groups, so $c(\mathcal{P}) \geq wg(g-s)/2 \geq \lceil n/g \rceil (g-s) \geq \mathrm{cap}(\mathcal{P})$, as the groups other than a largest one hold at most $\lceil n/g \rceil (g-s)$ variables. This choice has $U = wg(g-1) \leq 2n + 2g + g^2$ and loads $\sum_b c_{a,b} \leq 2\lceil n/g \rceil + g$. Hence any lower bound on $U$ that is derived only from inequalities $c(\mathcal{P}) \geq \log_2 \rho(\mathcal{P}) - \log_2 L$, with $\rho(\mathcal{P})$ bounded by the tensor rank of the grouping, is at most $2n + 2g + g^2$, which in Theorem D ($g = 3K^8$, $g^2 \leq 9n^{16/17}$) is $2n + o(n)$, against $U = O(kn \log \Delta)$ in the construction, even under the balanced loads that Theorem A requires.

*(d) What is left (argued).* Rank arguments, for any grouping of the nodes, thus cannot show that a kill needs more than about $2n$ shared bits, and in particular they cannot show that the factor $\log \Delta$, i.e. the factor $\log \log n$ that [1] states it does not know how to remove, is necessary.

A lower bound that forces it would have to use the graph and the bond dimensions together, for instance by showing that no integer function congruent to [$\xi$ satisfies $\Phi$] modulo $2^{n+1}$ is the value of a network on the complete graph with $O(n/g^2)$ bits per pair; such bond dimensions for a fixed graph are the graph-indexed ranks of Ye and Lim [34]. We have not checked what lower-bound methods are known for these ranks of explicit tensors. None of this gives a network with $O(n)$ shared bits; that question stays open.

**Comparison with a W[1] membership proof.** The cut-off of Theorem D can be compared with the counting in the proof that the uniform CSP parameterized by the solution size is in W[1] [33, Theorem 12]. There a nondeterministic program guesses an assignment $B$ of size $k$ and checks it with precomputed numbers indexed by small sets. For a constraint $\mathcal{C}_S$ with set of satisfying assignments $C_S$, a polynomial family $\tilde{C}_S$ of sets $T$ is defined; $I_T$ is the set of constraints with $T \in \tilde{C}_S$, and $f_{T,S}$ is the function on the proper supersets of $T$ in $C_S$ with $\sum_{W \subseteq U} f_{T,S}(W) = 1$ for every such $U$, given by Möbius inversion. With $l(T,W) = \sum_{\mathcal{C}_S \in I_T} f_{T,S}(W)$, the assignment $B$ satisfies the instance if and only if $\sum_{T \subset W \subseteq B} l(T,W) = |I_T|$ for every $T \subseteq B$ in $\tilde{C} = \bigcup_S \tilde{C}_S$ [33, Lemmas 10 and 11]: a satisfied constraint contributes exactly 1, and for some $T$ every violated constraint in $I_T$ contributes 0. The test is exact, with no truncation and no promise, because each constraint's contribution is computed from its own tuples; it is cheap because tuples have size at most $k$, so the intervals of the inversion have at most $2^k$ elements. In Theorem D the truncation at $\bar{n}$ violated clauses is exact only under the promise $|V(\xi)| \le \bar{n}$, a bound on the number of violated clauses of the whole formula that no node and no small set of clauses can refute, and the parity checks verify it. We argue that this is the difference that matters: the W[1] test compares counts for one guessed assignment, while a network sums over all $\xi$ a product of local factors, so its evidence against the many-violation assignments must itself be such a product, which is what the parity checks provide and what costs the factor $\log \Delta$. We know no local promise that would make the truncated sum of Theorem D exact and could be checked inside the node tables.

**Sparsification as a parameter.** Theorem D uses sparsification as a black box: a deterministic algorithm that, for $\varepsilon \in (0,1]$, splits a $k$-CNF on $n$ variables into at most $2^{\varepsilon n}$ formulas, in time $2^{\varepsilon n} n^{O(k)}$, such that every satisfying assignment satisfies exactly one output and no other assignment satisfies any, and every variable occurs in at most $\Delta(\varepsilon)$ clauses of each output. We call such an algorithm a *counting sparsifier with density* $\Delta(\varepsilon)$. The best known is Calabro's [25, Thm. 5.15, (5.4)] with the change of Dell et al. [26, App. A], with $\Delta(\varepsilon) \le 2(2 + 8192\varepsilon^{-2}\log_2^2(128/\varepsilon)) + 2$ for $k = 3$; we found no later improvement and no lower bound on $\Delta(\varepsilon)$ that grows as $\varepsilon \to 0$, and Calabro lists the removal of the $\varepsilon$-dependence as open [25, §5.5]. In the following, $(K_\ell, e_\ell)$ are the pairs of [1, (40)] with the pieces of [1, App. B], so $e_\ell = \Theta((\log K_\ell)^{-1/2})$.

**Proposition 3.15** (the factor $\log\log n$ is $\log \Delta$)**.** *Let $\mathcal{S}$ be a counting sparsifier with density $\Delta(\varepsilon)$. Replacing Calabro's sparsifier by $\mathcal{S}$ in Step 5 of the proof of [1, Theorem D(a)], the algorithm counts the satisfying assignments of a $k$-CNF on $n$ variables, $n \ge K_0^{17}$, $3 \le k \le (\log n)^{1/4}$, in time*

$$2^{O(ekn \log \bar{\Delta})}, \qquad \bar{\Delta} = \max\{2, \Delta(\varepsilon)\},\ \varepsilon \in [e, 2e],$$

*where $e = e_\ell$ for the largest $\ell$ with $K_\ell \le n^{1/17}$, provided $\bar{\Delta} \le (\log n)^{4kC_0}$. In particular, if $\Delta(\varepsilon) \le D$ for all $\varepsilon$ with a constant $D$, then the time is $2^{O(kn \log D/\sqrt{\log n})}$, without the factor $\log\log n$ of [1, Theorem D(b)].*

*Proof.* In the proof of Theorem D(a), Steps 1–4 treat one formula in which every variable occurs in at most $\Delta$ clauses, $2 \le \Delta \le (\log n)^{4kC_0}$, and show that its count takes $2^{O(ekn \log \Delta)}$ operations (l. 2356–2380). Step 5 applies the sparsifier with $\varepsilon \in [e, 2e]$ and runs Steps 2–4 on each of the at most $2^{2en}$ outputs, which gives $2^{2en} n^{O(k)} 2^{O(ekn \log \bar{\Delta})}$; the bound $\log \Delta \le 3k \log(8k/e) + 2$ and $e \ge (\log n)^{-C_0}$ are used there only to show that $\log \Delta = O(k \log \log n)$ and that $\Delta$ is within the range of Steps 1–4 (l. 2380–2411). With $\mathcal{S}$ in place of Calabro's algorithm the same argument gives the displayed time, using only the three properties of a counting sparsifier and the range condition. For $\Delta \le D$ and $e = \Theta((\log n)^{-1/2})$ (as $\log K_\ell = \Theta(\log n)$ by Theorem A(a)) the time is $2^{O(kn \log D/\sqrt{\log n})}$. □

So the factor $\log \log n$ of the general bound is $\log \Delta(e)$ of the sparsifier evaluated at the target exponent, and it disappears under an $\varepsilon$-independent density. It would also disappear if the assignments violating more than $\bar{n}$ clauses could be eliminated with $O(n)$ shared bits, since the remaining marks cost $O(kn)$ bits; this second route we only argue. Quantitatively the parameter is modest: Theorem D's own accounting floors $\Delta$ at 2, and even $\Delta = 2$ leaves $U/n \approx 210$ and divides $\log_2 n^*$ by at most 32 at $\rho = 1/\sqrt{2}$ (row G7: $5.0 \cdot 10^8 \to 1.6 \cdot 10^7$), with $\Delta = 1/\varepsilon$ already giving $6.2 \cdot 10^7$. A sharper analysis of Calabro's own lemmas lowers $\log_2 \Delta$ by about 4.7 bits but does not change the number of levels. The factor $\log \log n$ of [1] is stated there as one the authors do not know how to remove; the reformulation above names the two places where it could go.

# 4 Limits of rewrite-and-merge

Sections 2 and 3 looked at the model of Signed Network Sums from its two ends: at a fixed small number $K$ of nodes it goes below the floors of rank-one methods and contraction trees, and at astronomically large $K$ it gives the subexponential #3-SAT algorithm of [1], whose explicit thresholds Section 3 computes. This section explains why the two ends are separated, inside the paradigm on which the large end rests: rewrite a few configurations of equality tests by an exact identity, and evaluate every term of the identity by merging groups of nodes into the nodes of a smaller clique sum. The organising fact is a dichotomy (Section 4.1): what decides whether such an algorithm has a realistic threshold is not any of the constants that Section 3 improves, but whether the exponent $e(K)$ per edge bit decays polynomially or only polylogarithmically in $K$. We then define the narrowest class of algorithms that contains the scheme of [1] and all our constructions (Section 4.2), prove what follows from the definition (Sections 4.3 and 4.4), and prove that in the balanced homogeneous two-merge part of the class the decay is at most polylogarithmic, $1/e(K) \le (\log_2 K)^3$ (Section 4.5). At fixed $K$ we show what size arguments can and cannot decide (Section 4.6). Section 4.8 lists what is open; the one door that our arguments leave open in the full model is a configuration that meets three or more groups of a merge.

Throughout, a statement called Theorem, Lemma or Proposition is proved here, given the class definition and the cited facts about quantum functionals [5] and four-party rank inequalities [8]; values marked *computed* come from numerical searches and are lower estimates unless said otherwise; statements marked *argued* have an argument that we do not claim as a complete proof.

## 4.1 The threshold dichotomy

Before proving anything about algorithms we fix what a lower bound on $e(K)$ must look like to say something about thresholds. The answer is a dichotomy: the type of decay of $e(K)$ in $K$,

not its constants, decides whether the threshold is astronomical.

Fix a problem and a reduction that turns an instance with $n$ variables and $m$ clauses into weighted clique sums with $S = s\,n$ edge bits ($s$ edge bits per variable), and let $2^{\gamma n}$ be the running time of the best known algorithm for the problem. Write $\mathrm{bar} = \gamma/s$ for the *bar*: a clique-sum algorithm whose cost is $2^{eS}$ beats the known algorithm only if $e < \mathrm{bar}$. In Section 3 the number of nodes that is available at instance size $m$ is limited by the balancing of the reduction, $K \le m^{1/q}$ with a constant $q \ge 2$ (the balancing exponent of Section 3). We use only the following main-term reading of Section 3, in which all lower-order terms are dropped.

> **Threshold model.** An algorithm whose exponent on $K$-node instances is $e(K)$ gains at size $m$ only if $e(K) < \mathrm{bar}$ for some $K$ with $q \log_2 K \le \log_2 m$; up to lower-order terms, it gains at size $m$ if there is such a $K$. The threshold $m^*$ is the least size at which it gains.

**Proposition 4.1** (Threshold dichotomy)**.** *In the threshold model:*

*(a) if $e(K) \ge C(\log_2 K)^{-\alpha'}$ for all $K$, with constants $C, \alpha' > 0$, then $\log_2 m^* \ge q\,(C/\mathrm{bar})^{1/\alpha'}$;*

*(b) if $e(K) \le CK^{-\beta}$ for all $K$, with constants $C, \beta > 0$, then, up to the lower-order terms dropped by the model, $\log_2 m^* \le (q/\beta)\log_2(C/\mathrm{bar}) + q$.*

*Proof.* (a) A gain at size $m$ needs a $K$ with $C(\log_2 K)^{-\alpha'} \le e(K) < \mathrm{bar}$, i.e. $\log_2 K > (C/\mathrm{bar})^{1/\alpha'}$, and $\log_2 m \ge q \log_2 K$. (b) Take $K = \lceil (C/\mathrm{bar})^{1/\beta} \rceil + 1$; then $e(K) \le CK^{-\beta} < \mathrm{bar}$ and $q \log_2 K \le (q/\beta)\log_2(C/\mathrm{bar}) + q$ when $(C/\mathrm{bar})^{1/\beta} \ge 2$. □

So a polylogarithmic decay makes $\log m^*$ grow like a power of $1/\mathrm{bar}$, and a polynomial decay makes it grow like $\log(1/\mathrm{bar})$. The difference is large exactly where the bar is small. For general 3-SAT through the reduction of Section 3, $\gamma = 0.4092$ and about 1270 edge bits per variable (Section 3 counts both ends of an edge, $U/n \approx 2540$) give $\mathrm{bar} \approx 3.2 \cdot 10^{-4}$ per edge bit. This bar rests on the sharp count, in Section 3, of the bits of Theorem D of [1], an estimate (Table 5); a cheaper reduction would raise it. With $C = 1$ and $q = 2$, (a) gives $\log_2 m^* \ge 6.2 \cdot 10^3$ for $\alpha' = 1$ and $\ge 1.9 \cdot 10^7$ for $\alpha' = 1/2$, while (b) gives $\log_2 m^* \le 234$ for $\beta = 0.1$. The case $C = 1$, $\alpha' = 1$ is exactly what Theorem 4.13 below proves for grid algorithms (Corollary 4.15); the other two values only show the shape. With the actual constants of the recursion of [1] (where $\alpha' = 1/2$ for the pieces of its Appendix B), the thresholds of Section 3 are larger still. The constants that Section 3 improves (the bar through sparsification and the parity kill, the per-level overheads, the balancing exponent $q$) move $C$, bar and $q$, but not the type of decay. A method "subtler than the recursion" therefore means one thing precisely: an exponent that decays polynomially in $K$. A recursion with constant growth $K_{l+1} = gK_l$ and a constant per-level factor $\rho$ would be such a method, with $\beta = \log(1/\rho)/\log g$; the squaring $K_{l+1} = K_l^2$ of [1] gives only polylogarithmic decay. The rest of this section shows that the squaring is forced for grid levels and for balanced homogeneous two-merge levels of rewrite-and-merge.

## 4.2 The rewrite-and-merge class

A lower bound needs a model. We want the narrowest one that still contains every algorithm of this kind that we know, so that its lower bounds apply to them, and we want its cost to charge no more than [1] charges, so that the bounds also apply to that paper's own accounting. The definition below abstracts one level of the recursion: rewrite configurations of equality tests by

exact identities, then evaluate every term by merging groups of nodes. Three remarks after it say what the class contains, why balance is needed, and where it stops.

**Instances.** A *weighted clique sum* has nodes $[K]$, on each edge $e$ a string of $w_e\lambda$ bits ($w_e > 0$), and at each node $a$ a table $W_a$, an arbitrary function of $a$'s copies of the strings on its edges; its value is the sum over all copies of the product of the $W_a$ and of the equality tests $[\mathrm{eq}_e]$ between the two copies of each edge string. Its size is $S = \lambda \sum_e w_e$. The *exponent* of an algorithm is $\limsup_{\lambda\to\infty} \sup_W \log_2(\text{time})/S$, and $e_{\mathcal{C}}(K)$ is the infimum of the exponents over the algorithms of a class $\mathcal{C}$ on uniform instances ($w_e = 1$, $S = \lambda\binom{K}{2}$). Every algorithm reads a node table, and the largest one is indexed by $(K-1)\lambda = 2S/K$ bits; so $e(K) \geq 2/K$, the *input floor*.

**Definition 4.2** (Rewrite-and-merge level)**.** A level of a rewrite-and-merge algorithm on a weighted clique sum is given by the following data.

(L1) *Configurations.* A partition of the edge set into configurations $C$, with edge set $E(C)$ and node set $V(C)$; $f_C$ is the product of the equality tests of $E(C)$, a tensor whose legs are the nodes of $V(C)$ (the leg of $x$ is $x$'s copies of the strings of $E(C)$). A single edge is a configuration.

(L2) *Block identities.* Every string is cut into blocks of $m = m(\lambda)$ bits, $m \to \infty$ or $m = \lambda$. For each configuration $C$ and block there is an exact identity $f_C|_{\text{block}} = \sum_{i\in I_C} \pi_{C,i}$ (up to a nonzero scalar), with $|I_C| \leq 2^{o(m)}$, where each piece $\pi_{C,i}$ is a tensor with the same legs. The level's identity is the product of these sums over all (configuration, block) pairs; a *term* $Q$ chooses one piece for each pair, and every choice is a term (product identities).

(L3) *Merges.* Partitions $P_1, \ldots, P_p$ of $[K]$. Merge $k$ is usable only if every configuration meets at most two groups of $P_k$.

(L4) *Evaluation.* Each term $Q$ is evaluated under one usable merge $k(Q)$ of the algorithm's choice. A piece of a configuration that meets two groups $G, G'$ of $P_k$ is written by a fixed rank decomposition across $(V(C) \cap G \mid V(C) \cap G')$; its index becomes part of the merged edge $(G, G')$, and its *cost* $\mathrm{c}_k(\pi)$ is the $\log_2$ of that rank. Pieces inside one group are multiplied into that group's table. The merged instance is the weighted clique sum on the groups of $P_k$ whose edge $(G, G')$ carries the sum of the costs of the pieces on it.

(L5) *Cost.* The cost of the level is the sum over terms of the cost of building the merged node tables plus the cost of solving the merged instance. Building a table costs at least its number of entries; no other charge for building is made. The merged instance is solved by an algorithm of the class or by a base method, and the cost of an algorithm of the class depends only on the weights of its instance.

$\mathrm{RM}_p$ is the class of algorithms all of whose levels have at most $p$ merges. A partition is *balanced* if all its groups have the same size and it has between 2 and $K/2$ groups. A level is *homogeneous* if for each merge $P_k$ every pair of groups of $P_k$ carries the same multiset of configuration types (a type being the configuration together with the cells $G \cap H$, $G \in P_1$, $H \in P_2$, in which its nodes lie), and each type occurs on each pair in a number of (configuration, block) incidences that tends to infinity with $\lambda$. $\mathrm{RM}_2^{\mathrm{bh}}$ denotes the algorithms of $\mathrm{RM}_2$ all of whose levels are balanced and homogeneous.

The cost in (L4) is the paper's "costs $c$ bits" of [1, Sec. 2.2]: a function costs $c$ bits if it is a sum over $\theta \in \{0,1\}^c$ of products of a function of each side, and the least such $c$ is at least the $\log_2$ of the rank. So the class charges at most what the paper charges.

**Remark 4.3** (What the class contains)**.** The level of [1] is a $\mathrm{RM}_2^{\mathrm{bh}}$ level: the nodes form a $g \times g$ grid, the configurations are the rectangles (two diagonal edges $[x = v][y = u]$, every edge whose endpoints differ in both row and column being a diagonal of exactly one rectangle) and the edges within a row or column, the merges are rows and columns, there is one block per string ($m = \lambda$) and $|I| = \mathrm{poly}(\lambda)$, and every pair of rows carries $g$ straight edges and $g(g-1)/2$ rectangles. The homogeneity clause (many configurations of a type per pair of groups) holds as $g \to \infty$, or at fixed $g$ in the per-block form; at fixed $g$ with one block per string the paper's level is outside the clause as stated. One difference in accounting: the paper keeps the internal strings of a merged node as settings and sums them by brute force per table entry, which costs at least one operation per entry, so the paper's cost is at least the class's cost and every lower bound for the class applies to the paper's algorithm. Our constructions of Section 2 at $K = 4, 6, 9, 16$ are balanced homogeneous two-merge levels with a tree or a $K = 4$ method on top. The algorithm of Appendix D of [2] uses the polynomial method and is not in the class.

The balance condition in the definition is not a convenience: without it the size charge of (L5) can be gamed.

**Remark 4.4** (Balance is necessary)**.** Without balance the class bound fails. Take $P_1 = \{[K] \setminus \{u\}, \{u\}\}$. The group table of $K-1$ nodes has $2^{(K-1)\lambda} = 2^{2S/K}$ entries, and the merged instance has two nodes and $2S/K$ bits, so the modelled cost is $2^{2S/K}$ and $e_{\mathrm{RM}_2}(K) \le 2/K$, the input floor. The true cost of building that table is the cost of a $(K-1)$-node clique sum with external legs, which a size charge does not see. Balance excludes this; a stronger building charge for large groups would be the alternative.

The other restriction, (L3), limits which configurations a merge can use; it is the one that matters for what the class leaves open.

**Remark 4.5** (Higher-arity pieces are inside the class; hyperedges are not)**.** (L1) allows configurations of any size, subject only to (L3). So pieces of high arity are in the class as long as their configuration meets at most two groups of each merge. What (L3) excludes is a configuration that meets three or more groups of a merge, which would become a hyperedge of the merged instance. In particular, summing the terms of a product set of block choices gives coarser pieces of a coarser configuration inside the class, while summing terms over a set of choices that is not a product set gives a tensor on several blocks' legs, i.e. a piece of a configuration that may meet three groups.

### 4.3 Consequences of the definition

This subsection collects four facts that follow directly from the definition: every identity has a piece that is expensive across a given split (the pigeonhole floor), sharing work across terms cannot go below the sizes of the tables (the sharing cap), a configuration can at best split its cost evenly between two merges (the two-merge floor), and few edges lie in configurations that could trade at all (the count of four-cell edges).

Let $C$ be a configuration and $(A \mid V(C) \setminus A)$ a split of its nodes across which $r$ edges of $C$ cross. Then $f_C|_{\mathrm{block}}$ has rank $2^{rm}$ across the split, since it is the identity matrix on the crossing strings tensored with factors on one side.

**Proposition 4.6** (Pigeonhole floor)**.** *In every block identity of (L2), some piece has* $\log_2$*-rank at least* $rm - \log_2 |I_C|$ *across the split.*

*Proof.* Rank is subadditive and the pieces sum to a tensor of rank $2^{rm}$. □

The class charges a term by the sizes of its merged tables. One might hope to do better by sharing computation between terms; the next proposition shows that sharing cannot go below the table sizes.

**Proposition 4.7** (Sharing cap)**.** *Let a term* $T_Q$ *be evaluated in any way that is bilinear across some split* $(X \mid Y)$ *of the nodes, i.e. as* $\sum_\sigma t_X(\sigma)\, t_Y(\sigma)$ *where* $t_X$ *depends on the tables of* $X$ *and* $t_Y$ *on those of* $Y$*; for instance, by a contraction tree whose tables are materialised, computed by any algorithm and shared with other terms in any way. Then the number of strings* $\sigma$*, and hence the cost of writing the table* $t_X$*, is at least* $2^{\sum_\pi \mathrm{c}_X(\pi)}$*, where* $\mathrm{c}_X(\pi)$ *is the* $\log_2$*-rank of the piece* $\pi$ *across the split induced by* $(X \mid Y)$ *on its legs (raw edges counted as pieces of full rank).*

*Proof.* A term is a tensor product of its pieces on disjoint legs, so its rank across $(X \mid Y)$ is the product of the pieces' ranks across the induced splits. Pairing the term with arbitrary node tables of $X$ and $Y$ gives a matrix, indexed by the tables of $X$ and of $Y$, that factors through the strings $\sigma$; since products of node tables span all tensors on the legs, its rank is the rank of the term across $(X \mid Y)$. Writing $t_X$ costs at least its number of entries. □

In the class the merged edge weights of (L4) are exactly these log-ranks, so Proposition 4.7 says that sharing computation across terms can lower the cost of an algorithm at most to the cost of writing the materialised tables of its terms; the size game of Section 4.6 computes that floor. Summing terms before contracting does not escape it when the sum is over a product set of block choices, since that only produces a coarser Signed Network Sum to which the proposition applies again; a non-product grouping is a higher-arity piece (Remark 4.5), to which the proposition applies with a cost we cannot evaluate. A sharper form (the rank equals the number of strings in a minimal decomposition) is not needed here.

**Proposition 4.8** (Two-merge floor)**.** *Let* $C$ *meet two groups of each of two merges, with induced splits* $S_1, S_2$ *of* $V(C)$ *across which* $r_1, r_2$ *of its edges cross. For every* $\theta \in [0,1]$*, taking in a fraction* $\theta$ *of the blocks a piece of Proposition 4.6 for* $S_1$ *and in the rest one for* $S_2$*, every such term pays on* $C$ *at least* $\theta r_1 - o(1)$ *per* $\lambda$ *under merge 1 and at least* $(1-\theta) r_2 - o(1)$ *under merge 2. In particular the per-level factor of any two-merge identity on* $C$ *satisfies* $\rho \geq 1/2$*.*

*Proof.* Costs add over blocks, and $\log_2 |I_C| = o(m)$. When $m = \lambda$ the fractions are taken over the configurations of a type on a pair of groups, which homogeneity makes numerous. □

For the rectangle (two diagonal edges of a $2 \times 2$ grid) this is the observation of [1, App. H] that one piece needs $2\lambda - O(\log \lambda)$ bits row-wise, which leaves it at most about $(4\rho - 2)\lambda$ column-wise; the proposition is its general form, for every configuration.

**Four-cell reduction.** Under two merges $P_1, P_2$ the nodes of a configuration meeting at most two groups $G, G'$ of $P_1$ and $H, H'$ of $P_2$ fall into at most four cells, $11 = G \cap H$, $12 = G \cap H'$, $21 = G' \cap H$, $22 = G' \cap H'$. Both splits are unions of cells, so grouping each cell into one party changes neither the ranks of pieces across $S_1, S_2$ nor which decompositions exist. Hence $f_C$ is the graph tensor of a multigraph on the four cells, with multiplicities $d$ (on 11–22), $a$ (12–21),

$v_1$ (11–21), $v_2$ (12–22), $h_1$ (11–12), $h_2$ (21–22), times a factor internal to the cells. The split $S_1 = \{11, 12\} \mid \{21, 22\}$ crosses $d, a, v_1, v_2$ and $S_2 = \{11, 21\} \mid \{12, 22\}$ crosses $d, a, h_1, h_2$. Every question about trading cost between two merges is a question about this six-parameter tensor.

**Proposition 4.9** (Count of four-cell edges)**.** *Let $c_k$ be the largest group size of $P_k$. At most $K(c_1c_2+c_1+c_2)/2$ edges lie in configurations with four nonempty cells. For balanced partitions with $n_k$ groups, the fraction $t$ of edges in four-cell configurations satisfies $t \le (c_1c_2+c_1+c_2)/(K-1) = \big(K/(n_1n_2) + 1/n_1 + 1/n_2\big)\, K/(K-1)$.*

*Proof.* Fix an endpoint $u \in G \cap H$ of such an edge. If the other endpoint lies in another group of $P_1$ and of $P_2$, it lies in a cell $G' \cap H'$ of the same configuration with $G \cap H' \neq \emptyset$, so $H'$ is one of at most $c_1$ groups that meet $G$, and there are at most $c_1c_2$ such endpoints. Otherwise it lies in $H$ (at most $c_2$ choices) or in $G$ (at most $c_1$). A configuration with four nonempty cells meets two groups of each merge. □

For two partitions into groups of size at most $c$ this gives at most $K(c^2 + 2c)/2$ edges, a fraction $O(c^2/K)$. So a level in which a constant fraction of the bits can trade needs groups of size $\Omega(\sqrt{K})$, that is, at most $O(\sqrt{K})$ merged nodes: the squaring of [1], forced by the locality of the trade, *provided* that only four-cell configurations trade. The next section shows that this proviso is a statement about tensors that flattening ranks cannot decide, and proves the part of it that we need.

### 4.4 Which configurations can trade

The count of four-cell edges is useful only if configurations on fewer cells cannot trade cost between the two merges. This subsection proves that they cannot (Lemma 4.11). The example first shows why the natural tool, flattening ranks, is not enough; the proof uses quantum functionals instead.

**Example 4.10** (Flattening inequalities do not suffice)**.** Let $C$ be a triangle with one node in each of the cells $11, 12, 22$ ($d = h_1 = v_2 = 1$, cell 21 empty), with $N = 2^m$ values per block string. Then $S_1$ isolates the party 22 and $S_2$ isolates the party 11, and $f_C$ is the matrix multiplication tensor $\langle N, N, N\rangle$. Consider pieces of the two forms $A = \alpha(11) \otimes M(12, 22)$ and $B = \beta(22) \otimes M'(11, 12)$ with $M, M'$ of rank $N^2$: $A$ is full across $S_1$ (rank $N^2$) and free across $S_2$ (rank 1), $B$ the reverse. Each satisfies every three-party flattening-rank inequality (for $A$, $r_{22} \le r_{11}r_{12}$ holds with $r_{12} = N^2$), and a family containing one of each kind passes the pigeonhole test of Proposition 4.6 for both splits. If a family of $2^{o(m)}$ such pieces summed to $f_C$, the half-half mixture would make every term pay half of the triangle's bits under each merge, a trade on a configuration with no rectangle. So no argument that uses only the flattening ranks of the pieces can show that three-cell configurations do not trade; the invariant that excludes such families is the quantum functional.

We use the quantum functionals $F^\theta$ of Christandl, Vrana and Zuiddam [5], defined for complex tensors and a probability vector $\theta$ on the parties, with the following properties: $F^\theta$ is monotone under restriction, additive on direct sums, multiplicative on tensor products and normalised ($F^\theta$ of the rank-$r$ unit tensor is $r$); and $F^\theta(T) \le \prod_x R_x(T)^{\theta_x}$, where $R_x$ is the flattening rank of party $x$ against the rest. A sum $T + T'$ is a restriction of $T \oplus T'$ (each party adds its two components), so $F^\theta(\sum_i T_i) \le \sum_i F^\theta(T_i)$. On a graph tensor, in which every edge is an $N \times N$ identity matrix between its two parties, $F^\theta = N^{\sum_{\{x,y\}} (\theta_x + \theta_y)}$, the sum over edges between distinct parties (edges inside a party contribute a vector, i.e. a factor 1).

**Lemma 4.11** (Configurations with at most three cells do not trade)**.** *Under two merges, let $C$ be a configuration with at most three nonempty cells. For every block identity of (L2) on $C$ there is a piece that pays, under each merge whose split crosses $C$, its full cost up to $O(\log_2 |I_C|)$ per block:* $\mathrm{c}_1 \geq r_1 m - 2\log_2 |I_C|$ *and* $\mathrm{c}_2 \geq r_2 m - 2\log_2 |I_C|$.

*Proof.* With at most two nonempty cells, either $C$ crosses at most one merge, and Proposition 4.6 gives the piece, or the cells are opposite (11 and 22, or 12 and 21), the two splits coincide on $V(C)$, and again Proposition 4.6 gives it. With three nonempty cells, let the empty one be 21; the other cases are symmetric. On $V(C)$ the split $S_1$ is $\{11, 12\} \mid \{22\}$, which isolates $x = 22$, and $S_2$ is $\{11\} \mid \{12, 22\}$, which isolates $y = 11$; the third cell $z = 12$ is the corner. Then $r_1$ is the number of edges of $C$ at $x$ that leave $x$'s cell, and $r_2$ the same for $y$.

First fix the strings of all edges inside cells to one value; this restricts $f_C|_{\text{block}}$ to the graph tensor $f'$ of the three-party multigraph, and each piece $\pi$ to a tensor $\pi'$ with $\sum \pi' = f'$ and with ranks across $S_1, S_2$ at most those of $\pi$. Take $\theta = (1/2, 1/2, 0)$ on $(x, y, z)$. On $f'$ an $x$–$y$ edge contributes $N^1$ and an $x$–$z$ or $y$–$z$ edge $N^{1/2}$, so $F^\theta(f') = 2^{m(r_1+r_2)/2}$. For a piece, $F^\theta(\pi') \leq R_x(\pi')^{1/2} R_y(\pi')^{1/2}$, and the flattenings at $x$ and at $y$ are the splits $S_1$ and $S_2$; so $F^\theta(\pi') \leq 2^{(\mathrm{c}_1(\pi)+\mathrm{c}_2(\pi))/2}$. By sub-additivity some piece has $(\mathrm{c}_1 + \mathrm{c}_2)/2 \geq m(r_1 + r_2)/2 - \log_2 |I_C|$.

It remains to split this sum, and here the isolation of single cells is essential: in $f'$ the leg of $x$ has dimension exactly $2^{mr_1}$, so no restricted piece is over-full across $S_1$, $\log_2 R_x(\pi') \leq mr_1$, and likewise $\log_2 R_y(\pi') \leq mr_2$. With the bound on the sum this gives $\log_2 R_x(\pi') \geq mr_1 - 2\log_2 |I_C|$ and $\log_2 R_y(\pi') \geq mr_2 - 2\log_2 |I_C|$, and the costs of $\pi$ are at least these. □

The step without over-full pieces is the one that fails elsewhere. For the 4-cycle ($d = a = 0$, only $v$- and $h$-edges), the inequality of Song, Chen, Sun and Hu (Proposition 4.22 below) applied across the third split $\{11, 22\} \mid \{12, 21\}$ gives a piece with $\mathrm{c}_1 + \mathrm{c}_2 \geq (v + h)m$ up to $o(m)$, but the side $\{11, 12\}$ of $S_1$ has dimension $2^{m(v+2h_1)}$, so pieces can be over-full across $S_1$ or $S_2$, and a three-piece family that is consistent with every constraint we know defeats every mixture. So the 4-cycle is settled only for families without over-full pieces. The smallest open case of a general statement is $K_4$ minus a diagonal ($d = 1$, $a = 0$, $v_1 = v_2 = h_1 = h_2 = 1$): we argue that the known flattening inequalities and the functional with $\theta = 1/2$ on the two ends of the diagonal allow a family that saves one bit per block under each merge, and whether such a family exists is open. Neither case is needed below.

The conjectured general form (we call it the rectangle-content bound) would say that only $\min(d, a)$ bits of a four-cell configuration can be traded between the two merges, one for one, and that $v$-, $h$- and excess diagonal edges always pay in full; for $d = a = 1$ it is the floor $\rho \geq 1/2$ of Proposition 4.8. It would improve the constants of Theorem 4.12 below, not its structure.

Lemma 4.11, the two-merge floor and Proposition 4.9 together are a counting lemma with a proved premise: in $\mathrm{RM}_2$, only edges of four-cell configurations can trade, there are at most $K(c_1c_2 + c_1 + c_2)/2$ of them, and they save at most half of their bits under each merge.

## 4.5 Polylogarithmic decay in balanced homogeneous two-merge algorithms

This subsection turns the counting lemma into a bound on the decay of $e(K)$: first for balanced homogeneous two-merge algorithms, with exponent 3 (Theorem 4.12), then for grid algorithms with the exact exponent 1 (Theorem 4.13), which gives the threshold bound of Corollary 4.15, and finally for the recursion of [1] as written, with one block per string (Theorem 4.18).

Write $f(n) = 1/e_{\mathrm{RM}_2^{\mathrm{bh}}}(n)$. We need one condition on base methods: a base method used on $n$-node instances has exponent at least $(\log_2 n)^{-3}$. By the input floor this holds for every

method on at most $2^{11}$ nodes, since $2/n \geq (\log_2 n)^{-3}$ there (the ratio $(n/2)/(\log_2 n)^3$ is at most 1 on $[2, 2^{11}]$, with value 1 at $n = 2$ and $1024/1331$ at $n = 2^{11}$), and for larger $n$ it holds for every method whose exponent is at least $1/1331$, for instance every contraction tree, whose largest table crosses a cut with between $n/3$ and $2n/3$ nodes on one side.

Consider a level on $K$ nodes with two balanced merges, $n_k$ groups of size $c_k = K/n_k$ in $P_k$ ($2 \leq n_k \leq K/2$). The fraction of edges internal to the groups of $P_k$ is $(c_k - 1)/(K - 1) \leq 1/n_k$; write $a_k \geq 1 - 1/n_k$ for the fraction that crosses $P_k$, and $t$ for the fraction in four-cell configurations. A term is evaluated under one merge, and its merged instance under merge $k$, with $B_k$ bits, costs at least $2^{e(n_k)B_k}$ if it has uniform weights. The cost of the level is at least the largest, over terms, of the cheaper of its two merges. Two inequalities follow.

(F1) $f(K) \leq f(n_1)/a_1 + f(n_2)/a_2$. The adversary takes, for every configuration crossing both merges, the two pieces of Proposition 4.8 in proportions $\theta$ and $1 - \theta$, and for every configuration crossing one merge the piece of Proposition 4.6. Every merged edge of merge 1 then carries at least $\theta$ times its full weight, which is the same for all pairs of groups by balance, so, bounding the merged instance through its shortest edge as in the proof of Theorem 4.18, it costs at least the uniform instance with $\theta a_1 S$ bits; likewise for merge 2 with $(1 - \theta)a_2 S$. Equalising $e(n_1)\theta a_1 = e(n_2)(1 - \theta)a_2$ gives $e(K) \geq \big(f(n_1)/a_1 + f(n_2)/a_2\big)^{-1}$.

(F2) $f(K) \leq \max_k f(n_k)/(a_k - t)$. The adversary takes, for every configuration with at most three cells, the piece of Lemma 4.11, and for the four-cell configurations a piece depending only on the type. By homogeneity the merged instances are uniform, and under merge $k$ they carry at least $(a_k - t)S$ bits.

A level with one merge is (F2) with $t = 0$. All bounds hold up to factors $1 - o(1)$ as $\lambda \to \infty$, which do not affect exponents.

**Theorem 4.12.** *For every $K \geq 2$, every algorithm of $\mathrm{RM}_2^{\mathrm{bh}}$ whose base methods satisfy the base condition has exponent at least $(\log_2 K)^{-3}$ on uniform $K$-node instances; that is, $f(K) \leq (\log_2 K)^3$.*

*Proof.* Strong induction on $K$; write $L = \log_2 K$ and $l_k = \log_2 n_k \leq L - 1$. For $K \leq 2^{11}$ the input floor gives $f(K) \leq K/2 \leq L^3$. Let $L \geq 11$; an algorithm on $K$ nodes either is a base method, covered by the base condition, or starts with a level, to whose merged instances the induction hypothesis $f(n_k) \leq l_k^3$ applies, together with the input floor $f(n_k) \leq n_k/2$.

*Case 1: $t \leq 1/L$.* Use (F2). If $n_k \geq 4L$, then $a_k - t \geq 1 - 1/(4L) - 1/L = 1 - 1.25/L$ and $f(n_k)/(a_k - t) \leq (L-1)^3/(1 - 1.25/L) \leq L^3$, because $(1 - 1/L)^3 \leq 1 - 3/L + 3/L^2 \leq 1 - 1.25/L$ for $L \geq 2$. If $n_k < 4L$, then $f(n_k) \leq n_k/2 < 2L$ and $a_k - t \geq 1 - 1/2 - 1/11 > 0.4$, so $f(n_k)/(a_k - t) < 5L \leq L^3$.

*Case 2: $t > 1/L$.* By Proposition 4.9, $K/(n_1 n_2) + 1/n_1 + 1/n_2 > (K - 1)/(KL) \geq 0.99/L$, so one of the three terms exceeds $0.33/L$: $n_1 < 3.1L$, or $n_2 < 3.1L$, or $n_1 n_2 < 3.1KL$.

*Case 2a: $n_1 < 3.1L$* (the case $n_2 < 3.1L$ is symmetric). Use (F1). The first term is $f(n_1)/a_1 \leq (n_1/2)/(1 - 1/n_1) = n_1^2/(2(n_1 - 1)) \leq n_1 < 3.1L$. If $n_2 \geq 4L$, the second is at most $(L-1)^3 \cdot 4L/(4L - 1) \leq (L-1)^3(1 + 0.3/L) \leq L^3 - 2.7L^2 + 3.3L$, and the total is at most $L^3 - 2.7L^2 + 6.4L \leq L^3$. If $n_2 < 4L$, the second is at most $n_2 < 4L$ and the total is below $7.1L \leq L^3$.

*Case 2b: $n_1, n_2 \geq 3.1L$ and $n_1 n_2 < 3.1KL$.* Then $l_1 + l_2 < L + l_0$ with $l_0 = \log_2(3.1L)$, and $l_k \in [l_0, L - 1]$. By (F1) and $1/a_k \leq 1 + \delta$, $\delta = 1/(3.1L - 1)$, $f(K) \leq (1 + \delta)(l_1^3 + l_2^3)$. The convex

function $l_1^3 + l_2^3$ on this polygon is largest at a vertex, which is $(L-1, l_0+1)$ or its mirror image, so $f(K) \le (1+\delta)\big((L-1)^3 + (l_0+1)^3\big)$. Since $(L-1)^3 \le L^3 - 3L^2 + 3L$, this is at most $L^3$ provided

$$(1+\delta)(l_0+1)^3 + 3L + \delta L^3 \le 3L^2.$$

At $L = 11$ the left side is $232.9 + 33 + 40.2 = 306.1$ and the right side 363; the difference of the two sides grows with $L$. □

The exponent 3 is an artefact of the proof. The square grid with ideal pieces (pieces full in one direction and free in the other, which reach $\rho = 1/2$) attains $f(K) = \log_2 K$, so no bound of the form $f(K) \le (\log_2 K)^\gamma$ with $\gamma < 1$ can hold if such pieces exist; the true exponent for $\mathrm{RM}_2^{\mathrm{bh}}$ lies between 1 and 3 in that case, and the pieces of [1, App. B], which exist, give $e(K)$ of order $(\log K)^{-1/2}$.

**Grid algorithms: the exact exponent.** For the levels that all known constructions use, the induction closes with exponent 1, because the bookkeeping of a grid is exact. A *grid level* on $K = ab$ nodes ($a, b \ge 2$) places the nodes on an $a \times b$ grid; $P_1$ is the partition into the $a$ rows, $P_2$ into the $b$ columns; the configurations are the raw row edges, the raw column edges, and the rectangles (the two diagonal edges of each $2 \times 2$ subgrid), each rectangle expanded by its own exact per-block identity, with $\lambda/m \to \infty$ blocks per string. A *grid algorithm* uses only grid levels (with any $a, b$ and any piece families, which may change from level to level) and base methods of exponent at least $1/\log_2 n$ on $n$ nodes. Contraction trees and brute force satisfy the base condition for all $n \ge 2$, since their exponent is at least $\lceil n/3 \rceil (n - \lceil n/3 \rceil)/\binom{n}{2} \ge 1/\log_2 n$. The recursion of [1] with its brute-force or tree base and its rectangle identities applied per block, and the per-block constructions of Section 2 (the $K = 4$ constructions are $2 \times 2$ levels), are grid algorithms once the grouping of the merged nodes (their Lemma 8) and the padding to a $(K, L)$-clique are admitted as steps, which Theorem 4.18 below does; the recursion as written, with one block per string, is covered by that theorem, and the rank-counted grouping of Section 2, which acts on the whole string, is not covered.

**Theorem 4.13.** *Every grid algorithm has exponent at least* $1/\log_2 K$ *on* $K$*-node instances:* $f(K) \le \log_2 K$.

*Proof.* Strong induction on $K$. If the algorithm is a base method, this is the base condition. Otherwise it starts with an $a \times b$ grid level. The fraction of edges inside rows is $i_1 = (b-1)/(K-1)$, inside columns $i_2 = (a-1)/(K-1)$, and the rectangle diagonals make up the rest, $t = (a-1)(b-1)/(K-1)$; so $i_1 + i_2 + t = 1$ exactly. Let the adversary take, in every rectangle and in the same proportions $\theta$ and $1-\theta$ of its blocks, the piece full across the row split and the piece full across the column split (Proposition 4.8). Under the row merge every merged edge then carries the same number of bits, its column edges in full and $\theta$ of its rectangle bits, so by restriction the merged instance on $a$ nodes costs at least $2^{e(a)(i_2+\theta t)S}$; under the column merge, $2^{e(b)(i_1+(1-\theta)t)S}$. Write $x_1 = e(a)$, $x_2 = e(b)$. If the $\theta$ that equalises the two exponents lies in $[0,1]$, the common value is $x_1x_2(i_1+i_2+t)/(x_1+x_2) = x_1x_2/(x_1+x_2)$, so $f(K) \le f(a) + f(b) \le \log_2 a + \log_2 b = \log_2 K$. Otherwise, say it exceeds 1; then at $\theta = 1$ the row merge is the cheaper one and $f(K) \le f(a)/(1-i_1)$. Since $\log_2 c/\log_2 K$ is concave in $c$ and agrees with $(c-1)/(K-1)$ at $c = 1$ and $c = K$, we have $i_1 = (b-1)/(K-1) \le \log_2 b/\log_2 K$, hence $1 - i_1 \ge \log_2 a/\log_2 K$ and $f(K) \le \log_2 a \cdot \log_2 K/\log_2 a = \log_2 K$. The case below 0 is symmetric. □

The bound cannot be improved by this argument alone, because it would be attained by pieces whose existence is open.

**Remark 4.14** (tightness)**.** For $K = 4^{2^j}$ and in the cost model that charges tables by their size only, the bound is attained by square grids with pieces that are full in one direction and free in the other, if such pieces exist: on a $g \times g$ grid the equalised value gives $f(g^2) = 2f(g)$ exactly when the pieces reach $\rho = 1/2$.

Through the dichotomy, Theorem 4.13 turns into a lower bound on the threshold. This is the bound behind the size $2^{6000}$ of the introduction. It holds for every choice of pieces and every schedule; only a cheaper reduction, which raises the bar bar, could lower it.

**Corollary 4.15.** *In the threshold model, every grid algorithm, in particular the recursion of [1] with its rectangle identities applied per block or as written (Corollary 4.19 below) and any piece families, has* $\log_2 m^* \geq q/\mathrm{bar}$*; for general 3-SAT through the reduction of Section 3 (*$\mathrm{bar} \approx 3.2 \cdot 10^{-4}$ *per edge bit,* $q \geq 2$*) this is* $\log_2 m^* \geq 6.2 \cdot 10^3$*. For the single-block form of the recursion as written the bound holds up to a factor* $1 - 10^{-21}$ *(Corollary 4.19 below).*

*Proof.* Proposition 4.1(a) with $C = 1$, $\alpha' = 1$ and Theorem 4.13. □

For comparison only (no lower bound of this form is proved for them), the decay that the pieces actually known achieve, $\alpha' = 1/2$, would correspond to $\log_2 m^* \approx q/\mathrm{bar}^2 \approx 1.9 \cdot 10^7$; Section 3 has the exact numbers. So a usable threshold needs levels that are not grids.

**One block per string.** When a whole string is one block ($m = \lambda$, as in the recursion of [1], whose pieces depend on $\lambda$), the mixture of Proposition 4.8 must be taken across rectangles instead of across blocks, and a row pair and a column pair share only one rectangle. The merged instance is then only nearly uniform, so we state the induction for weighted instances and use only their shortest edge. Write $\beta(I)$ for the least number of bits on an edge of an instance $I$.

**Definition 4.16** (Single-block grid algorithms)**.** A *single-block grid algorithm* is built from the following steps, each of which solves the merged instances of its terms by a single-block grid algorithm that may depend on the term and on $\lambda$:

(i) a *grid level* as above with one block per string ($m = \lambda$);

(ii) a *grouping level*: one balanced merge into $n$ groups of $c = K/n$ nodes, with single edges as configurations (Lemma 8 of [1], case (ii));

(iii) *padding*: lengthening edge strings by bits that the tables ignore;

(iv) a *base method* satisfying the weighted base condition: on every $n$-node instance $I$ it costs at least $2^{\beta(I)\binom{n}{2}/\log_2 n}$.

We assume a function $\sigma(\lambda) = o(\lambda)$ with $\log_2 |I_C| \leq \sigma(\lambda)$ for every identity used at parameter $\lambda$. Let $h_A$ be the largest number of grid and grouping levels on a chain of nested steps of $A$, and define $\Pi_A$ recursively: $\Pi_A = 1$ for a base method and for a $2 \times 2$ level; a padding step or a grouping level has the largest $\Pi$ of the algorithms it runs; an $a \times b$ grid level with $a, b \geq 3$ has that largest value divided by $1 - \bar{\varepsilon}_{a,b}$, where $\bar{\varepsilon}_{a,b} = 2/a^2 + 2/b^2$.

Brute force and contraction trees satisfy the weighted base condition, by the bound on their largest table used for the base condition above, since every edge carries at least $\beta(I)$ bits; and every algorithm satisfies it on 4 nodes, since a node table of $I$ is indexed by at least $3\beta(I) = \beta(I)\binom{4}{2}/\log_2 4$ bits (the input floor).

**Lemma 4.17** (Rounding)**.** *For integers $\mu, \nu \geq 1$ and $\theta \in [0,1]$, the matrix $X_{ij} = \lfloor (i+j+1)\theta \rfloor - \lfloor (i+j)\theta \rfloor$ ($0 \leq i < \mu$, $0 \leq j < \nu$) has entries in $\{0,1\}$, row sums between $\lfloor \nu\theta \rfloor$ and $\lceil \nu\theta \rceil$, and column sums between $\lfloor \mu\theta \rfloor$ and $\lceil \mu\theta \rceil$.*

*Proof.* The entries are 0 or 1 because $0 \leq \theta \leq 1$. The sum of row $i$ telescopes to $\lfloor (i+\nu)\theta \rfloor - \lfloor i\theta \rfloor$, and $\lfloor x+y \rfloor - \lfloor x \rfloor$ is $\lfloor y \rfloor$ or $\lfloor y \rfloor + 1$, and equals $y$ when $y$ is an integer. Columns are the same with $\mu$ in place of $\nu$. □

With the rounding lemma the adversary can spread the two kinds of pieces over the rectangles almost evenly, and the proof of Theorem 4.13 goes through for weighted instances, at a loss that is small when the grid sides are large.

**Theorem 4.18** (One block per string)**.** *Let $A$ be a single-block grid algorithm all of whose grid levels have both sides at least 3 or are $2 \times 2$. For every $\lambda$ and every instance $I$ on $K$ nodes,*

$$\log_2 \mathrm{cost}(A, I) \;\geq\; (\beta(I) - h_A \sigma(\lambda)) \binom{K}{2} \Big/ (\Pi_A \log_2 K).$$

*In particular $1/e(K) \leq \Pi_A \log_2 K$ on uniform instances, and if the grid levels on every chain of $A$ satisfy $\sum_l \bar{\varepsilon}_{a_l,b_l} \leq \delta < 1$, then $1/e(K) \leq \log_2 K/(1-\delta)$.*

*Proof.* Induction on the structure of $A$; write $\beta = \beta(I)$, $\sigma = \sigma(\lambda)$, $h = h_A$, and assume $\beta > h\sigma$, since otherwise there is nothing to prove. Each level costs at least the cost of solving the merged instance of any one of its terms under the merge chosen for that term. We use Proposition 4.6 in the form: a configuration whose edges crossing a split carry $r$ bits has a piece of $\log_2$-rank at least $r - \sigma$ across it. For a base method the claim is the weighted base condition, and for a $2 \times 2$ level it is the input floor shown above. Padding raises $\beta$, so the claim for the algorithm that is run gives it.

*Grouping level.* Every one of the $c^2$ edges between two groups pays at least $\beta - \sigma$, so every merged edge has at least $c^2(\beta - \sigma)$ bits, and the induction hypothesis (with at most $h-1$ levels and a factor at most $\Pi_A$) gives $\log_2 \mathrm{cost} \geq (\beta - h\sigma)c^2\binom{n}{2}/(\Pi_A \log_2 n)$, using $c^2 \geq 1$. Here $c^2\binom{n}{2} = (1-i)\binom{K}{2}$ with $i = (c-1)/(K-1)$, and $i \leq \log_2 c/\log_2 K$, since $\log_2 x/\log_2 K$ is concave and agrees with $(x-1)/(K-1)$ at $x = 1$ and $x = K$. So $(1-i)/\log_2 n \geq 1/\log_2 K$, which is the claim.

*Grid level, $a, b \geq 3$.* Let $\Pi'$ be the largest $\Pi$ of the algorithms run on its merged instances, so $\Pi_A = \Pi'/(1 - \bar{\varepsilon}_{a,b})$, and put $x_1 = 1/(\Pi' \log_2 a)$, $x_2 = 1/(\Pi' \log_2 b)$. With $i_1, i_2, t$ as in the proof of Theorem 4.13, let $\varepsilon_1 = 2\binom{a}{2}/\binom{K}{2}$ and $\varepsilon_2 = 2\binom{b}{2}/\binom{K}{2}$. For every rectangle there is a piece of $\log_2$-rank at least $2\beta - \sigma$ across the row split and one across the column split. Index the $\mu = \binom{a}{2}$ row pairs by $i$ and the $\nu = \binom{b}{2}$ column pairs by $j$; given $\theta \in [0,1]$, let the term $Q$ take in rectangle $(i,j)$ the first piece if $X_{ij} = 1$ (Lemma 4.17) and the second otherwise. Every choice of pieces is a term, so $Q$ is one. Under the row merge a merged edge carries the $b$ column edges between its rows and at least $\lfloor \theta\nu \rfloor \geq \theta\nu - 1$ rectangles with the first piece, so it has at

least $b(\beta-\sigma)+\lfloor\theta\nu\rfloor(2\beta-\sigma)\ge b(\beta-\sigma)+2\lfloor\theta\nu\rfloor(\beta-\sigma)\ge(b+2\theta\nu-2)(\beta-\sigma)$ bits. Since $b+2\theta\nu-2\ge 1$, the induction hypothesis gives

$$\log_2 \mathrm{cost}_{\mathrm{row}} \ge (\beta-h\sigma)\binom{K}{2}R(\theta), \qquad R(\theta)=x_1(i_2+\theta t-\varepsilon_1),$$

because $b\binom{a}{2}=i_2\binom{K}{2}$, $2\nu\binom{a}{2}=t\binom{K}{2}$ and $2\binom{a}{2}=\varepsilon_1\binom{K}{2}$. In the same way, with at least $\mu-\lceil\theta\mu\rceil\ge(1-\theta)\mu-1$ rectangles of the second piece on each column pair, $\log_2\mathrm{cost}_{\mathrm{col}}\ge(\beta-h\sigma)\binom{K}{2}C(\theta)$ with $C(\theta)=x_2(i_1+(1-\theta)t-\varepsilon_2)$. The bounds hold for every $\theta$, because $x_1,x_2$ do not depend on the term. $R$ increases and $C$ decreases. If they meet at some $\theta\in[0,1]$, the common value $V$ satisfies $V/x_1+V/x_2=i_1+i_2+t-\varepsilon_1-\varepsilon_2=1-\varepsilon_1-\varepsilon_2$, so $V=(1-\varepsilon_1-\varepsilon_2)/(\Pi'\log_2 K)$. If $R(1)<C(1)$, take $\theta=1$: since $1-i_1=b(a-1)/(K-1)$ we have $\varepsilon_1=(2/b^2)(1-i_1)$, and $1-i_1\ge\log_2 a/\log_2 K$ as above, so $R(1)\ge(1-2/b^2)/(\Pi'\log_2 K)$; the case $C(0)<R(0)$ is symmetric with $2/a^2$. Finally $\varepsilon_1+\varepsilon_2\le\bar\varepsilon_{a,b}$, since $a(a-1)/(K(K-1))\le a^2/K^2=1/b^2$ and likewise for $b$. So in every case $\min(R,C)\ge(1-\bar\varepsilon_{a,b})/(\Pi'\log_2 K)=1/(\Pi_A\log_2 K)$, which is the claim. The last statement follows from $\prod_l(1-\bar\varepsilon_l)\ge 1-\sum_l\bar\varepsilon_l$. □

Applied to the recursion of [1] as it is written, where the grid sides are astronomically large, the loss is negligible.

**Corollary 4.19.** *The recursion of [1] as written (single block, Theorem A with the grids $g_l=2^{3l+x_0}K_l$ of its (40), $x_0\ge 18$) has exponent at least $(1-\delta)/\log_2 K_l$ on $K_l$-node instances for every $l$, with $\delta=\sum_{l\ge0}4/g_l^2<8.5\cdot10^{-22}$. Hence Corollary 4.15 holds for it with $\log_2 m^*\ge(1-\delta)\,q/\mathrm{bar}$, and for general 3-SAT still $\log_2 m^*\ge 6.2\cdot10^3$.*

*Proof.* Each level of that recursion is a $g_l\times g_l$ grid level with one piece per rectangle and $|I|\le(2\lambda)^c$ (so $\sigma=c\log_2(2\lambda)$), followed, under the merge with the smaller total, by the grouping of the $g_l$ merged nodes into $K_l$ groups of $2^{3l+x_0}$ nodes (its Lemma 8, case (ii)) and the padding to a $(K_l,L)$-clique; the base is brute force. Keeping the internal strings of a merged node as settings costs at least the size charge (Remark 4.3). So it is a single-block grid algorithm with $\bar\varepsilon=4/g_l^2$ at level $l$, and every chain meets each level once. By Theorem A(a) of [1], $\log_2 g_l=3l+x_0+(3x_0+6)2^l-6l-2x_0-6$, which is 36 at $l=0$ for $x_0=18$ and grows faster than $2^l$, so $\delta\le 4\cdot2^{-72}(1+2^{-100})<8.5\cdot10^{-22}$. The loss $h\sigma=O(l\log\lambda)$ is $o(\lambda)$. The threshold statement is Proposition 4.1(a) with $C=1-\delta$ and $\alpha'=1$. □

For general single-block grid algorithms the factor depends on the grid sides. If every grid level has both sides at least $g_0$, a chain meets at most $\log_2 K/\log_2 g_0$ grid levels, so $\delta\le 4\log_2 K/(g_0^2\log_2 g_0)$; at $\log_2 K=1/\mathrm{bar}\approx3.1\cdot10^3$, the range that matters for Corollary 4.15, this is about $10^{-3}$ for $g_0=1024$, but for $g_0\le16$ the product $\Pi_A$ is no longer close to 1, so the corollary in the single-block form is proved here only for large sides. The loss is real for unbalanced grids: on an $a\times b$ grid with $a\ll b$ a column pair holds only $\binom{a}{2}$ rectangles, and one of them is lost to rounding. Grids with a side 2 are excluded except at $K=4$, where the input floor gives the bound for every algorithm. On a $2\times b$ level with $b\ge3$ every column pair holds a single rectangle, so a term gives each merged column edge either all of the rectangle's bits or none, and a bound in terms of the shortest edge gains nothing from mixing; the method gives only $1/e(2b)\lesssim 2/e(b)$. We do not know whether the bound $1/e(K)\le\log_2 K$ can fail for such levels; that would need an algorithm that is markedly cheaper on an instance with two edge lengths than on the uniform instance of the mean length, which is the robustness question of Remark 4.20 for grids.

**Remark 4.20** (Homogeneity and the general balanced case)**.** (F1) holds for every balanced level, homogeneous or not. (F2) needs homogeneity: four-cell configurations can sit on a few pairs of groups while $t$ stays small. For instance, with $P_1$ consisting of groups $G_1, \dots, G_{n_1}$ of size $c$ and $P_2$ of pairs $\{u, v\}$ with $u \in G_a$, $v \in G_{a+1}$ (cyclically), every pair $(G_a, G_{a+1})$ is entirely in four-cell configurations while $t$ is about $3c/K$; under merge 1 the merged instance is a clique with a Hamiltonian cycle of light edges. The class has no lower bound for such near-uniform instances except restriction to a uniform sub-clique, which loses a constant factor in the exponent per level, too much for a polylogarithmic induction. The general balanced case of Theorem 4.12 therefore needs a robustness lemma for near-uniform weights (Section 4.8).

For balanced homogeneous two-merge algorithms, Theorem 4.12 gives the following consequences, weaker in the threshold but valid for a larger class than the grid algorithms.

**Corollary 4.21.** *Let a problem be reduced to uniform clique sums with $s$ edge bits per variable and solved by an algorithm of $\mathrm{RM}_2^{\mathrm{bh}}$ whose base methods satisfy the base condition. Then:*

*(a) on $K$-node instances the running time is at least $2^{S/(\log_2 K)^3}$ with $S = sn$; with $K \le m$ and $m$ polynomial in $n$, this is $2^{\Omega(sn/\log^3 n)}$, so no running time $2^{n^{1-c'}}$ with $c' > 0$ arises when $s$ is bounded below by a positive constant;*

*(b) in the threshold model of Section 4.1, $\log_2 m^* \ge q\,\mathrm{bar}^{-1/3}$; the threshold grows exponentially in a power of $1/\mathrm{bar}$, not polynomially as under polynomial decay.*

*Proof.* (a) is Theorem 4.12; (b) is Proposition 4.1(a) with $C = 1$ and $\alpha' = 3$. □

For general 3-SAT with bar $\approx 3.2 \cdot 10^{-4}$, (b) gives only $\log_2 K^* \ge \mathrm{bar}^{-1/3} \approx 15$, i.e. $\log_2 m^* \ge 29$ at $q = 2$, which is not galactic; the quantitative statement is the one for grid algorithms, Corollary 4.15. Among balanced homogeneous two-merge levels the proved exponent is 3; lowering it to 1 for the whole class is the quantitative open problem of this section.

## 4.6 Size arguments at fixed $K$

Theorem 4.12 is asymptotic. At a fixed small $K$ the natural lower-bound tool is the size of the tables that the terms must materialise, which by Proposition 4.7 bounds every evaluation of them, shared or not. We define the *size game* of a one-level layout: the edges of $K$ are raw or grouped into crossings (a crossing is the configuration of two disjoint edges $[x = v][y = u]$ on four nodes, the rectangle of the grid); every crossing is rewritten by a per-block family of pieces with given $\log_2$-ranks across the splits of its four legs; a term chooses any contraction tree; the value is the maximum over the adversary's mixtures of the minimum over trees of the largest table, as an exponent of $N = 2^\lambda$, and the exponent per edge bit is the value divided by $\binom{K}{2}$. Only sizes enter: no entropy of naming the pieces and no bilinear cost of building the tables. Two families are used: *ideal pieces*, costs $(2, 0, 2)$ and $(0, 2, 2)$ across (rows, columns, internal split), which satisfy every constraint we know; and the pieces of [1, App. B], with costs $(\sqrt{2+u}, \sqrt{2-u})$ and the least internal cost $|p - q|$ that the rank inequality below allows.

At $K = 6$ the $2 \times 3$ grid with ideal pieces has value between 6.667 and 6.725 ($e = 0.444$–$0.448$), with the pieces of Appendix B 6.988 ($e = 0.466$), against 8 for the best tree ($e = 8/15 = 0.533$) and 5 for the input floor. Among 280 random layouts of one to seven crossings at $K = 6$ with ideal pieces the best value found is 7.0, none below the grid. So among the layouts tried the grid

is the best. As discussed after Table 3, even with ideal pieces one-level layouts move away from the input floor as $K$ grows, while staying a constant factor below the tree.

By Proposition 4.7, shared computation can lower the cost of a construction at most to its size-game value. For the pieces of Appendix B at $\omega = 2$ the constructions of Section 2 cost 3.547 (per-block accounting; 3.566 in the grid model used at larger $K$), at 7.357–7.360 and 15.66–15.73 at $K = 4, 6, 9$, against size floors 3.414, 6.988 and 14.69: relative gaps of the construction's cost of 3.7%, 5.0%, and 6.2–6.6%: a saving of bounded relative size, not one that grows with $K$ as polynomial decay would need.

**Why no size argument decides a fixed $K \geq 6$ (argued).** A term is a tensor product of pieces on disjoint legs, and the flattening rank of such a product across any cut is the product of the factors' ranks across the induced splits. Hence every linear inequality between flattening log-ranks (subadditivity, complement symmetry, the inequality of Proposition 4.22, the inequalities of Cadney, Huber, Linden and Winter [7], under any grouping of the nodes into parties) holds for every term as soon as it holds for every piece under the induced grouping of its four legs. The only non-multiplicative constraint is the pigeonhole, which is per block. Functional pigeonholes obtained by restricting to a sub-clique give floors of at most $K' - 1 \leq 5$ at $K = 6$, below 6.667. So the size question at any fixed $K$ reduces to the question which families of four-leg pieces exist, and no constraint at $K \geq 6$ excludes what $K = 4$ allows. At $K = 4$ the ideal pieces reach the input floor exactly, so every size argument is tight there; whether they exist is the question of the per-level factor in the window $\rho \in [1/2, 1/\sqrt{2}]$ (Proposition 4.8 and [1, App. B]), equivalently of the $K = 4$ cost in $[3, 3.547]$ at $\omega = 2$. A one-parameter family $\{(2, x, 2 - x), (x, 2, 2 - x)\}$, $0 < x < 1$, is also admissible and might give less than 6.667 at $K = 6$; it has not been computed.

**One level of pair pieces (argued).** With the mixture 1/2 on every crossing, every split crossing costs at least 1; any contraction tree over $K$ leaves has a node with between $K/3$ and $2K/3$ leaves, and at that cut at most $5K^2/36$ of the roughly $K^2/4$ crossings are unsplit, so the largest table is at least $K^2/9 - K/4$ and $e(K) \geq 2/9 - O(1/K)$ for every one-level layout with ideal pair pieces, against the tree's limit 4/9. One level is worth at most a factor 2, the two-merge floor seen from the other side.

**The third cut at $K = 4$.** A crossing on four nodes has three splits; the two merges use two of them. The following is the four-party rank inequality conjectured by Cadney, Huber, Linden and Winter [7, Hypothesis 1] and proved by Song, Chen, Sun and Hu [8, Thm. 9], read for a pure four-party state, i.e. a four-tensor.

**Proposition 4.22** ([8])**.** *For every complex tensor $T$ with four legs $A, B, C, D$,* $\operatorname{rank}(T; BC \mid AD) \leq \operatorname{rank}(T; AB \mid CD) \cdot \operatorname{rank}(T; AC \mid BD)$.

For a piece with costs $p, q$ on the two crossing splits and $r$ on the internal split this gives $r \geq |p - q|$: a piece that is full on one merge split and free on the other is full on the third. With this inequality the three-cut size game at $K = 4$ (each of the three matchings of $K_4$ raw or rewritten, a term choosing one of the three cuts) is no longer conditional: over all two-piece families on a grid of step 1/4 in the parameters, the value is at least 3, the two-cut ideal value, so the third cut never lowers the size floor below the input floor (computed on the grid, not proved for all real parameters). Without the per-matching structure, flattening inequalities alone admit families below the input floor, and what excludes them is again a quantum functional, as in Example 4.10.

### 4.7 Three escapes and the splits that block them

Three ways around the obstructions of this section fail for one reason: each leaves intact a split of the strings of a configuration across which $f_C|_{\text{block}}$ has full rank, so Proposition 4.6 (whose proof applies verbatim to any split of the strings) gives a piece full across it, and the escape has no evaluation in which that piece is cheap. In the first two this gives $\rho = 1$, in the third $\rho \geq 2/3$.

**Identity search.** In $AG(2,3)$ with rows, columns and diagonals $x - y = \text{const}$ as merges $k = 0, 1, 2$, two of the lines $x + y = \text{const}$, $\{a_i\}$ and $\{b_i\}$ ($i \in \mathbb{Z}_3$), give a configuration $C$ of six edges that merge $k$ meets in the groups $\{a_i, b_{i+k}\}$, each pair of groups carrying one edge of each triangle: the candidate for $\rho < 1/2$ with three orthogonal merges (Section 4.8). $C$ meets three groups of each merge, outside (L3); the identities searched for it, and every rewrite of its crossings one at a time, are of the following form. A *bundle* is the pair of edges of $C$ between two groups of $P_k$; a piece is *$k$-local* if it is a tensor product of one factor on the strings of each bundle, and it is charged $\mathrm{c}_k(\pi)$, the sum of the factors' $\log_2$-ranks across their pairs of groups, as (L4) would charge each bundle on its own. Let $X_{ij}$, $Y_{ij}$ be the strings of $a_i a_j$, $b_i b_j$ at $a_i$, $b_i$, call them forward if $j = i + 1$ and backward if $j = i - 1$, and let $Q$ split the twelve strings into forward and backward ones.

**Proposition 4.23.** *(a) The rank of $f_C|_{block}$ across $Q$ is $N^6$. (b) Every $k$-local piece $\pi$ has $\log_2 \operatorname{rank}(\pi; Q) = \mathrm{c}_k(\pi)$. (c) Hence, if every piece of a block identity on $C$ is $k$-local for some $k$, some piece costs at least $6m - \log_2 |I_C|$ under every merge for which it is local, and the term taking it in every block pays the full $6\lambda - o(\lambda)$ bits of $C$ under whichever merge evaluates it: $\rho = 1$ on $C$. An identity of products of factors on pairs of strings inside groups has at least $N^6$ pieces.*

*Proof.* The directions $j - i$ and $i - j$ of $X_{ij}$ and $X_{ji}$ differ since 3 is odd, and likewise for $Y$; so all six equalities cross $Q$, which gives (a). The bundle of merge $k$ on $\{a_i, b_{i+k}\}$, $\{a_j, b_{j+k}\}$ has the strings $X_{ij}, Y_{i+k,j+k}$ of direction $j - i$ on one group and $X_{ji}, Y_{j+k,i+k}$ of direction $i - j$ on the other: shifts preserve direction, so $Q$ restricted to a bundle is the split between its groups, and ranks across $Q$ multiply over the factors, (b). (c) is Proposition 4.6 across $Q$ with (b), since a term evaluated under merge $k$ has only $k$-local pieces on $C$; pair factors inside groups have cost 0. □

Accordingly a search of the last form ($N = 2$, $L$ products per merge) finds the identity of [1, App. H] on the rectangle at $L = 1$ and nothing on $C$ (*computed* residuals 0.98, 0.95 at $L = 1, 2$; $3L < 64$). The proof applies to two copies of $K_n$ glued by the $n$ cyclic shifts for every odd $n$ (for even $n$ the edges of direction $n/2$ do not cross $Q$). Left open on $C$ are pieces coupling different bundles, hyperedges under the merge, for which Section 4.8 gives $\rho \geq 1/2$.

**Structured merged tables (argued).** A level grows $K$ by a constant factor only if one merge has groups of constant size $g$; on a grid the other merge then has $g$ groups and the input floor $2/g$. Using that merged tables are contractions (leaving (L5)) does not help within rewrite-and-merge. The term with, in every rectangle, a piece full across the row split pays it in full in every step bilinear across a cut inducing that split (Proposition 4.7); so a cheap evaluation keeps the two corners in each column of each rectangle together, and hence whole columns. Building the column tables is the column merge with its floor; keeping them as networks leaves each column's internal clique ($1/g$ of the bits) untouched and an instance on all $K$ nodes, so the recursion does not advance.

**The shape of App. B.** A graded space $V = \bigoplus_E V_E$ is given, $D(E) = \log_2 \dim V_{\le E}$, and $h$ is the least $E$ with $D(E) \ge \lambda$; the strings $x, v$ index operator labels $a, b$ and $y, u$ index states, all of weight at least $h$; for every integer $s \in [-H, W]$ the piece $A_s$ factors across the row split $(x, y \mid u, v)$ through $V_{\le s+H}$ and across the column split through $V_{\le W-s}$, with counted bits $c_s^{\mathrm{row}} \ge D(s+H) - o(\lambda)$, $c_s^{\mathrm{col}} \ge D(W-s) - o(\lambda)$. In App. B, $W = \operatorname{wt} b + \operatorname{wt} y = 2h$ ($\operatorname{wt} y$ the weight of $y$'s state) and $H = \operatorname{wt} a + \operatorname{wt} b = 2h$ (Lemma 9). Let $\rho_B = \max_s (c_s^{\mathrm{row}} + c_s^{\mathrm{col}})/(4\lambda)$, the $\rho$ of [1, Def. 7].

**Proposition 4.24.** *In this shape let $(W+H)/2 \ge \mu h$, $\mu \ge 1$, let $D$ be concave on $[h, W+H]$ and $\log_2 |I| = o(\lambda)$. Then $\rho_B \ge \frac{1}{2}(1 + \frac{\mu-1}{2\mu-1}) - o(1)$; in particular $\rho_B \ge 2/3 - o(1)$ when $W, H \ge 2h$, as in App. B.*

*Proof.* Put $M = (W+H)/2$. The pieces sum to $d\,[x = v][y = u]$, of rank $2^{2\lambda}$ across the row split, and $A_s$ has rank at most $2^{D(2M)}$ there, so $D(2M) \ge 2\lambda - \log_2 |I|$ (Proposition 4.6). The shift $s^* = \lfloor (W-H)/2 \rfloor$ has both windows at least $M' = \lfloor M \rfloor$. With $t = (M'-h)/(2M-h)$, concavity gives $D(M') \ge (1-t)\lambda + tD(2M)$, so $4\lambda\rho_B \ge c_{s^*}^{\mathrm{row}} + c_{s^*}^{\mathrm{col}} \ge 2\lambda(1+t) - o(\lambda)$; and $(M-h)/(2M-h)$ increases in $M$, so $t \ge (\mu-1)/(2\mu-1) - 1/h$. □

Equality in the concavity step holds exactly for $D$ affine on $[h, 2M]$ (for $M = 2h$: slope $\lambda/(3h)$, a third of the mean slope below $h$). For $D(E) = cE^\alpha$ and $W, H \ge 2h$ the middle piece gives $\rho_B \ge \max(2^{\alpha-1}, 2^{-\alpha}) \ge 1/\sqrt{2}$, with equality only at $\alpha = 1/2$, so App. B is optimal among power profiles. Within the shape $\rho_B \in [2/3, 1/\sqrt{2}]$, and below $1/\sqrt{2}$ one needs a *kinked* profile (fast growth up to the weight of the strings, slower beyond) with Lemmas 9 and 10 of [1]; we know no such algebra. A window $H$ as small as the weight of one factor ($\mu = 3/2$) would give 5/8.

**Fermionic Fock space (argued).** The exterior algebra on $n$ modes, $D(E) = \log_2 \sum_{j \le E} \binom{n}{j}$, has a nearly linear, then flat profile, but even in the shape above with Lemma 9's windows its counted factor is at best 0.763 (*computed*, $n = 2000$; 0.660 needs $H = h$). And it has no such pieces: App. B's $Q_s$ factor in both orders (Lemma 9(a)), a consequence of vertex-algebra locality, while Wick ordering $\psi_T \psi_S^* = \sum_{U \subseteq S \cap T} \pm \psi_{S \setminus U}^* \psi_{T \setminus U}$ ($x, v$ encoded by $h$-sets $T, S$, $y, u$ by states of weight $h$; the term $U = S = T$ is $[x = v][y = u]$) gives pieces that factor in one order only. The reordered terms with $|U| = j$ factor across the row split through the shared $U$ and a state of weight $j$, at $\log_2$-rank up to $2\log_2 \binom{n}{j} = 2\lambda - O(\log(n/h))$ for $j = h-1$, and are dense across the column split. So one piece is nearly full across both splits, unlike the pieces of Example 4.10, and $\rho \to 1$. The infinite free-fermion vertex algebra is local but has, by the boson–fermion correspondence, the partition profile; truncating its modes at $n$ kinks the profile only at weights of order $n^2$, where locality fails.

## 4.8 What remains open

The first three items below would sharpen or extend the bounds of this section without changing their message; the last is the only place where the message itself could change.

**The general four-cell case.** The rectangle-content bound of Section 4.4, that only $\min(d, a)$ bits of a four-cell configuration can be traded, is open. Its smallest open case is $K_4$ minus a diagonal; the 4-cycle is open when over-full pieces are allowed. It would sharpen the exponent of Theorem 4.12, not its structure.

**Non-homogeneous balanced levels.** Theorem 4.12 for all balanced two-merge levels needs a robustness lemma: a clique sum whose weights are uniform except on a sparse set of pairs (such as the cycle of Remark 4.20) costs nearly as much as the uniform one, in a form that loses only $1 - o(1)$ in the exponent per level. We believe it, but it requires the level analysis for weighted instances.

**Three or more merges.** With $p$ merges the count applies to every pair of them. In the $d$-dimensional grid with line merges, an edge whose endpoints differ in $r$ coordinates lies in one $r$-subcube, and the facet cuts of that subcube are unions of groups of all but one of its merges; the pigeonhole then forces every identity to pay at least $1 - 1/r$ of the subcube's bits, so line merges are worse than the two-dimensional grid. Three pairwise orthogonal merges can have a floor as low as $1/3$, but orthogonality again forces squaring. Below $1/2$ one would need identities beyond those we know: we argue that pieces that are products of factors on disjoint pairs of strings cannot go below $\rho = 1/2$ on the smallest configuration that escapes the known obstructions, the translation matching of the affine plane $AG(2, 4)$ with three of its other parallel classes as merges; and for every configuration whose nodes split into two halves, each meeting every block of every merge at most once, the uniform quantum functional proves $\rho \geq 1/2$ for all pieces. What is not covered is a piece that is not a product over pairs of strings on a configuration without such halves: on $AG(2, 4)$ this means a hyperedge over four blocks of the third merge, whose cost the model does not yet define. With $p$ merges the analogue of Lemma 4.11 concerns $2^p$ cells. Even a success here would be a lever on $\rho$, i.e. on $\alpha'$, not a change of the type of decay.

**The open door: hyperedge configurations.** Inside $\mathrm{RM}_2$ every configuration meets at most two groups of each merge, and Remark 4.5 shows that this is exactly what a non-product grouping of terms across different merged edges violates. We state the role of such configurations as a necessary condition, not a sufficient one. Within the class, and modulo the counting lemma's premise in the general four-cell case, polynomial decay of $e(K)$ needs a size-game value that decays polynomially, and by Sections 4.4 and 4.6 only pieces of growing arity whose configurations meet three or more groups of a merge (equivalently, non-product groupings of terms) could give one. Such pieces would also have to be realisable, which no current tool decides, since rank inequalities for six or more parties are incomplete; their tables would have to be buildable in time close to their size, which is a bilinear-complexity question; a block would have to need few of them, since their naming entropy is paid per block; and the constants $C$ and $\beta$ would have to be moderate, since by Proposition 4.1 the threshold is $\log_2 m^* \approx (q/\beta) \log_2(C/\mathrm{bar})$ plus the lower-order constants of the reduction. No construction gives evidence of polynomial decay, and for the full model of Signed Network Sums no barrier is known: the tools used here (ranks, flattenings, quantum functionals) bound pieces, not the way they are combined beyond pairs of groups.

# 5 Open problems

The problems are ordered from the small end to the large end. The first four would sharpen the picture of Sections 2 and 4; the fifth is the only one whose solution could give a usable threshold inside Signed Network Sums; the last two would lower the thresholds of Section 3 without changing their scale.

1. *The window at four nodes.* At $K = 4$ every lower-bound tool we know stops at the input floor $N^3$, and the best construction at $\omega = 2$ uses Appendix B pieces of [1]. Is there a four-leg piece family with per-merge factor $\rho$ below $1/\sqrt{2}$ whose tables are built near their size? Equivalently, close the window $\rho \in [1/2, 1/\sqrt{2}]$; at fixed $K$ every table-size argument reduces to this question (Section 4). Inside the shape of the pieces of Appendix B of [1], every concave profile gives $\rho \geq 2/3$ (Proposition 4.24), so $\rho \in [1/2, 2/3)$ needs pieces of a different shape, and $\rho \in [2/3, 1/\sqrt{2})$ a kinked profile.

2. *Every-K separation.* Do one-level constructions beat every contraction tree at every $K \geq 4$, on the current matrix-multiplication curve and not only at $\omega = 2$? At $K = 6$ the construction beats the best tree on the current curve (8.017 against 9) but not the width bound 8.

3. *The four-cell lemma.* Does a configuration on all four cells of two merges trade at most its rectangle content? The smallest open case is $K_4$ minus one diagonal. It would only sharpen constants in the class barrier.

4. *Robustness and more merges.* Extend the class barrier to balanced but non-homogeneous partitions (a robustness lemma for near-uniform weights; for grid levels with both sides at least 3 the near-uniform case is settled by Theorem 4.18, and $2 \times b$ levels with $b \geq 3$ remain open) and to three or more merges (the four-cell lemma for $2^p$ cells; the open case is non-pair pieces on pairwise orthogonal designs).

5. *The open door.* Is there an exact identity whose configurations meet three or more groups of a merge and whose pieces are cheap? A usable threshold for general 3-SAT inside Signed Network Sums needs one; it would also have to be realisable, buildable near its size, and use few pieces per block.

6. *A linear-size kill.* Can the assignments that violate many clauses be eliminated with $O(n)$ edge bits? This would remove the factor $\log \log n$ from the general bound and lower the general threshold by orders of magnitude, though not to a usable size.

7. *Clustering for the variable network.* Prove the clustering lemma for the variable network of Theorem D of [1]; it is the condition of row G4, and one of the two conditions of row G7, of the general table of Section 3.

# References


[1] Claude (Anthropic), *A $2^{o(n)}$ time algorithm for 3-SAT*, 26 October 2026, Hexagon preprint hexagon:2610.00577v1 (27 October 2026), `https://hexagonmath.org/2610.00577v1`; also `https://github.com/anthropics/formal-math/blob/16a5ce41304fcdaea4e0be7c132e1c361dc26da5/eth/paper.pdf`.

[2] OpenAI, *The Exponential Time Hypothesis is false*, 17 October 2026, revised 25 October 2026, `https://github.com/openai/math/blob/main/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026/paper.pdf`.

[3] S. Szeider, *OpenAI's claimed refutation of ETH: exhaustive search below* $2^{131071}$, Zenodo, 27 October 2026, with an addendum on [1], `https://doi.org/10.5281/zenodo.23688832`.

[4] V. Tulsyan, *Crossover: where the ETH-refutation algorithms first beat brute force*, note in the repository *Hamiltonian cycles after the ETH refutations*, 29 October 2026, `https://github.com/xennygrimmato/eth-hamiltonicity/blob/1fae6d9b7eb8d5ca2a5f758430752baf87e88242/research/crossover.md`.

[5] M. Christandl, P. Vrana, J. Zuiddam, *Universal points in the asymptotic spectrum of tensors*, arXiv:1709.07851 / STOC 2018; and *Asymptotic tensor rank of graph tensors: beyond matrix multiplication*, Computational Complexity 28 (2019), arXiv:1609.07476.

[6] P. Austrin, P. Kaski, K. Kubjas, *Tensor network complexity of multilinear maps*, Theory of Computing 18(16) (2022) 1–54; ITCS 2019; arXiv:1712.09630.

[7] J. Cadney, M. Huber, N. Linden, A. Winter, *Inequalities for the ranks of multipartite quantum states*, Linear Algebra Appl. 452 (2014) 153–171, arXiv:1308.0539.

[8] Z. Song, L. Chen, Y. Sun, M. Hu, *A complete picture of the four-party linear inequalities in terms of the 0-entropy*, arXiv:2105.07679 (2021).

[9] V. Vassilevska Williams, Y. Xu, Z. Xu, R. Zhou, *New bounds for matrix multiplication: from alpha to omega*, SODA 2024.

[10] Amanov, van den Berg, Zuiddam, arXiv:2610.30054 (2026), preprint.

[11] P. Uznański, arXiv:2610.10494 (2026), preprint.

[12] C. Calabro, R. Impagliazzo, R. Paturi, *A duality between clause width and clause density for SAT*, CCC 2006.

[13] S. DeSalvo, I. Pak, *Log-concavity of the partition function*, Ramanujan J. 38 (2015).

[14] D. H. Lehmer, *On the remainders and convergence of the series for the partition function*, Trans. AMS 46 (1939).

[15] J. Alman, B. Li, *Asymptotic rank speedup theorems, revisited*, arXiv:2605.21738 (2026), preprint.

[16] OpenAI, *An upper bound of 9/4 for the matrix multiplication exponent*, 2 October 2026, `https://github.com/openai/math` (item 107), and the companion *Complex matrix multiplication below 2.258 and rectangular bounds*, 24 September 2026; not peer-reviewed, Lean formalisation claimed.

[17] P. Bürgisser, M. Clausen, M. A. Shokrollahi, *Algebraic Complexity Theory*, Springer, 1997.

[18] L. Lovász, *On the ratio of optimal integral and fractional covers*, Discrete Math. 13 (1975) 383–390; S. K. Stein, *Two combinatorial covering theorems*, J. Combin. Theory Ser. A 16 (1974) 391–397.

[19] J. Peng, Z. Sheng, M. Xiao, *New algorithms for #2-SAT and #3-SAT*, IJCAI 2025, arXiv:2507.14504.

[20] K. Kutzkov, *Complementary branching for #SAT: bounded-degree bounds and faster #3-SAT*, Proc. IJCAI 2026, 2282–2290.

[21] K. Kutzkov, *New upper bound for the #3-SAT problem*, Inform. Process. Lett. 105 (2007) 1–5.

[22] S. C. Liu, *Chain, generalization of covering code, and deterministic algorithm for k-SAT*, ICALP 2018, arXiv:1804.07901.

[23] T. Jiang, S. Cai, *A better analysis for PPSZ for 3-SAT*, arXiv:2607.10697 (2026), preprint.

[24] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976 (Theorem 14.5).

[25] C. Calabro, *The Exponential Complexity of Satisfiability Problems*, PhD thesis, UC San Diego, 2009.

[26] H. Dell, T. Husfeldt, D. Marx, N. Taslaman, M. Wahlén, *Exponential time complexity of the permanent and the Tutte polynomial*, ACM Trans. Algorithms 10 (2014), arXiv:1206.1775.

[27] M. Xiao, *An exact MaxSAT algorithm: further observations and further improvements*, Proc. IJCAI 2022, `https://www.ijcai.org/proceedings/2022/0262.pdf`.

[28] K. Brilliantov, V. Alferov, I. Bliznets, *Improved algorithms for maximum satisfiability and its special cases*, Proc. AAAI 2023, 3898–3905.

[29] A. Lincoln, V. Vassilevska Williams, R. Williams, *Tight hardness for shortest cycles and paths in sparse graphs*, Proc. SODA 2018; arXiv:1712.08147.

[30] T. Sakai, K. Seto, S. Tamaki, J. Teruyama, *Improved exact algorithms for mildly sparse instances of Max SAT*, Proc. IPEC 2015, LIPIcs 43, doi:10.4230/LIPIcs.IPEC.2015.90.

[31] J. Alman, V. Vassilevska Williams, *OV graphs are (probably) hard instances*, Proc. ITCS 2020, LIPIcs 151, 83.

[32] R. Impagliazzo, R. Paturi, F. Zane, *Which problems have strongly exponential complexity?*, J. Comput. System Sci. 63 (2001), 512–530.

[33] R. Majdoddin, *Uniform CSP parameterized by solution size is in W[1]*, arXiv:1810.04190 (2018).

[34] K. Ye, L.-H. Lim, *Tensor network ranks*, arXiv:1801.02662 (2018).

| lever | effect | status |
|---|---|---|
| base case | brute force $e_0 = \frac{1}{2} \to$ tree 0.2554: 22264 $\to$ 5396 (4.1×) | proved (Lemma 3.7) |
| exact edge-bit total | $U < 2km \to U = 2(km - n)$, bar 0.0881 $\to$ 0.1321: 5396 $\to$ 1538 (3.5×; 1.07× at $\Delta = 4.26$) | proved (count in [1, §6]) |
| group size per level | paper's $2^{3\ell+x_0} \to$ least admissible: 33858 $\to$ 28761 (1.2×) | proved (Prop. 3.2) |
| initial size $x_0$ | 18 $\to$ 7: 28761 $\to$ 22264 (1.3×) | proved ($c = 30$) |
| top balancing $q$ | 9 $\to$ 2.03: 22264 $\to$ 5037 (4.4×); with tree base and exact $U$: 1538 $\to$ 417 | proved (Lemma C, Prop. 3.10) |
| slack schedule $\zeta_\ell$ | paper's $2^{-\ell-5}$ is within 2% of optimal in $\log K$ | proved shape (Rem. 3.3) |
| sharper tails in balancing | $\tau = 9$ (weights in a band of 1/3): 22526 $\to$ 18666 (1.2×) | hypothetical, needs proof |
| overhead constants | $3\zeta, 3/K \to \zeta, 1/K$: 18666 $\to$ 13549 (1.4×) | hypothetical |
| no balancing slack | $s = 1$: 22526 $\to$ 3168 (7×) | hypothetical |
| per-level factor $\rho$ | $1/\sqrt{2} \to \frac{1}{2}$: 22264 $\to$ 2091 (11×) | hypothetical; below 2/3 excluded in App. B's shape for concave profiles (Prop. 4.24), below $\frac{1}{2}$ for two merges (Section 4) |
| growth of $K$ per level | $K_{\ell+1} = 16K_\ell$ instead of squaring: 33858 $\to$ 378 | hypothetical; excluded in rewrite-and-merge (Section 4) |
| general 3-SAT ($\log_2 n^*$ from row G7 of Table 5, $5.0 \cdot 10^8$) | | |
| sparsification density | Calabro $\to \Delta = 1/\varepsilon$: $6.2 \cdot 10^7$; $\to \Delta = 2$: $1.6 \cdot 10^7$ (32×) | parameter (Prop. 3.15) |
| parity-check kill | bits per variable $U/n$ (row G7: about 1400) $\to$ 800: $2.5 \cdot 10^8$; 200: $1.6 \cdot 10^7$; 50: $9.7 \cdot 10^5$; 25: $2.4 \cdot 10^5$ | 800 is the floor of the parity paradigm; below: no construction known |
| $\rho$ (general) | $1/\sqrt{2} \to 2/3 \to \frac{1}{2}$: $5.0 \cdot 10^8 \to 6.2 \cdot 10^7 \to 2.4 \cdot 10^5$ | hypothetical; 2/3 is the floor of the shape of App. B (Prop. 4.24) |

Table 7: Levers of the recursion, density one (first two blocks) and general 3-SAT (last block); main-term values of $\log_2 m^*$ resp. $\log_2 n^*$.

| $K$ (layout) | input floor | ideal pieces | App. B pieces | tree (raw) |
|---|---|---|---|---|
| 4 (2 × 2) | 3 | 3 | 3.414 | 4 |
| 6 (2 × 3) | 5 | [6.667, 6.725] | 6.988 | 8 |
| 9 (3 × 3) | 8 | $\geq 13.90$ | $\geq 14.69$ | 18 |

Table 8: Size-game values (exponents of $N$; divide by $\binom{K}{2}$ for the exponent per edge bit). The ideal value at $K = 6$ is a search value (6.667) and a grid bound with a Lipschitz estimate (6.725); the other values with nontrivial pieces at $K = 6, 9$ are search values, i.e. lower estimates of the maximum over mixtures. Computed by `calc/r123_s2_sizegame.py`.

# Appendices

## A Values of the one-level constructions at $K = 9$ and $K = 16$

Table 9 lists, for each construction, the certified interval over all term profiles (lower end: the value at the worst profile found in closed form; upper end: the certified bound) on three curves. G is the eight-square expansion grouped by naming bits, E the grouping counted by ranks, EM the rank count for at most half column-choice blocks and G above, B the Appendix B pieces with $m \to \infty$. The finite-$m$ rows use exact partition numbers. Scripts: calc/r122d_certify.py, r122d_tables9.py, r122d_subst.py, r122d_bfinite.py.

| $K$ | construction | $\omega = 2$ | current curve | VXXZ |
|---|---|---|---|---|
| 9 | best tree found | 18 | 20.25 | 21.344 |
| 9 | G | $[17.333, 17.721]$ | $[19.125, 19.50]$ | $[20.158, 20.554]$ |
| 9 | EM | $[17.269, 17.657]$ | $[18.908, 19.428]$ | $[19.929, 20.478]$ |
| 9 | E | $[17.204, 17.593]$ | $[18.908, 19.356]$ | $[19.929, 20.402]$ |
| 9 | B, $m \to \infty$ | $[15.656, 15.725]$ | $[16.295, 17.614]$ | $[17.176, 18.566]$ |
| 9 | B, $m = 256$ / 512 | 16.456 / 16.187 | – / 18.019 | – |
| 16 | best tree found | 60 | 64 | 65.376 |
| 16 | G | 60 | $[60, 62.25]$ | $[61.28, 64.42]$ |
| 16 | EM | $[59.229, 59.615]$ | $[59.229, 61.865]$ | $[60.50, 64.04]$ |
| 16 | E | 59.229 | $[59.229, 61.480]$ | $[60.50, 63.65]$ |
| 16 | B, $m \to \infty$ | $[49.94, 57.95]$ | $[49.94, 59.95]$ | $[51.01, 60.92]$ |
| 16 | B, $m = 512$ | 59.718 | 61.72 | – |

Table 9: Exponents of $N$ at $K = 9$ (width bound 18) and $K = 16$ (width bound 60).

## B Computed schedules and the constant $c$

### B.1 The optimal admissible schedules

Table 10 lists, level by level, the schedules behind columns (X) and (T) of Table 4 at densities 1 and 4.26: $\log_2 K_\ell$, $e_\ell$, and the chosen $\log_2 \zeta_\ell$ and $\log_2 s_\ell$, where $s_\ell$ is the least value with (S1) and (S2). The exponents decrease along every schedule, so (S4) holds, and (S3) holds since $x_0 \geq 4$ and $c = 30$. Values are rounded; the model is that of `calc/r123_levers.py` (main term, $\rho = 1/\sqrt{2}$, $\tau = 1$, overheads $3\zeta$ and $3/K$).

### B.2 The requirements on $c$ on the exact range

Table 11 gives, for selected $\lambda$, the least $c$ that each condition of Definition 7 of [1] requires for the pieces of its Theorem B, with partition numbers computed exactly; $\Delta$ is the number of partitions of $0, \ldots, 4h$ and the last column is the maximum before the $+2$ of the truncation.

## C Size-game computations

The values of Table 8 were computed by `calc/r123_s2_sizegame.py`. A layout lists the crossings; a term's profile assigns to each crossing type the adversary's mixture parameter; for each profile

| density 1, brute-force base, $x_0 = 7$ | | | | | density 1, tree base, $x_0 = 8$ | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $\ell$ | $\log_2 K_\ell$ | $e_\ell$ | $\log_2 \zeta_\ell$ | $\log_2 s_\ell$ | $\ell$ | $\log_2 K_\ell$ | $e_\ell$ | $\log_2 \zeta_\ell$ | $\log_2 s_\ell$ |
| 0 | 7 | 0.500 | −4.5 | 12.2 | 0 | 8 | 0.255 | −4.5 | 12.3 |
| 1 | 38.5 | 0.443 | −6.0 | 17.0 | 1 | 40.7 | 0.226 | −6.0 | 17.1 |
| 2 | 111 | 0.334 | −7.0 | 20.4 | 2 | 116 | 0.171 | −7.5 | 21.5 |
| 3 | 263 | 0.244 | −8.5 | 24.6 | 3 | 274 | 0.123 | −9.0 | 25.6 |
| 4 | 575 | 0.175 | −9.5 | 27.7 | 4 | 600 | 0.0880 | | |
| 5 | 1205 | 0.124 | −11.0 | 31.7 | | | | | |
| 6 | 2474 | 0.0880 | | | | | | | |
| density 4.26, brute-force base, $x_0 = 7$ | | | | | density 4.26, tree base, $x_0 = 8$ | | | | |
| 0 | 7 | 0.500 | −5.5 | 14.3 | 0 | 8 | 0.255 | −5.5 | 14.4 |
| 1 | 42.7 | 0.410 | −6.5 | 18.2 | 1 | 44.9 | 0.209 | −7.0 | 19.3 |
| 2 | 122 | 0.304 | −8.0 | 22.5 | 2 | 128 | 0.153 | −8.0 | 22.6 |
| 3 | 289 | 0.218 | −9.0 | 25.7 | 3 | 302 | 0.110 | −9.5 | 26.8 |
| 4 | 629 | 0.156 | −9.5 | 27.8 | 4 | 657 | 0.0781 | −10.0 | 28.9 |
| 5 | 1313 | 0.111 | −10.5 | 30.8 | 5 | 1372 | 0.0555 | −11.0 | 31.9 |
| 6 | 2687 | 0.0785 | −11.5 | 33.9 | 6 | 2808 | 0.0393 | −13.0 | 36.9 |
| 7 | 5442 | 0.0556 | −12.5 | 36.9 | 7 | 5689 | 0.0278 | | |
| 8 | 10957 | 0.0393 | −14.0 | 40.9 | | | | | |
| 9 | 21996 | 0.0278 | | | | | | | |

Table 10: Admissible schedules behind Table 4; the last row of each block is the top clique, $\log_2 m^* = 9 \log_2 K_L$ (main term, $q = 9$).

the script finds the contraction tree minimising the largest table by dynamic programming over subsets of nodes, and the value is the maximum over profiles, found by search. Search values are therefore lower estimates of the maximum over mixtures, and the margins below the tree derived from them are upper bounds. Checks: for raw layouts the value equals the best tree (4, 8, 18 at $K = 4, 6, 9$), and ideal pieces at $K = 4$ give the input floor 3. For the $2 \times 3$ grid with ideal pieces a grid evaluation over profiles with a Lipschitz bound gives the upper value 6.725 (grid maximum 6.65). The worst profiles found are listed in the output file.

| $\lambda$ | $h$ | $\log_2 \Delta$ | $\max(c^{\mathrm{row}} + c^{\mathrm{col}})$ | (b) | (c1) | (c2) | (c3) |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 6.1 | 8 | 5.17 | 3.17 | 6.97 | **27.20** |
| 2 | 4 | 9.8 | 14 | 4.17 | 2.04 | 4.38 | 20.76 |
| 4 | 8 | 15.4 | 21 | 3.23 | 1.68 | 3.56 | 15.31 |
| 8 | 17 | 24.4 | 34 | 2.84 | 1.53 | 3.20 | 11.44 |
| 12 | 29 | 33.4 | 46 | 2.63 | 1.50 | 3.12 | 10.05 |
| 25 | 86 | 61.3 | 85 | 2.53 | 1.49 | 3.09 | 8.37 |
| 50 | 271 | 113.7 | 158 | 2.50 | 1.52 | 3.12 | 7.43 |
| 100 | 927 | 216.3 | 303 | 2.64 | 1.55 | 3.18 | 6.84 |
| 200 | 3361 | 419.1 | 589 | 2.70 | 1.59 | 3.24 | 6.49 |
| 271 | 5988 | 562.3 | 792 | 2.81 | 1.60 | 3.27 | 6.39 |

Table 11: Requirements on $c$ per condition of Definition 7 (`calc/r123_trace_c.py`). The maximum over $1 \le \lambda \le 271$ is 27.198 at $\lambda = 1$; beyond, at most 9.59 (Lemma 3.4).

| crossings | layouts | least value | median value |
|---|---|---|---|
| 1 | 40 | 8.000 | 8.000 |
| 2 | 40 | 7.000 | 7.000 |
| 3 | 40 | 7.000 | 7.500 |
| 4 | 40 | 7.000 | 7.500 |
| 5 | 40 | 7.286 | 7.667 |
| 6 | 40 | 7.500 | 8.000 |
| 7 | 40 | 7.885 | 8.137 |

Table 12: Random layouts of crossings at $K = 6$ with ideal pieces (search values; the least value 6.9999 for four crossings is rounded). None is below the $2 \times 3$ grid (6.667–6.725). Computed by `calc/r123_s2_layouts.py`.